Motion-Robust Accelerated MRI using Deep Learning

Laurens Beljaards

Colophon

About the cover:

The cover depicts an MRI machine and motion patterns estimated with low temporal resolution (red line) and high temporal resolution (green line). The bottom left panel showcases a motion-corrupted MRI scan, while the bottom right panel displays the same scan after applying the proposed correction approach. The composition was placed manually and style transferred to an old comic book style using diffusion models from Copilot and ChatGPT.

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Co-promotor:	Dr. N. Pezzotti
Leden promotiecommissie:	Prof. dr. ir. B.P.F. Lelieveldt (secretaris) Prof. dr. J.A. Schnabel *Technical University of Munich, München* Dr. M.W.A. Caan *Amsterdam University Medical Center, Amsterdam* Dr. D.H.J. Poot *Erasmus Medical Center, Rotterdam*

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1

Introduction

Magnetic resonance imaging (MRI) is a non-invasive imaging technique for acquiring clinical images of the internal anatomy of patients. In order to generate an image of the relevant body part, MRI measures the subject's raw signal in the frequency domain, which is called k-space [1, 2]. Each point in k-space represents how strongly a particular spatial frequency is present in the MR signal of the object being imaged. A mathematical operation called the inverse Fourier transform converts the measured k-space into a 2D or 3D image, as illustrated in Figure 1.1. In this process, each point in k-space is mapped back into image space, where it contributes a corresponding sinusoidal basis function as shown in Figure 1.2, which are then summed to reconstruct the final image. During an MRI scan, magnetic field gradients and radiofrequency pulses are used to sample k-space line by line along a certain trajectory. Filling more of the center of k-space yields more information about coarse structures, while acquisition of k-lines in the periphery provides the finer details. Therefore, choices about how much of k-space to sample and the way to sample it directly determine image quality, resolution, and scan time [3]. Two different ways to sample k-space are shown in Figure 1.3. Moreover, the interplay between the magnetization, RF-pulses and the gradients determines the contrast of the MRI images. Examples of MRI contrasts are T1-weighted, T2-weighted, proton density-weighted, fluid attenuated inversion recovery (FLAIR) and diffusion-weighted MRI, but many more contrasts are possible. Modern MRI systems acquire data using an array of receiver coils placed around the anatomy. These multi-coil acquisitions record a unique k-space simultaneously per coil. Each coil has a unique spatial sensitivity profile, meaning it receives the MR signal more strongly from nearby tissue and more weakly from distant regions. These coil-specific measurements increase signal-to-noise ratio (SNR) and provide additional spatial encoding information for image reconstruction.

The demand for radiological exams and MRI in particular is higher than ever before, driven by an aging population, expanding clinical indications, wider use of imaging in screening and follow-up examinations, and increasingly complex protocols. Scanner hours, trained MR technicians and radiologist reporting capacity are limited, so small inefficiencies can multiply into long waiting lists and delayed diagnoses. Image quality failures in particular, most commonly caused by patient motion, may force



Figure 1.1: A table of frequencies called k-space is obtained from the MRI scanner. We can transform the data from the frequency domain to the image domain using the inverse fast Fourier transform (IFFT), and from the image domain back to the frequency domain using the fast Fourier transform (FFT).

repeat acquisitions that waste time and result in scheduling issues to accommodate upcoming patients [4].

Artificial intelligence (AI) is often defined as the capability of computer systems to perform tasks that typically require human intelligence, such as reasoning, learning, perception and language understanding [5], though not all of these capabilities are required. AI can address several of the bottlenecks that limit imaging efficiency and reliability. AI-based reconstruction models can recover high-quality images while sampling less MR data, shortening scan times without sacrificing diagnostic content [6, 7]. AI models could also perform automated quality control by detecting patient motion or other acquisition problems in real time. This would allow MR technicians to intervene immediately or trigger adaptive protocols rather than discovering failures only after reconstruction when the patient already left the imaging room or even the hospital [8]. Moreover, models that can retrospectively correct artifacts in motion-corrupted data and restore images of diagnostic quality reduce the need for repeated

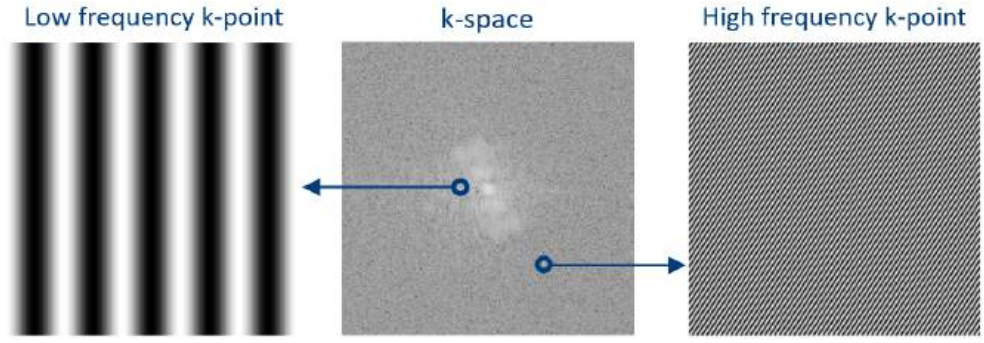


Figure 1.2: Each point in k-space corresponds to a spatial frequency that appears as a sinusoidal pattern in image space. Summing all such patterns reconstructs the final image.

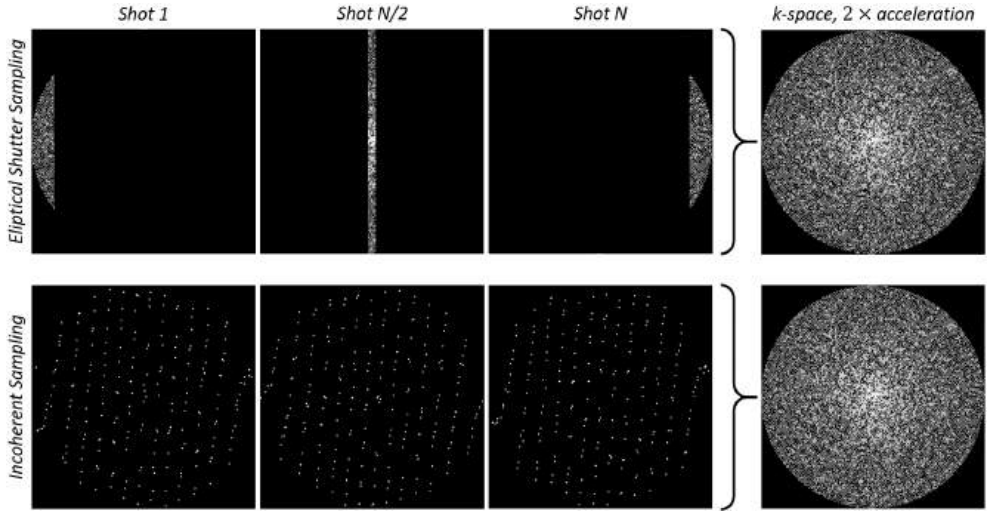


Figure 1.3: Top: A 3D k-space being filled conventionally from left to right, which each point in the 2D slice representing a line of frequencies. Since about half of the lines are never acquired, the acquisition is accelerated by $2\times$. Bottom: k-space can be filled in different manners depending on when we sample which line, with each approach having its own advantages and downsides. The bottom pattern guarantees a varied distribution of frequencies in each shot, though it requires large phase-encoding gradient jumps.

scans and thus decrease the burden to the patient and the whole system. Beyond reconstruction, AI-driven registration and segmentation approaches can improve downstream tasks such as radiotherapy treatment planning and quantification [9].

Machine Learning is a subset of AI in which a model learns to perform tasks by example or through trial and error without being explicitly programmed for those tasks [5]. **Deep learning**, a subset of machine learning, focuses on training deep artificial neural networks composed of multiple layers of interconnected neurons. These mathematical models excel at learning to transform complex inputs (for example, raw MRI signals or images) into useful outputs (clean images, labels, or motion scores) by training on large sets of annotated examples [10]. After each prediction, the network's internal parameters are updated according to the ground truth annotations, so that the prediction will be more accurate next time. Because training is performed on large amounts of data, this intensive process is executed on dedicated computing hardware, commonly GPUs (graphics processing units). After training, the network's weights are frozen, and the model can be used for inference, which is the act of performing predictions or reconstruction on unseen data.

Segmentation of the target organ and surrounding organs at risk is a key task in radiotherapy, enabling treatment planning and objective volume and shape mea-

surements. Manually delineating organs is a labor intensive process, but recently deep learning-based approaches have shown promise in reducing this workload by automatically classifying each image voxel as a specific organ or background [11, 12]. Modern deep networks can produce segmentations in less than a second and generalize across patient anatomies and scanner settings when trained on diverse examples.

MRI acceleration reduces scan time by acquiring fewer k-space samples than a fully sampled protocol and relying on reconstruction algorithms to recover the missing information. Traditional approaches include parallel imaging, which exploits differences in sensitivity profiles of multiple receiver coils, and compressed sensing, which reconstructs the most plausible image consistent with the acquired data and a sparsity prior. Deep learning methods extend these ideas by learning reconstruction priors directly from the data, particularly mappings from undersampled k-space or a zerofilled image to the corresponding fully sampled ground truth image [13]. These models are often structured as unrolled models, which alternate reconstruction neural networks with data consistency operations [7, 14, 15, 16]. This enables reconstruction of images without reductions in image quality while being based on less k-space data, thereby speeding up examinations. Alternatively, higher resolution images can be obtained within the original amount of scan time.

Subject motion is one of the most frequent and troublesome failure modes of MRI as typical scan durations are long (2-10 minutes), and young or vulnerable patients may find it especially difficult to lie still for such prolonged amounts of time [17, 4]. Motion during an acquisition can result in phase shifts, inconsistent sampling of spatial frequencies and mixing of signals from different object positions [18]. All these effects can visibly degrade image quality, causing blurring and ghosting artifacts that reduce diagnostic confidence, as exemplified in Figure 1.4. Particularly, image degradation can reduce diagnostic sensitivity and specificity, increasing the risk of false negatives (missed lesions) and false positives (artifacts misinterpreted as disease processes), and therefore repeat acquisitions are frequently considered to be necessary [19]. Motion artifacts may also complicate quantitative measurements such as volumetry or physiological measurements. Addressing motion is therefore essential to maintain image quality, patient throughput and accurate clinical decision-making.

Motion correction strategies fall into two broad categories in practice. Prospective correction monitors motion during the scan, for example with external trackers or navigator signals, and updates gradients or slice positions in real time to keep the anatomy stationary relative to the encoding frame [17]. This can prevent many artifacts but requires low-latency tracking and complicated scanner integration. Retro-

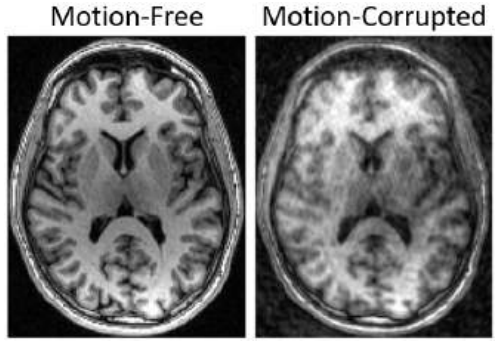


Figure 1.4: Subject motion can severely degrade image quality in MRI.

spective correction operates after acquisition of all raw data of a scan, by estimating motion from motion-corrupted data and correcting it during reconstruction [20, 21]. Retrospective methods are more flexible but must solve a harder inverse problem because the corrupted measurements are already affected by motion. After motion parameters that describe how the subject moved during acquisition are estimated, they can be used to reconstruct an image that is consistent with the measured k-space while compensating for the estimated motion.

Image registration is the process of estimating a mapping between images so that corresponding anatomy overlaps. The spatial transformations can be rigid (translations and rotations), affine (translations, rotations, shearing and scaling) or deformable (represented by a deformation vector field). The choice of transformation will depend on the underlying motion pattern as well as the clinical need [22, 23]. For example, head motion within one scan can be modeled as rigid since the brain does not deform significantly, while abdominal motion can only be accurately captured by deformable transformations. Registration is generally interscan, to align images from different patients or to line up scans of one patient from different visits or modalities. In this thesis, we will employ intrascan registration to obtain motion parameters for different scan segments.

1.1 Thesis outline

The goal of this thesis is to

> *leverage novel artificial intelligence-based approaches to detect, quantify and correct for motion in MRI.*

Restoring motion-corrupted scans to diagnostic quality without requiring a repeat acquisition has the potential to alleviate workflow pressure on MR technologists and improve the patient experience, particularly for vulnerable populations who struggle to remain still during imaging. To ensure practical viability, this thesis also aims to demonstrate applicability on in vivo motion-corrupted scans acquired on clinical MRI scanners. To this end we distinguish the following steps outlined in chapters 2 to 5, with Figure 1.5 offering a schematic overview of how these chapters relate to each other:

Chapter 2 describes a joint interscan registration and segmentation approach for follow-up abdominal CT scans. A cross-stitch multi-task network jointly learns registration and segmentation to improve contour propagation for adaptive radiotherapy, demonstrating how shared feature representations benefit both tasks and improve registration accuracy over single-task approaches [24, 25]. Although Chapter 2 considers CT rather than MRI, it establishes the use of AI-based learning and image registration as key methodological components that form the basis for the approaches developed later in the thesis for motion estimation in MRI.

Chapter 3 introduces an AI-based motion artifact severity estimator for undersampled MRI, enabling early detection of motion while the patient is still in the scanner. In addition, two unrolled MRI reconstruction networks are trained, one on undersampled motion-free data and one on undersampled motion-corrupted data. The latter model can reduce motion artifacts in motion-corrupted scans but will thereby deviate from the measurements and is therefore less desirable for reconstructing motion-free data. It is demonstrated how motion artifact severity prediction can guide selection of reconstruction models to improve image quality under motion while preserving quality on motion-free scans [26].

Chapter 4 proposes intrascan motion estimation by reconstructing and registering fractions of the MRI scan. Each k-space fraction, called a "segment", consists of multiple shots to ensure it contains enough information for an approximate reconstruction of the anatomy. The approximate reconstructions are registered using groupwise registration to estimate the motion parameters of each segment. The motion parameters enable data-consistent correction of motion-corrupted scans, leading to improved data fidelity and image quality [27].

Chapter 5 introduces a technique that enables more fine-grained intrascan motion states to be estimated. After initial motion-correction, it is proposed to iteratively

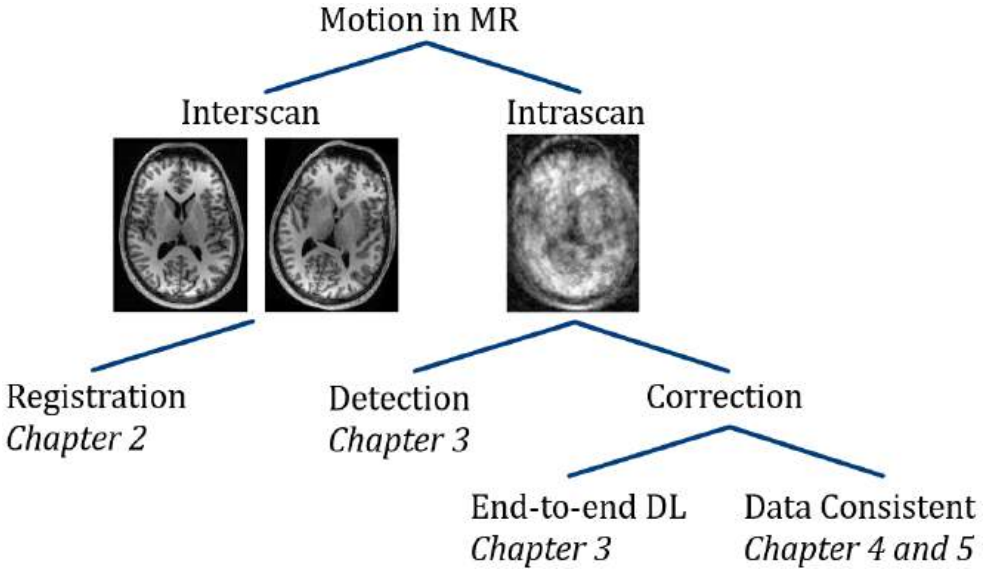


Figure 1.5: Overview of the structure of this thesis.

regroup fractions of k-space, which may have non-adjacent acquisition times, for further correction. The technique enables the motion estimation approach introduced in Chapter 4 to achieve a higher temporal resolution, producing higher fidelity images on scans with severe motion, as validated on volunteer data [28].

Chapter 6 summarizes the findings, discusses limitations and outlines directions for potential future research in AI-driven motion-robust accelerated MRI.

2

Joint Registration and Segmentation via Multi-Task Learning for Adaptive Radiotherapy of Prostate Cancer

This chapter was adapted from:

Laurens Beljaards, Mohamed S. Elmahdy, Fons Verbeek, and Marius Staring. "A Cross-Stitch Architecture for Joint Registration and Segmentation in Adaptive Radiotherapy". In: Proceedings of the Third Conference on Medical Imaging with Deep Learning, vol. 121, 2020. Proceedings of Machine Learning Research. PMLR, pp. 62–74. url: https://proceedings.mlr.press/v121/beljaards20a/beljaards20a.pdf

Mohamed Elmahdy, Laurens Beljaards, Sahar Yousefi, Hessam Sokooti, Fons Verbeek, Uulke Heide, and Marius Staring. "Joint Registration and Segmentation via Multi-Task Learning for Adaptive Radiotherapy of Prostate Cancer". In: IEEE Access, vol. 9, pp. 95551-95568, 2021. doi: 10.1109/ACCESS.2021.3091011

Abstract

Medical image registration and segmentation are two of the most frequent tasks in medical image analysis. As these tasks are complementary and correlated, it would be beneficial to apply them simultaneously in a joint manner. In this paper, we formulate registration and segmentation as a joint problem via a Multi-Task Learning (MTL) setting, allowing these tasks to leverage their strengths and mitigate their weaknesses through the sharing of beneficial information. We propose to merge these tasks not only on the loss level, but on the architectural level as well. We studied this approach in the context of adaptive image-guided radiotherapy for prostate cancer, where planning and follow-up CT images as well as their corresponding contours are available for training. At testing time the contours of the follow-up scans are not available, which is a common scenario in adaptive radiotherapy. The study involves two datasets from different manufacturers and institutes. The first dataset was divided into training (12 patients) and validation (6 patients), and was used to optimize and validate the methodology, while the second dataset (14 patients) was used as an independent test set. We carried out an extensive quantitative comparison between the quality of the automatically generated contours from different network architectures as well as loss weighting methods. Moreover, we evaluated the quality of the generated deformation vector field (DVF). We show that MTL algorithms outperform their Single-Task Learning (STL) counterparts and achieve better generalization on the independent test set. The best algorithm achieved a mean surface distance of 1.06 ± 0.3 mm, 1.27 ± 0.4 mm, 0.91 ± 0.4 mm, and 1.76 ± 0.8 mm on the validation set for the prostate, seminal vesicles, bladder, and rectum, respectively. The high accuracy of the proposed method combined with the fast inference speed, makes it a promising method for automatic re-contouring of follow-up scans for adaptive radiotherapy, potentially reducing treatment related complications and therefore improving patients quality-of-life after treatment. The source code is available at https://github.com/moelmahdy/JRS-MTL.

2.1 Introduction

Medical image analysis aims to extract clinically useful information that aids the diagnosis, prognosis, monitoring and treatment of diseases [29, 30]. Two of the most common tasks in such analyses are image registration and segmentation [31]. Image segmentation aims to identify and cluster objects that prevail similar characteristics into distinctive labels, where these labels can be used for diagnosis or treatment planning. Image registration is the task of finding the geometrical correspondence between images that were acquired at different time steps or from different imaging modalities. These two tasks are complementary, as for example image atlases warped by image registration algorithms are often used for image segmentation [32, 33], while image contours can be used to guide the image registration method in addition to the intensity images [34, 35, 36]. Contours are also used for evaluating the quality of the registration [37, 38]. However, each of these tasks has its own strengths and weaknesses. For instance, image segmentation algorithms can directly delineate images based on texture and surrounding anatomy, and may therefore be robust to large organ deformations. However it sometimes has difficulties with low contrast areas and irregularly shaped organs. On the other hand, image registration algorithms have the ability to encode prior knowledge of the patient's anatomy and therefore may perform better on low quality images. However, such methods sometimes have difficulty with large deformations. Therefore, coupling of image registration and segmentation tasks and modeling them in a single network could leverage their strengths and mitigate their weaknesses through the sharing of beneficial information.

Adaptive image-guided radiotherapy is an exemplar application where the coupling of image registration and segmentation is vital. In radiotherapy, treatment radiation dose is delivered over a course of multiple inter-fraction sessions. In an adaptive setting, re-imaging of the daily anatomy and automatic re-contouring is crucial to compensate for patient misalignment, to compensate for anatomical variations in organ shape and position, and an enabler for the reduction of treatment margins or robustness settings [39, 40]. These have an important influence on the accuracy of the dose delivery, and improve the treatment quality, potentially reducing treatment related side-effects and increasing quality-of-life after treatment [41]. Automatic contouring can be done by direct segmentation of the daily scan, or by registration of the annotated planning scan with the daily scan followed by contour propagation. Image registration has the advantage of leveraging prior knowledge from the initial planning CT scan and the corresponding clinical-quality delineations, which may especially be helpful for challenging organs. On the other hand, image segmentation methods may better delineate organs that vary substantially in shape and volume between treatment fractions, which is often the case for the rectum and the bladder.

In this study, we propose to fuse these tasks at the network architecture level as well as via the loss function. Our key contributions in this paper are as follows:

1. We formulate image registration and segmentation as a multi-task learning problem, which we explore in the context of adaptive image-guided radiotherapy.

2. We explore different joint network architectures as well as loss weighting methods for merging these tasks.

3. We adopt the cross-stitch network architecture for segmentation and registration tasks and explore how these cross-stitch units facilitate information flow between these tasks.

4. Furthermore, we compare MTL algorithms against single-task networks. We demonstrate that MTL algorithms outperform STL networks for both segmentation and registration tasks. To the best of our knowledge this is the first study to investigate various MTL algorithms on an architectural level as well as on a loss weighing level for joint registration and segmentation tasks.

5. We thoroughly investigate the internals of the STL and MTL networks and pinpoint the best strategy to merge this information to maximize the information flow between the two tasks.

Initial results of this work were presented in [24], focusing on the cross-stitch unit in a proposed joint architecture. In the current paper we extend this study to the architectural fusion of these tasks as well as different loss weighting mechanisms. Moreover, an extensive analysis of the different methodologies was performed, detailing the effect of architectural choices, information flow between the two tasks, etc.

The remainder of this paper is organized as follows: Section 2.2 introduces single-task networks, multi-task networks, and loss weighting approaches. In Section 2.3 we introduce the datasets and details about the implementation as well as the experiments. In Sections 2.5 and 2.6, we discuss our results, provide future research directions, and present our conclusions.

2.1.1 Related Work

In the last decade, researchers have been exploring the idea of fusing image segmentation and registration. Lu *et al.* [42] and Pohl *et al.* [43] proposed modeling these tasks using a Bayesian framework such that these tasks would constrain each other. Yezzi [44] proposed to fuse these tasks using active contours, while Unal *et al.* [45] proposed to generalize the previous approach by using partial differential equations without shape priors. Mahapatra *et al.* [36] proposed a Joint Registration and Segmentation (JRS) framework for cardiac perfusion images, where the temporal intensity images

are decomposed into sparse and low rank components corresponding to the intensity change from the contrast agent and the motion, respectively. They proposed to use the sparse component for segmentation and the low rank component for registration. However, most of the aforementioned methods require complex parameter tuning and yield long computation times.

Recently, deep learning-based networks have shown unprecedented success in many fields especially in the medical image analysis domain [46, 47, 48, 49, 50, 51], where deep learning models perform on par with medical experts or even surpassing them in some tasks [52, 53, 54, 55]. Several deep learning-based approaches have been proposed for joint registration and segmentation. The joining mechanisms in the literature can be classified in two categories, namely joining via the loss function and via the architecture as well as the loss function. Selected exemplar methods of the first approach are Hue *et al.* [56], who proposed to join segmentation and registration via a multi-resolution Dice loss function. Elmahdy *et al.* [34] proposed a framework that is a hybrid between learning and iterative approaches, where a CNN network segments the bladder and feeds it to an iterative-based registration algorithm. The authors integrated domain-specific knowledge such as air pocket inpainting as well as contrast clipping, moreover they added an extra registration step in order to focus on the seminal vesicles and rectum. Elmahdy *et al.* [35] and Mahapatra *et al.* [57] proposed a GAN-based (Generative Adversarial Network) approach, where a generative network predicts the correspondence between a pair of images and a discriminator network for giving feedback on the quality of the deformed contours. Exemplar methods of the second category are Xu *et al.* [58], who presented a framework that simultaneously trains a registration and a segmentation network. The authors proposed to jointly learn these tasks during training, however the networks can be used independently during test time. This enables prediction of only the registration output, when the labels are not available during test time. Estienne *et al.* [59] proposed to merge affine and deformable registration as well as segmentation in a 3D end-to-end CNN network. Recently Liu *et al.* [60] proposed an end-to-end framework called JSSR that registers and segments multi-modal images. This framework is composed of three networks: a generator network, that synthesizes the moving image to match the modality of the fixed image, a registration network that registers the synthesized image to the fixed image, and finally a segmentation network that segments the fixed, moving, and synthesized images.

All the previous methods explored the idea of joining segmentation and registration, where to the best of our knowledge none have explored how these tasks are best connected and how to optimize the information flow between them on both the loss and architectural levels.

2.2 Methods

2.2.1 Base Network Architecture

The base architecture for the networks in this paper is a 3D CNN network inspired by the U-Net and BIRNet architectures [61, 62]. Figure 2.1a shows the architecture of the base network. The network encodes the input through $3 \times 3 \times 3$ convolution layers with no padding. LeakyReLU [63] and batch normalization [64] are applied after each convolutional layer. We used strided convolutions in the downsampling path and trilinear upsampling layers in the upsampling path. Through the downsampling path, the number of feature maps increases while the size of the feature maps decreases, and vice versa for the upsampling path. The network has three output resolutions and is deeply supervised at each resolution. Each resolution is preceded by a $1 \times 1 \times 1$ fully convolutional layer (Fconv) so that at coarse resolution, the network can focus on large organs as well as large deformations, while vice versa at fine resolution. In order to extract the groundtruth for different resolutions, we perform cropping of different sizes as well as strided sampling so that for every input patch of size n^3, the sizes of the coarse, mid, and fine resolution are $(\frac{n}{4} - 7)^3$, $(\frac{n}{2} - 18)^3$, and $(n - 40)^3$, respectively.

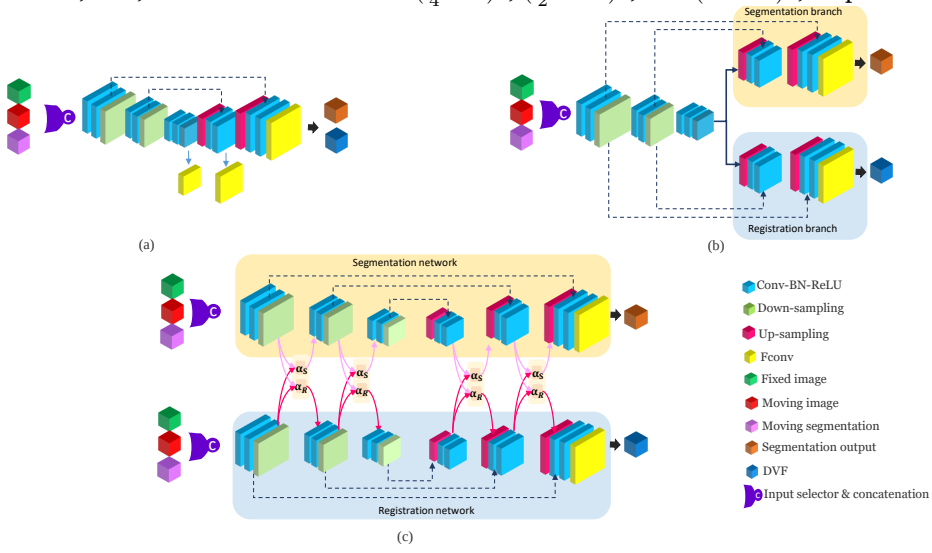


Figure 2.1: The proposed network architectures introduced in the paper. (a) is the base STL network architecture for either segmentation or registration, but also represents the dense parameter sharing MTL network architecture; (b) is the architecture with a shared encoder, while (c) is the Cross-stitch network architecture. Details about the number of feature maps are presented in the implementation details section.

2.2.2 Single Task Learning

Single-task networks are designed to solve one task and therefore require a large amount of labeled training samples, which are scarce in the medical domain since it takes time and trained medical personnel to contour these images. The segmentation and registration networks have the same architecture as the base network depicted in Figure 2.1a, but differ in the input and output layers. Here, single-task networks are considered baseline networks for comparing with the performance of the proposed multi-task networks.

2.2.2.1 Segmentation Network

The input to the segmentation network is the daily CT scan, referred to as the fixed image I_f, where the network predicts the corresponding segmentation S_f^{pred}. S_f^{pred} represents the probability maps for the background, target organs, and organs-at-risk. The network was trained using the Dice Similarity Coefficient (DSC) loss, which quantifies the overlap between the network prediction S_f^{pred} and the groundtruth S_f as follows:

$$\mathcal{L}_{\text{DSC}} = 1 - \frac{1}{K}\sum_{k=1}^{K}\frac{2 * \sum_x S_k^{\text{pred}}(x) \cdot S_k(x)}{\sum_x S_k^{\text{pred}}(x) + \sum_x S_k(x)}, \tag{2.1}$$

where K is the number of structures to be segmented, x is the voxel coordinate, S_k is the ground truth segmentation, and S_k^{pred} the predicted probabilities. The network has 779,436 trainable parameters.

2.2.2.2 Registration Network

The input to the registration network is the concatenation of the planning scan, referred to as the moving image I_m and the daily scan I_f. The network predicts the geometrical correspondence between the input images. This correspondence is represented by the displacement vector field (DVF), referred to as ϕ^{pred}. This DVF is then used to warp I_m. In an ideal scenario, the warped moving image I_m^{warped} would be identical to I_f. The network is trained using Normalized Cross Correlation (NCC) in order to quantify the dissimilarity between I_m^{warped} and I_f. Since the images are from a single imaging modality (CT) with a similar intensity distribution, NCC is an obvious choice abundantly used in the registration literature. Moreover, the implementation is straightforward and efficient when using plain convolution operations. NCC is defined by the following equation:

$$\mathcal{L}_{\text{NCC}} = 1 - \frac{\sum_x [(I_f(x) - \overline{I_f}) \cdot (I_m^{\text{warped}}(x) - \overline{I_m^{\text{warped}}})]}{\sigma_{I_f}\sigma_{I_m^{\text{warped}}}}, \tag{2.2}$$

where x is the voxel coordinate, and σ_{I_f} and $\sigma_{I_m^{\text{warped}}}$ are the standard deviation of the fixed and warped images, respectively. In order to encourage the network to predict a

smooth DVF, a bending energy penalty term is added for regularization:

$$\mathcal{L}_{\mathrm{BE}} = \frac{1}{N} \sum_{x} \|H(\phi^{pred}(x))\|_2^2, \tag{2.3}$$

where H is the Hessian matrix. Now the total registration loss becomes:

$$\mathcal{L}_{\mathrm{Registration}} = \mathcal{L}_{\mathrm{NCC}} + w \cdot \mathcal{L}_{\mathrm{BE}}, \tag{2.4}$$

where w is the bending energy weight. For more details on the selection of w, see Section 2.4.1. The network has 779,733 trainable parameters.

2.2.3 Multi Task Learning

In Multi-Task Learning (MTL), related tasks regularize each other by introducing an inductive bias, thus making the model agnostic to overfitting compared to its STL counterparts [65]. MTL can also be considered as an implicit data augmentation strategy, since it effectively increases the training sample size while encouraging the model to ignore data-dependent noise. Because different tasks have different noise patterns, modeling these tasks simultaneously enables the model to generalize well [66]. Moreover, in MTL models, some features can be more easily learned by one task than another, thus encouraging information cross-talk between tasks [67].

Also, in real-world scenarios, physicians usually incorporate knowledge from different imaging modalities or previous tasks in order to come up with a diagnosis or better understanding of the underlying problem. This illustrates that the knowledge embedded in one task can be leveraged by other tasks and hence it is beneficial to jointly learn related tasks.

Choosing the architecture of an MTL network is based on the following two factors [68]: *what to share* and *how to share*. *What to share* defines the form in which knowledge is shared between tasks. This knowledge sharing can be done through hand-crafted features, input images, and model parameters. *How to share* determines the optimal manner in which this knowledge is shared. In this paper, we focus on parameter-based sharing.

In the following sections, we investigate different MTL network architectures in order to best understand how segmentation and registration tasks share information on the architectural level. The investigated networks predict two sets of contours, one set resulting from the segmentation task and one from the registration task. In this paper, we select the best set of contours as the final output, based on the validation results. More sophisticated strategies are discussed in Section 2.5.

2.2.3.1 Joint Registration and Segmentation via the Registration network

The network in this method, dubbed JRS-reg, has the same architecture as the STL registration network from Section 2.2.2.2, except that this network is optimized using a joint loss as presented in Eq. 2.6.

2.2.3.2 Dense Parameter Sharing

In this architecture both segmentation and registration tasks are modeled using a single network, where both tasks share all parameters except for the task-specific parameters in the output layer, see Figure 2.1a. The network architecture is the same as the base network (see Section 2.2.1) except for the input and output layers. This dense sharing eliminates overfitting issues since it enforces the parameters to model all the tasks at once, however it does not guarantee the best representation for individual tasks [68]. The input to the network is the concatenation of I_m, I_f, and S_m. The network predicts the ϕ^{pred} between input images as well as S_f^{pred}. The network has 781,164 trainable parameters.

2.2.3.3 Encoder Parameter Sharing

Since the input to the segmentation and registration tasks are both CT scans, this means they both encode similar features in the down-sampling path of the network. Therefore in this network both tasks share the encoding path and then splits into two upsampling task specific decoder paths. We call this network the Shared Encoder Double Decoder (SEDD) network. Figure 2.1b shows the architecture of the network. The input to the network is the concatenation of I_m, I_f, and S_m. The network predicts ϕ^{pred} between the input images from the registration path while predicting S_f^{pred} from the segmentation path. The network has 722,936 trainable parameters.

2.2.3.4 Cross-stitch network

A flexible approach to share parameters is via a Cross-Stitch (CS) network [69]. In contrast to the heuristic approach of manually choosing which layers are shared and which are task-specific, the CS network introduces a learning-based unit to determine the amount of feature sharing between tasks. The CS units learn to linearly combine feature maps from the two networks, one for segmentation and one for registration, as shown in Figure 2.1c. The unit itself is defined as:

$$\begin{bmatrix} \bar{X}_S^{\ell,k} \\ \bar{X}_R^{\ell,k} \end{bmatrix} = \begin{bmatrix} \alpha_{SS}^{\ell,k} & \alpha_{SR}^{\ell,k} \\ \alpha_{RS}^{\ell,k} & \alpha_{RR}^{\ell,k} \end{bmatrix} \begin{bmatrix} X_S^{\ell,k} \\ X_R^{\ell,k} \end{bmatrix}, \tag{2.5}$$

where $X_S^{\ell,k}$ and $X_R^{\ell,k}$ represent the feature maps k at layer l for the segmentation and registration networks, respectively. $\alpha_{SS}^{\ell,k}$, $\alpha_{SR}^{\ell,k}$, $\alpha_{RS}^{\ell,k}$, and $\alpha_{RR}^{\ell,k}$ represent the learnable parameters of the CS unit. $\bar{X}_S^{\ell,k}$ and $\bar{X}_R^{\ell,k}$ are the output feature maps for the segmentation and registration networks, respectively. The advantage of CS units is that the network can dynamically learn to share the feature maps in case this is beneficial in terms of the final loss value. In case there is no benefit, an identity matrix can be learned, so that the feature maps become task-specific. This allows the network to learn a smooth sharing between the tasks at a negligible increase in the

number of parameters. As suggested by the original paper, we placed the CS units after the downsampling and upsampling layers resulting in a total of 4 CS units. The CS network has 779,000 trainable parameters.

2.2.4 Loss Weighting

The loss function for the MTL networks is defined by:

$$\mathcal{L} = w_0 \cdot \mathcal{L}_{\mathrm{NCC}} + w_1 \cdot \mathcal{L}_{\mathrm{DSC-R}} + w_2 \cdot \mathcal{L}_{\mathrm{DSC-S}} + w_3 \cdot \mathcal{L}_{\mathrm{BE}}, \tag{2.6}$$

where w_i are the loss weights. They are chosen based on the relative contribution of their corresponding tasks, so that different tasks would learn at the same pace. These weights can be chosen manually based on empirical knowledge, or automatically. A simple choice would be to weigh the losses equally with a fixed weight of 1. Following are some exemplar algorithms for choosing the loss weights automatically. Chen *et al.* proposed GradNorm [70] to weigh different tasks by dynamic tuning of the gradient magnitudes of the tasks. This tuning is achieved by dynamically changing the learning rate for each task so that all tasks would be learning at the same speed. The drawback of this approach is that it requires access to the internal gradients of the shared layers which could be cumbersome. Moreover, one needs to choose which shared layer to back propagate to in case of multiple shared layers. Kendall *et al.* [71] proposed to weigh each task by considering the homoscedastic uncertainty of that task, so that tasks with high output variance will be weighted less than tasks with low variance. This approach only adds few trainable parameters, namely equal to the number of loss functions. Inspired by GradNorm, Liu *et al.* proposed Dynamic Weight Averaging (DWA) [72], where each task is weighted over time by considering the rate of change of the relative loss weights. Contrary to GradNorm, DWA only requires the numerical values of the loss functions rather than their derivatives. In this paper, we compared equal weights versus homoscedastic uncertainty and DWA. For all the experiments, we set the weight of the bending energy to a fixed value of 0.5 (for more details see Section 2.4.1) instead of a trainable one. This is to prevent the network to set it too low in order to improve the DSC of the deformed contours on the account of the smoothness of the predicted DVF.

2.2.4.1 Homoscedastic Uncertainty

Homoscedastic uncertainty was proposed as a loss weighting method by Kendall *et al.* [71]. This is a task-dependent uncertainty which is not dependent on the input data but rather varies between tasks. The authors derived their finding by maximizing the Gaussian likelihood while considering the observational noise scalar σ that represents the homoscedastic uncertainty term related to each task. The following equation describes the weight loss using homoscedastic uncertainty, where σ is a trainable

parameter:

$$\mathcal{L}_{\text{homoscedastic}} = \sum_{i=1}^{T} \frac{1}{\sigma_i^2} \mathcal{L}_i + \log\ \sigma_i, \tag{2.7}$$

where T is the number of tasks. The higher the uncertainty of task i, the lower the contribution of its associated loss $\mathcal{L}_i$ to the overall loss. The $\log$ term can be viewed as a regularization term, so that the network would not learn a trivial solution by setting the uncertainty of all tasks to extreme values.

2.2.4.2 Dynamic Weight Averaging

Dynamic Weight Averaging (DWA) was proposed by Liu *et al.* [72]. Similar to GradNorm [70], DWA weights the losses via the rate of change of the loss of each task over the training iterations t. In contrast to GradNorm, DWA does not require access to the internal gradients of the network, but only requires the numerical loss values. According to DWA, the weight w of the loss $\mathcal{L}$ associated with the task k is defined as:

$$w_k(t) = \frac{K\ \exp(r_k(t-1)/tmp)}{\sum_i \exp(r_i(t-1)/tmp)},\ r_k(t-1) = \frac{\mathcal{L}_k(t-1)}{\mathcal{L}_k(t-2)}, \tag{2.8}$$

where r_k is the relative loss ratio and tmp is the temperature that controls the smoothness of the task weighting. Here, we set $tmp = 1$ as suggested by the original paper. For the initial two iterations, $r_k(t)$ is set to 1.

2.3 Datasets, Implementation, and Evaluation

2.3.1 Datasets

This study involves two datasets from two different institutes and scanners for patients who underwent intensity-modulated radiotherapy for prostate cancer. The first dataset is from Haukeland Medical Center (HMC), Norway. The dataset has 18 patients with 8-11 daily CT scans, each corresponding to a treatment fraction. These scans were acquired using a GE scanner and have 90 to 180 slices with a voxel size of approximately 0.9 $\times$ 0.9 $\times$ 2.0 mm. The second dataset is from Erasmus Medical Center (EMC), The Netherlands. This dataset consists of 14 patients with 3 daily CT scans each. The scans were acquired using a Siemens scanner, and have 91 to 218 slices with a voxel size of approximately 0.9 $\times$ 0.9 $\times$ 1.5 mm. The target structures (prostate and seminal vesicles) as well as organs-at-risk (bladder and rectum) were manually delineated by radiation oncologists. All datasets were resampled to an isotropic voxel size of 1 $\times$ 1 $\times$ 1 mm. All scans and corresponding contours were affinely registered beforehand using `elastix` [73], so that corresponding anatomical structures would fit in the network's field of view. The scan intensities were clipped to [-1000, 1000] .

2.3.2 Implementation and Training Details

All experiments were developed using Tensorflow (version 1.14) [74]. The convolutional layers were initialized with a random normal distribution ($\mu = 0.0$, $\sigma = 0.02$). All parameters of the Cross-stitch units were initialized using a truncated normal distribution ($\mu = 0.5$, $\sigma = 0.25$) in order to encourage the network to share information at the beginning of the training. In order to ensure fairness regarding the number of parameters in all the networks, the number of filters for the Cross-stitch network were set to [16, 32, 64, 32, 16], while for the other networks the numbers were scaled by $\sqrt{2}$ resulting in [23, 45, 91, 45, 23] filtermaps. This results in approximately 7.8×10^5 trainable parameters for each network. The networks were trained using the RAdam optimizer [75] with a fixed learning rate of 10^{-4}. Patches were sampled equally from the target organs, organs-at-risk and torso. All networks were trained for 200K iterations using an initial batch size of 2. The batch size is then doubled by switching the fixed and moving patches so that the network would warp the fixed patch to the moving patch and vice versa at the same training iteration.

The networks were trained and optimized on the HMC dataset, while the EMC dataset was used as an independent test set. Training was performed on a subset of 111 image pairs from 12 patients, while validation and optimization was carried out on the remaining 50 image pairs from 6 patients.

From each image, 1,000 patches of size 96 $\times$ 96 $\times$ 96 voxels were sampled. The size of the patch was chosen so that it would fit in the GPU memory, while still producing a patch size of 17^3 at the lowest resolution, which is a reasonable size to encode the deformation from the surrounding region. Losses from the deeply supervised resolutions were weighted equally, $\frac{1}{3}$ each. Training was performed on a cluster equipped with NVIDIA RTX6000, Tesla V100, and GTX1080 Ti GPUs with 24, 16 and 11 GB of memory, respectively. The source code is available at https://github.com/moelmahdy/JRS-MTL.

2.3.3 Evaluation Metrics

The automatically generated contours are evaluated geometrically by comparing them against the manual contours for the prostate, seminal vesicle, rectum, and bladder. The Dice similarity coefficient (DSC) measures the overlap between contours:

$$\mathrm{DSC} = \sum \frac{2 \mid S_f \cap S_g \mid}{\mid S_f \mid + \mid S_g \mid}, \tag{2.9}$$

where S_g is the generated contour from either the segmentation or the registration network. The distance between the contours is measured by the Mean Surface Distance

(MSD) and Hausdorff Distance (HD) defined as follows:

$$\text{MSD} = \frac{1}{2}\left(\frac{1}{N}\sum_{i=1}^{n} d\left(a_i, S_g\right) + \frac{1}{M}\sum_{i=1}^{m} d\left(b_i, S_f\right)\right), \tag{2.10}$$

$$\text{HD} = \max\left\{\max_i \left\{d\left(a_i, S_g\right)\right\}, \max_j \left\{d\left(b_i, S_f\right)\right\}\right\}, \tag{2.11}$$

where $\{a_1;\ a_2;\ \ldots;\ a_n\}$ and $\{b_1;\ b_2;\ \ldots;\ b_m\}$ are the surface mesh points of the manual and generated contours, respectively, and $d\left(a_i, S_g\right) = \min_j \|b_j - a_i\|$. For all the experiments, we apply the largest connected component operation on the network prediction.

In order to evaluate the quality of the deformations, we calculate the determinant of the Jacobian matrix. A Jacobian of 1 indicates that no volume change has occurred; a Jacobian > 1 indicates expansion, a Jacobian between 0 and 1 indicates shrinkage, and a Jacobian ≤ 0 indicates a singularity, i.e. a place where folding has occurred. We can quantify the smoothness and quality of the DVF by indicating the fraction of foldings per image and by calculating the standard deviation of the Jacobian alongside the MSD of the segmentation.

A repeated one-way ANOVA test was performed using a significance level of $p = 0.05$. P-values are only stated for the comparisons between the best network with the other networks.

2.4 Experiments and Results

			Prostate		Seminal vesicles		Rectum		Bladder	
Network	Input	Output path	$\mu \pm \sigma$	median	$\mu \pm \sigma$	median	$\mu \pm \sigma$	median	$\mu \pm \sigma$	median
Seg	I_f		$1.49 \pm 0.3^*$	1.49	2.50 ± 2.6	2.09	3.39 ± 2.2	2.73	$1.60 \pm 1.1^*$	1.13
	$I_f \oplus S_m$		1.31 ± 0.4	1.23	1.63 ± 0.9	1.26	2.88 ± 3.4	2.06	1.12 ± 0.5	0.97
	$I_f \oplus I_m$		$3.06 \pm 0.6^*$	3.01	5.36 ± 4.4	3.71	$14.57 \pm 9.4^*$	11.58	1.46 ± 1.3	1.12
	$I_f \oplus I_m \oplus S_m$		1.26 ± 0.4	1.20	2.08 ± 2.2	1.27	2.79 ± 1.6	2.45	1.05 ± 0.4	0.97
Reg	$I_f \oplus I_m$		$1.43 \pm 0.8^*$	1.29	$1.71 \pm 1.4^*$	1.37	$2.44 \pm 1.1^*$	2.17	$3.40 \pm 2.3^*$	2.71
	$I_f \oplus I_m \oplus S_m$		1.91 ± 1.3	1.59	1.92 ± 1.5	1.44	2.58 ± 1.1	2.33	3.88 ± 2.5	3.16
JRS-reg	$I_f \oplus I_m$		1.16 ± 0.3	1.16	1.32 ± 0.6	**1.11**	2.08 ± 1.0	1.82	2.57 ± 2.0	2.04
	$I_f \oplus I_m \oplus S_m$		1.20 ± 0.4	1.13	1.35 ± 0.7	1.16	2.08 ± 1.0	1.82	2.63 ± 2.3	1.90
Cross-stitch	$I_f \,\|\|\, I_f \oplus I_m$	Segmentation	$1.47 \pm 0.3^*$	1.48	$2.93 \pm 3.0^*$	2.08	$2.93 \pm 2.0^*$	2.25	1.19 ± 1.0	0.89
		Registration	1.10 ± 0.3	1.07	1.38 ± 0.7	1.17	2.12 ± 1.0	1.89	2.55 ± 2.1	1.89
	$I_f \,\|\|\, I_f \oplus I_m \oplus S_m$	Segmentation	$\mathbf{1.06 \pm 0.3}$	**0.99**	$\mathbf{1.27 \pm 0.4}$	1.15	$\mathbf{1.76 \pm 0.8}$	**1.47**	$\mathbf{0.91 \pm 0.4}$	**0.82**
		Registration	1.10 ± 0.3	1.06	1.30 ± 0.6	1.13	2.00 ± 1.0	1.75	2.45 ± 2.1	1.81
	$I_f \oplus S_m \,\|\|\, I_f \oplus I_m \oplus S_m$	Segmentation	$2.05 \pm 0.7^*$	2.00	$3.66 \pm 4.4^*$	2.19	$2.44 \pm 1.0^*$	2.35	$1.09 \pm 0.5^*$	0.93
		Registration	1.40 ± 0.4	1.35	1.31 ± 0.6	1.17	2.27 ± 1.0	2.02	2.56 ± 1.9	1.96
	$I_f \oplus I_m \oplus S_m \,\|\|\, I_f \oplus I_m \oplus S_m$	Segmentation	1.08 ± 0.3	1.05	$1.54 \pm 0.9^*$	1.28	1.88 ± 1.0	1.61	1.01 ± 0.7	0.82
		Registration	1.20 ± 0.3	1.18	1.35 ± 0.7	1.16	2.12 ± 1.1	1.87	2.54 ± 2.2	1.80

Table 2.1: The effect of network input for the different architectures on the validation set (HMC) in terms of MSD (mm). Lower values are better. Here, $\oplus$ is the concatenation operation, and $\cdot\|\cdot$ represents the inputs to the segmentation network (left of $\|$) and the inputs to the registration network (right of $\|$). Stars denote one-way ANOVA statistical significance with respect to the Cross-stitch network with $I_f \,||\, I_f \oplus I_m \oplus S_m$ as inputs.

In the paper we present two single-task networks dubbed *Seg* and *Reg* networks (see Sections 2.2.2.1 and 2.2.2.2 for more details). Moreover, we investigated multiple

multi-task networks, namely JRS-reg, dense, SEDD, and Cross-stitch (see Sections 2.2.3.1, 2.2.3.2, 2.2.3.3, and 2.2.3.4 for more details). We compared our proposed methods against three state-of-the-art methods that were developed for prostate CT contouring. These methods represent three approaches, namely an iterative conventional registration method, a deep learning-based registration method, and a hybrid method. For the iterative method, we used `elastix` software [73] with the NCC similarity loss using the settings proposed by Qiao *et al.* [76]. In the deep learning method proposed by Elmahdy *et al.* [35], a generative network is trained for contour propagation by registration, while a discrimination network evaluates the quality of the propagated contours. Finally, we compare our methods against the hybrid method proposed by Elmahdy *et al.* [34], where a CNN network segments the bladder and then feeds it to the iterative registration method as prior knowledge.

Following, we optimize some of the network settings on the validation set (HMC), in order to investigate the influence of the bending energy weight, network inputs, weighting strategy and network architecture on the results. Then, on the independent test set, we present the final results comparing with methods from the literature.

2.4.1 Bending Energy Weight

We compared the single-task registration, the JRS-reg method and the Cross-stitch network for a set of bending energy weights, see Equations (2.4) and (2.6), while the weights of the other loss functions are set to 1. Figure 2.2 shows the performance of the aforementioned methods using different bending energy weights. The optimal performance of the registration network occurs at a bending weight of 0.5, while the optimal bending weight for both JRS-reg and Cross-stitch network is much lower but with higher standard deviation of the Jacobian. Therefore, for the remainder of the paper we set the weight of the bending energy to 0.5 since it achieves the best compromise between the contour performance in terms of MSD and the registration performance in terms of the std. of the Jacobian determinant.

2.4.2 Optimization of the Networks Inputs

During training, validation, and testing, we have access to the fixed image I_f, the moving image I_m, and the moving segmentation S_m. In Table 2.1 we compared different sets of inputs on the validation dataset. This experiment helps to better understand how these network interpret and utilize these inputs and how this would reflect on the network outcome represented by the MSD metric. For this experiment we used equal loss weights for the MTL networks.

Feeding S_m to the segmentation network improves the results substantially compared to only feeding I_f, especially for the seminal vesicles, while feeding I_m deteriorates the results. For the registration and JRS-reg networks, feeding S_m alongside I_f and I_m resulted in a similar performance compared to not feeding it. Since the

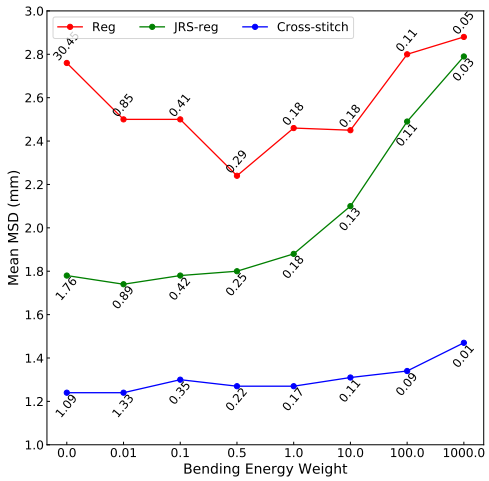


Figure 2.2: The performance of the registration, JRS-registration and Cross-stitch networks with different bending energy weights on the validation set (HMC), in terms of mean MSD averaged over the four organs. The annotation at each point represents the standard deviation of the determinant of the Jacobian.

			Prostate		Seminal vesicles		Rectum		Bladder	
Network	Weight	Output path	$\mu \pm \sigma$	median	$\mu \pm \sigma$	median	$\mu \pm \sigma$	median	$\mu \pm \sigma$	median
JRS-reg	Equal	Registration	1.20 ± 0.4	1.13	1.35 ± 0.7	1.16	2.08 ± 1.0	1.82	$2.63 \pm 2.3^{*}$	1.90
	Homoscedastic	Registration	1.20 ± 0.3	1.20	1.22 ± 0.5	1.07	2.05 ± 1.0	1.81	2.34 ± 2.2	1.60
	DWA	Registration	1.22 ± 0.3	1.18	$1.37 \pm 0.7^{*}$	1.20	$2.29 \pm 1.1^{*}$	2.04	$3.18 \pm 2.4^{*}$	2.43
Dense	Equal	Segmentation	1.14 ± 0.4	1.06	1.73 ± 2.1	1.12	1.91 ± 0.9	1.64	1.04 ± 0.7	0.87
		Registration	1.20 ± 0.3	1.11	$1.33 \pm 0.7^{*}$	1.10	2.16 ± 1.1	1.85	2.56 ± 1.9	1.90
	Homoscedastic	Segmentation	1.09 ± 0.3	1.04	1.51 ± 1.2	1.13	1.86 ± 0.8	1.69	0.99 ± 0.4	0.91
		Registration	1.17 ± 0.3	1.15	1.31 ± 0.6	1.13	2.17 ± 1.0	1.96	$2.63 \pm 2.0^{*}$	1.95
	DWA	Segmentation	$1.12 \pm 0.3^{*\dagger}$	1.04	1.74 ± 2.0	1.13	$1.99 \pm 0.9^{*}$	1.77	1.00 ± 0.4	0.85
		Registration	1.14 ± 0.3	1.14	1.27 ± 0.6	**1.07**	$2.24 \pm 1.1^{*}$	1.97	2.72 ± 1.9	2.13
SEDD	Equal	Segmentation	$1.47 \pm 0.6^{*\dagger}$	1.31	2.81 ± 4.6	1.34	1.97 ± 1.0	1.59	1.21 ± 1.0	0.94
		Registration	$1.28 \pm 0.4^{*}$	1.19	$1.50 \pm 0.9^{*}$	1.26	$2.26 \pm 1.1^{*}$	1.94	$2.61 \pm 2.1^{*}$	1.83
	Homoscedastic	Segmentation	$1.15 \pm 0.3^{\dagger}$	1.14	1.47 ± 1.0	1.22	2.12 ± 1.1	1.91	0.99 ± 0.2	0.94
		Registration	1.19 ± 0.3	1.21	1.23 ± 0.5	1.13	2.15 ± 1.0	1.92	2.31 ± 2.0	1.64
	DWA	Segmentation	$1.22 \pm 0.3^{*\dagger}$	1.18	1.44 ± 0.8	1.21	2.12 ± 1.4	1.73	1.10 ± 0.6	0.93
		Registration	1.22 ± 0.3	1.22	$1.32 \pm 0.6^{*}$	1.10	$2.30 \pm 1.1^{*}$	2.01	$2.86 \pm 1.9^{*}$	2.41
Cross-stitch	Equal	Segmentation	$\mathbf{1.06 \pm 0.3}$	**0.99**	$\mathbf{1.27 \pm 0.4}$	1.15	$\mathbf{1.76 \pm 0.8}$	**1.47**	$\mathbf{0.91 \pm 0.4}$	**0.82**
		Registration	$1.10 \pm 0.3^{*}$	1.06	1.30 ± 0.6	1.13	$2.00 \pm 1.0^{*}$	1.75	2.45 ± 2.1	1.81
	Homoscedastic	Segmentation	$1.23 \pm 0.3^{\dagger}$	1.16	1.51 ± 1.2	1.17	2.37 ± 1.0	2.09	0.92 ± 0.2	0.89
		Registration	1.24 ± 0.3	1.24	1.32 ± 0.6	1.13	2.12 ± 1.0	1.89	2.45 ± 1.9	1.97
	DWA	Segmentation	$1.34 \pm 0.4^{*\dagger}$	1.27	1.75 ± 1.7	1.29	$2.32 \pm 0.9^{\dagger}$	2.11	$1.17 \pm 0.8^{*}$	0.91
		Registration	1.22 ± 0.3	1.19	1.27 ± 0.6	1.09	$2.21 \pm 1.0^{*}$	2.00	$2.93 \pm 2.3^{*}$	2.27

Table 2.2: MSD (mm) values for the different networks and loss weighting methods for the HMC dataset. Lower values are better. Stars and daggers denote one-way ANOVA statistical significance for intra-network experiments with respect to Homoscedastic weights and inter-network experiments with respect to Cross-stitch with Equal weights, respectively. Grey numbers represent the values of the worst path between the segmentation and registration paths, while bold numbers represent the best results.

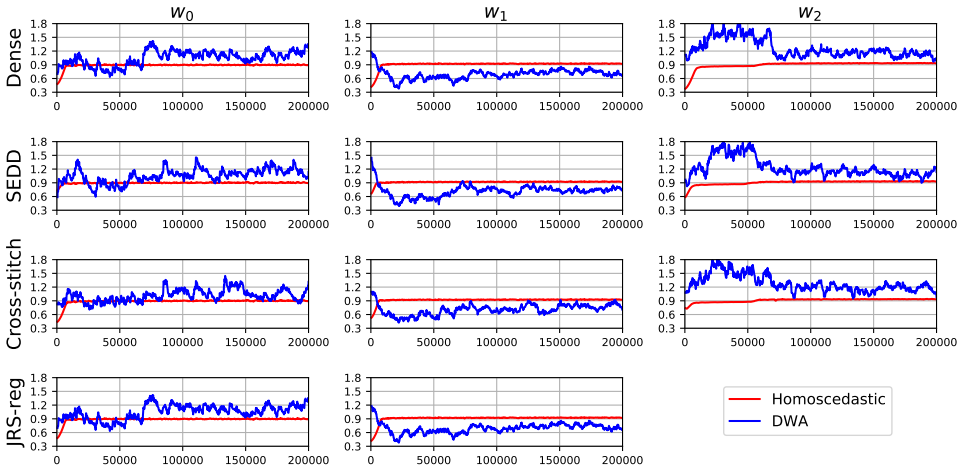


Figure 2.3: The evolution of the loss weights during training for different multi-task networks on the validation dataset (HMC).

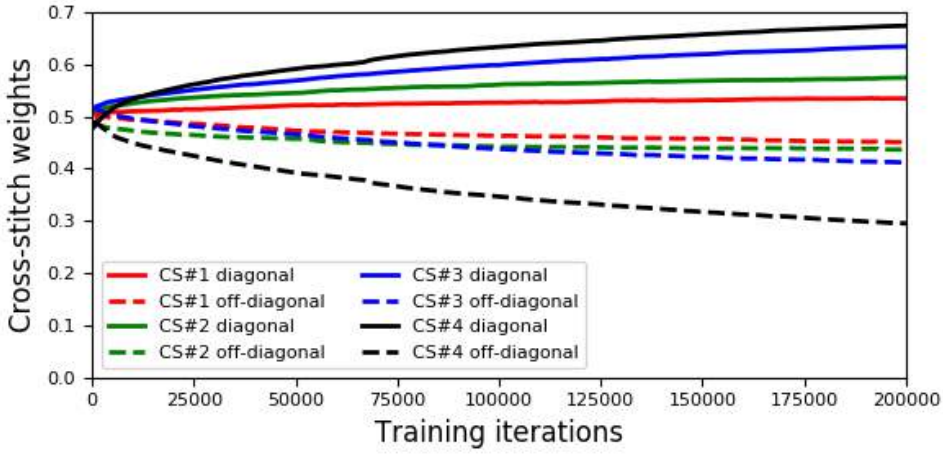


Figure 2.4: The evolution of the Cross-stitch units weights during training using equal weights. CS#1 and CS#2 are placed in the down-sampling path, while CS#3 and CS#4 are placed in the upsampling path. The solid lines represent the mean of the weights across the diagonal of the CS unit, while the dashed lines represent the mean of the off-diagonal weights.

Cross-stitch network is composed of two networks, one for segmentation and the other for registration, we experimented with various combinations of inputs. The results are

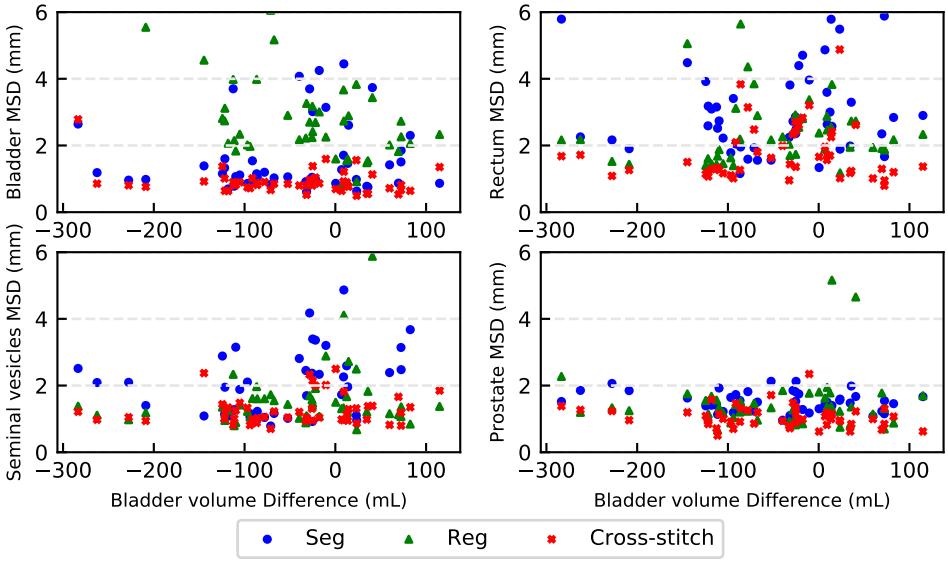


Figure 2.5: The effect of the bladder volume deviation from the planning volume on the performance of the Seg, Reg, and Cross-stitch networks for the validation set (HMC).

very consistent with our previous findings on the single-task networks on the effect of using S_m as an input.

For the remainder of this paper, we chose to use I_f as input for the segmentation network, and I_f and I_m as inputs for the registration network. Although adding S_m proved to be better especially for the segmentation network, here we exclude it, since these two methods act as a baseline and this is the standard setting in single-task networks. For dense, SEDD, and JRS-reg networks, we select a concatenation of I_m, I_f, and S_m for the final network. For the Cross-stitch network, we select I_f for the segmentation network and the concatenation of I_m, I_f, and S_m for the registration network.

2.4.3 Optimization of loss weighting strategy

In this experiment we investigate the performance of the various loss weighting strategies introduced in Section 2.2.4 in order to select the best weighting method for the underlying tasks.

Table 2.2 shows the results of the different weighting strategies for the MTL networks in terms of MSD. For the JRS-reg network architecture, weighting the losses with homoscedastic uncertainty achieved comparable results to using equal

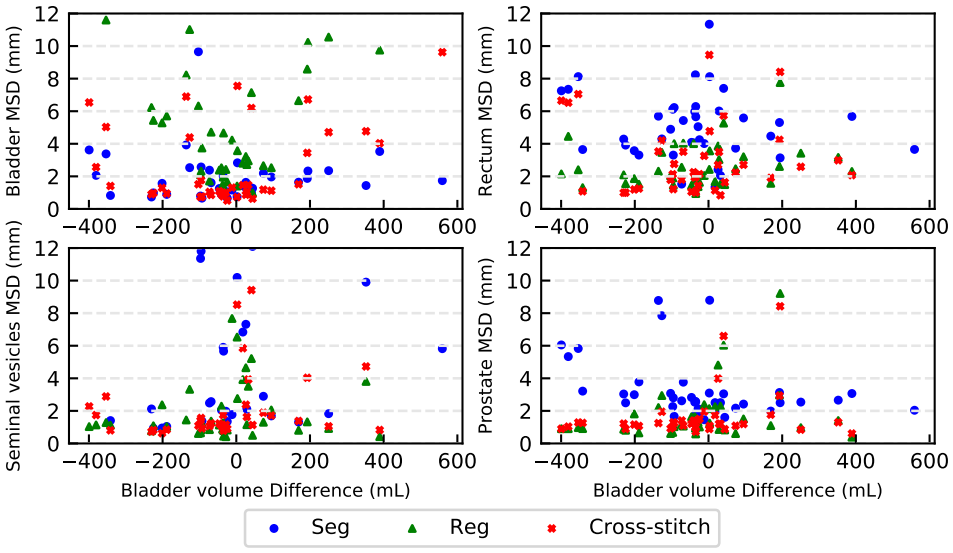


Figure 2.6: The effect of the bladder volume deviation from the planning volume on the performance of the STL and the Seg, Reg, and Cross-stitch networks for the independent test set (EMC).

weights, while DWA scored somewhat less. For the dense and SEDD architectures, homoscedastic weighting achieved a slightly better performance, while equal weights was best for the Cross-stitch network. For these architectures (dense, SEDD, and Cross-stitch), the segmentation output path showed improvement over the registration output path.

Figure 2.3 illustrates the evolution of the loss weights w_i during training, for different multi-task network architectures and weighting strategies.

For the remainder of this paper and based on the previous findings, we chose the homoscedastic uncertainty weighting strategy for the JRS-reg, dense and SEDD networks, while using equal weights for the Cross-stitch network.

2.4.4 Analysis of Cross-stitch units

Analysis of the behavior of the Cross-stitch units during training facilitates the understanding of how the segmentation and registration networks interacts in the MTL settings. Figure 2.4 shows the mean of the CS units across the diagonal and off-diagonal (See Equation (2.5)). Higher weights on the diagonal means that the network tends to separate the task-specific feature maps, while higher weights off-diagonal means that the network tends to share the corresponding feature maps.

2.4.5 Effect of the bladder filling

For the HMC dataset, which was used for training and validation, a bladder filling protocol was in place, meaning that the deformation of the bladder between daily and planning scans is not large. However, this is not the scenario for the EMC dataset, the test set.

Figure 2.5 and 2.6 illustrate the effect of the bladder volume variation from the planning scan on the performance of the Seg, Reg, and Cross-stitch networks. The Cross-stitch network is resilient to bladder filling for both the HMC and EMC datasets.

2.4.6 Evaluation of the Quality of the DVF

The smoothness of the predicted DVF is an important parameter to evaluate the predicted deformation field. Table 2.5 shows a detailed analysis of the DVF in terms of the standard deviation of the determinant of the Jacobian as well as the folding fraction for the registration path of the different networks.

		Prostate		Seminal vesicles		Rectum		Bladder	
Network	Output path	$\mu \pm \sigma$	median	$\mu \pm \sigma$	median	$\mu \pm \sigma$	median	$\mu \pm \sigma$	median
Seg	Segmentation	1.49 ± 0.3	1.49	2.50 ± 2.6	2.09	3.39 ± 2.2	2.73	1.60 ± 1.1	1.13
Reg	Registration	1.43 ± 0.8	1.29	1.71 ± 1.4	1.37	2.44 ± 1.1	2.17	3.40 ± 2.3	2.71
JRS-reg	Registration	1.20 ± 0.3	1.20	1.22 ± 0.5	**1.07**	2.05 ± 1.0	1.81	2.34 ± 2.2	1.60
Dense	Segmentation	1.09 ± 0.3	1.04	1.51 ± 1.2	1.13	1.86 ± 0.8	1.69	0.99 ± 0.4	0.91
	Registration	1.17 ± 0.3	1.15	1.31 ± 0.6	1.13	2.17 ± 1.0	1.96	2.63 ± 2.0	1.95
SEDD	Segmentation	1.15 ± 0.3	1.14	1.47 ± 1.0	1.22	2.12 ± 1.1	1.91	0.99 ± 0.2	0.94
	Registration	1.19 ± 0.3	1.21	1.23 ± 0.5	1.13	2.15 ± 1.0	1.92	2.31 ± 2.0	1.64
Cross-stitch	Segmentation	$\mathbf{1.06 \pm 0.3}$	**0.99**	$\mathbf{1.27 \pm 0.4}$	1.15	$\mathbf{1.76 \pm 0.8}$	**1.47**	$\mathbf{0.91 \pm 0.4}$	**0.82**
	Registration	1.10 ± 0.3	1.06	1.30 ± 0.6	1.13	2.00 ± 1.0	1.75	2.45 ± 2.1	1.81
Elastix [76]	Registration	1.73 ± 0.7	1.59	2.71 ± 1.6	2.45	3.69 ± 1.2	3.50	5.26 ± 2.6	4.72
Hybrid [34]	Registration	1.27 ± 0.3	1.25	1.47 ± 0.5	1.32	2.03 ± 0.6	1.85	1.75 ± 1.0	1.26
JRS-GAN [35]	Registration	1.14 ± 0.3	1.04	1.75 ± 1.3	1.44	2.17 ± 1.1	1.89	2.25 ± 1.9	1.54

Table 2.3: MSD (mm) values for the different networks on the validation set (HMC). Lower values are better.

		Prostate		Seminal vesicles		Rectum		Bladder	
Network	Output path	$\mu \pm \sigma$	median	$\mu \pm \sigma$	median	$\mu \pm \sigma$	median	$\mu \pm \sigma$	median
Seg	Segmentation	3.18 ± 1.8	2.57	9.33 ± 10.1	5.82	5.79 ± 3.4	5.18	1.88 ± 1.5	1.50
Reg	Registration	2.01 ± 2.5	1.18	2.86 ± 5.2	1.18	2.89 ± 2.5	2.23	5.98 ± 4.7	4.44
JRS-reg	Registration	1.94 ± 2.6	1.16	2.48 ± 4.8	1.01	2.67 ± 2.4	2.05	4.80 ± 4.6	2.12
Dense	Segmentation	2.01 ± 2.6	**1.15**	4.08 ± 7.2	1.23	3.70 ± 5.4	**2.03**	2.75 ± 3.1	1.23
	Registration	1.93 ± 2.5	1.15	2.53 ± 4.7	1.01	2.67 ± 2.3	2.13	5.08 ± 4.4	3.01
SEDD	Segmentation	1.99 ± 2.4	1.24	6.26 ± 8.9	3.01	4.21 ± 4.9	2.12	2.43 ± 2.9	1.04
	Registration	1.92 ± 2.5	1.19	2.43 ± 4.5	1.07	2.72 ± 2.4	2.17	4.86 ± 4.4	2.22
Cross-stitch	Segmentation	1.88 ± 1.9	1.30	2.76 ± 3.5	1.28	4.87 ± 6.8	2.49	$\mathbf{1.66 \pm 1.7}$	**0.85**
	Registration	1.91 ± 2.3	1.23	2.41 ± 4.5	**0.95**	2.78 ± 2.4	2.16	4.90 ± 4.0	2.84
Elastix [76]	Registration	$\mathbf{1.42 \pm 0.7}$	1.17	2.07 ± 2.6	1.24	3.20 ± 1.6	3.07	5.30 ± 5.1	3.27
Hybrid [34]	Registration	1.55 ± 0.6	1.36	$\mathbf{1.65 \pm 1.3}$	1.22	$\mathbf{2.65 \pm 1.6}$	2.36	3.81 ± 3.6	2.26

Table 2.4: MSD (mm) values for the different networks on the independent test set (EMC). Lower values are better. Results for JRS-GAN are not available for this dataset.

2.4.7 Comparison against the state-of-the-art

Table 2.3 and 2.4 show the results for the validation set (HMC) and test set (EMC), respectively. The first two networks in each table are single-task networks. For both

	Validation set (HMC)		Independent test set (EMC)	
Network	Std. Jacobian	Folding fraction	Std. Jacobian	Folding fraction
Reg	0.2935 ± 0.1022	0.0049 ± 0.0039	0.4129 ± 0.2258	0.0112 ± 0.0115
JRS-reg	0.2543 ± 0.0505	0.0030 ± 0.0014	0.3148 ± 0.1106	0.0066 ± 0.0062
Dense	0.2062 ± 0.0431	0.0018 ± 0.0012	0.2558 ± 0.0899	0.0036 ± 0.0027
SEDD	0.2626 ± 0.1167	0.0019 ± 0.0016	0.4287 ± 0.3000	0.0066 ± 0.0074
Cross-stitch	0.2241 ± 0.0784	0.0024 ± 0.0018	0.3301 ± 0.1869	0.0071 ± 0.0070

Table 2.5: Analysis of the determinant of the Jacobian for the validation and the independent test sets. Lower values are better.

sets, the registration network outperformed the segmentation network for all organs except the bladder. The mean MSD for the independent test set is higher than the corresponding numbers in the validation set for most organs. However, the median values are on par. For the MTL networks, the segmentation path of the networks achieved better performance than the registration path on both datasets except for the seminal vesicles. The Cross-stitch network achieved the best results compared to the other MTL networks.

The proposed STL and MTL networks were compared against other state-of-the-art methods that were evaluated using the HMC dataset. For the validation set, the STL network achieved comparable results, while the Cross-stitch network outperformed these methods for both output paths. On the test set, *elastix* [76] and the Hybrid method [34] performed better except for the bladder, although the median values of the MTL networks were better.

For the quality of the predicted contours, Figure 2.7 and 2.8 show example contours from the HMC and EMC datasets for the Seg, Reg, and Cross-stitch networks. The examples show that the Cross-stitch network achieves better results compared to the Seg and Reg networks especially for the seminal vesicles and rectum with large gas pockets.

2.5 Discussion

In this study, we proposed to merge image registration and segmentation on the architectural level as well as the loss, via a multi-task learning setting in order to leverage their strengths and mitigate their weaknesses through the sharing of beneficial information. We studied different network architectures and loss weighting methods in order to explore how these tasks interact, and thereby leverage the shared knowledge between them. Moreover, we carried out extensive quantitative analysis in the context of adaptive radiotherapy, and compared the proposed multi-task methods to their single-task counterparts. In this paper, a substantial number of experiments were executed, where we explored the following methodological choices: the bending energy weight, the input to the STL and MTL networks, and the loss weighting

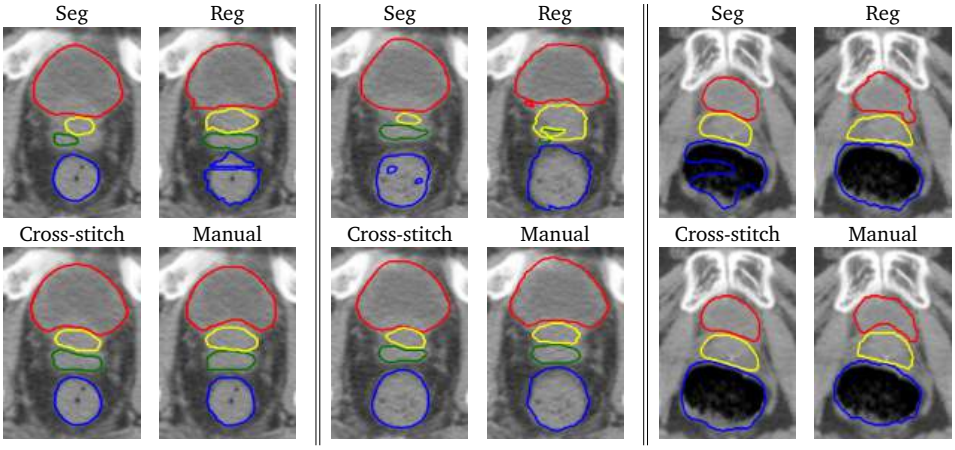


Figure 2.7: Example contours from the validation dataset (HMC) generated by the proposed STL and MTL networks. From left to right, the selected cases are the first, second, and third quartile in terms of the prostate MSD of the Cross-stitch network. The contours of the bladder, prostate, seminal vesicles, and rectum are colored in red, yellow, green, and blue, respectively.

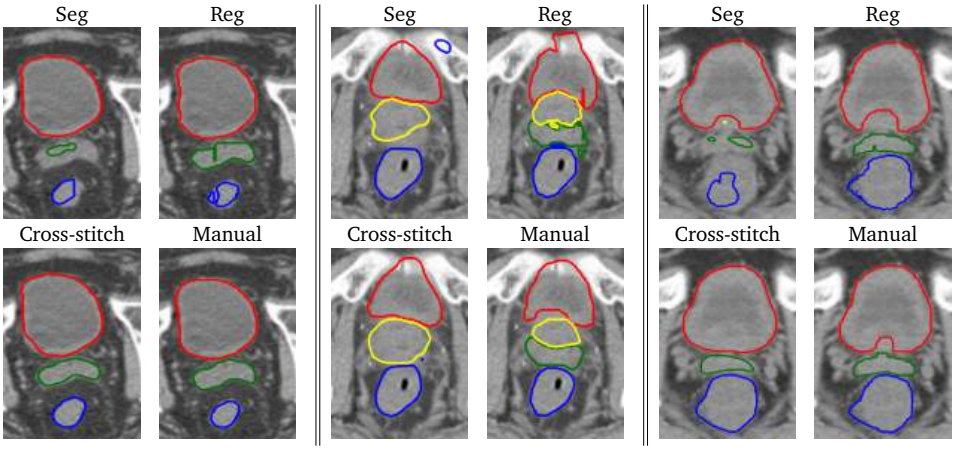


Figure 2.8: Example contours from the independent test set (EMC) generated by the proposed STL and MTL networks. From left to right, the selected cases are the first, second, and third quartile in terms of the prostate MSD of the Cross-stitch network.

method. We also performed a thorough analysis on how Cross-stitch units and loss weights evolve during training. Finally, we compared our proposed methods against state-of-the-art methods.

In all the experiments we fixed the weight of the bending energy weight so that the network would not set it too low in order to improve the DSC of the deformed

contours on the account of the smoothness of the predicted DVF. As shown in Figure 2.2 low bending energy weights result in better contour quality on the account of the smoothness of the predicted DVF.

For the inputs to the STL networks, additionally feeding S_m to the segmentation network resulted in a statistically significant improvement especially for the seminal vesicles. Apparently the network considers S_m as an initial estimation for S_f and subsequently uses it as a guidance for its final prediction. When feeding I_m the results deteriorated; this may confuse the network as I_f and I_m have the same anatomy but with different shapes and local positions. The addition of both I_m and S_m performed similar to the addition of only S_m, which indicates that the networks learned to ignore I_m. For the registration network, the addition of S_m resulted in a sub-optimal result, since the S_m contours on its own does not represent the underlying deformation well.

For the inputs to the MTL networks, in the JRS-reg network, feeding S_m alongside I_f and I_m resulted in a similar performance compared to not feeding it. This indicates that the incorporation of S_m via the DSC loss, already enables the JRS-reg network to exploit this extra information, and that additionally adding S_m as a network input does not provide further benefits. In the Cross-stitch network, we found that adding S_m to the registration network results in a statistically significant improvement. Furthermore, feeding S_m to one of the networks is sufficient, proving that segmentation and registration networks communicate their knowledge efficiently through the Cross-stitch units.

We selected the STL networks with I_f (for segmentation) and I_f alongside I_m (for registration) as input to our baseline methods. Between these two networks, the registration network performed better overall, since the registration network leverages prior knowledge from the organs in the moving image. For the bladder, the segmentation network achieved better results; Apparently the registration network had difficulties finding the correspondence between the bladder in the fixed and moving images, since it tends to deform considerably between visits. However, the segmentation network failed to segment the seminal vesicles for five cases. That is explained by the fact that the seminal vesicles are a difficult structure to segment, due to its relatively small size, undefined borders, and poor contrast with its surroundings. The registration network on the other hand is able to employ the surrounding anatomy as context, to accurately warp the seminal vesicles.

For the multi-task networks, we demonstrated that fusing segmentation and registration tasks is performing better than its single-task counterparts. Merging these tasks using Cross-stitch network achieved the best results on both the validation and testing datasets.

Different loss weighting methods achieved comparable results as shown in Table 2.2. In Figure 2.3, homoscedastic uncertainty tended to weigh all losses equally, using

almost a fixed weight of 0.9 during most of the training iterations. On the contrary, DWA tended to fluctuate during training as the weights are updated based on the ratio of the loss from previous iterations, which fluctuates due to the batch-based training. Since the fixed and moving images are affinely registered beforehand, DWA tended to down-weigh the registration loss and the associated DSC at the beginning of the training, while weighting the segmentation network loss more in order to improve its prediction. Later during training, all the weights stabilized around 0.9 similar to homoscedastic uncertainty. Although both methods stabilized by the end of the training around the same value (0.9), the homoscedastic uncertainty achieved slightly better results compared to DWA and equal weighting methods, except for the Cross-stitch network. Our reasoning behind this is that homoscedastic uncertainty, unlike other methods, is learnable during the training and highly dependent on the underlying task uncertainty.

By analyzing the performance of the Cross-stitch units as demonstrated in Figure 2.4, we found that the Cross-stitch units tended to average feature maps for the down-sampling path, while preferring to be more task-specific for the upsampling path. This somewhat mimics the shared encoder double decoder (SEDD) network, but in contrast to this network, the Cross-stitch network does not completely split the decoder paths. This finding confirms that the segmentation and registration tasks are correlated and thereby encode similar features.

We carried out an experiment to study the effect of the bladder filling protocol between the HMC and EMC datasets. As shown in Figure 2.5, the HMC dataset has a bladder filling protocol so the volume of the bladder changes slightly around 100 mL between different sessions, which is not the case for the EMC dataset as shown in Figure 2.6. Since the registration-based networks and joint networks were trained on small bladder deformations, they failed on large deformations, however the segmentation network was not affected since it does not depend on the deformation but rather the underlying texture to segment the bladder.

In terms of the smoothness of the predicted DVF shown in Table 2.5, MTL networks achieved lower numbers for the standard deviation of the Jacobian as well as for the folding fraction, compared to the STL network (Reg), on both the test and validation set. Our reasoning is that joining the segmentation task to the registration task works as an additional regularization to the registration network. Due to the fact that the higher the quality of the predicted DVF, the higher the quality of the propagated contours and subsequently the lower the DSC loss. The numbers on the test set are slightly higher than the validation set, but this is due to the variance between the deformations between both sets and the fact that the network has not seen the test set before. This can be addressed using transfer learning as suggested by Elmahdy *et al.* [50] or by using synthetic deformations that mimic the one presented in the EMC

dataset.

In the paper, we compared our algorithm against different algorithms from various categories: non-learning (`elastix` [73], a popular conventional tool); hybrid [34], and GAN-based [35]. The presented multi-task networks outperformed these approaches on the validation set and performed on par to these methods for the test set. However, the test time for the hybrid and `elastix` methods are in the order of minutes, while the presented methods have the advantage of fast prediction in less than a second. This enables online automatic re-contouring of daily scans for adaptive radiotherapy. Moreover, in our hybrid study [34] we carried out an extensive dosimetric evaluation alongside the geometric evaluation. The predicted contours from that study met the dose coverage constraints in 86%, 91%, and 99% of the cases for the prostate, seminal vesicles, and lymph nodes, respectively. Since our multi-task networks outperformed the geometrical results in that study, we expect that our contours would achieve a higher success rate in terms of the dose coverage. This could potentially reduce treatment related complications and therefore improve patient quality-of-life after treatment.

A promising direction for future research is the addition of a third task, potentially radiotherapy dose plan estimation. Hence, we can generate contours that are consistent with an optimal dose planning. Further studies could also focus on sophisticated MTL network architectures similar to sluice networks [77] or routing networks [78]. Moreover, we can study how to fuse the contours from the segmentation and registration paths in a smarter way rather than simply selecting one of them based on the validation set.

2.6 Conclusion

In this paper, we propose to formulate the registration and segmentation tasks as a multi-task learning problem. We presented various approaches in order to do so, both on an architectural level and via the loss function. We experimented with different network architectures in order to investigate the best setting that maximizes the information flow between these tasks. Moreover, we compared different loss weighting methods in order to optimally combine the losses from these tasks.

We showed that multi-task learning approaches outperform their single-task counterparts. Using an adaptive parameter sharing mechanism via Cross-stitch units gives the networks freedom to share information between these two tasks, which resulted in the best performance. An equal loss weighting approach had similar performance to more sophisticated methods.

The cross stitch network with equal loss weights achieved a median MSD of 0.99 mm, 0.82 mm, 1.13 mm and 1.47 mm on the validation set and 1.09 mm, 1.24 mm, 1.02 mm, and 2.10 mm on the independent test set for the prostate, bladder,

seminal vesicles, and rectum, respectively. That is equal or less than slice thickness (2 mm). Due to the fast inference of the methods, the proposed method is highly promising for automatic re-contouring of follow-up scans for adaptive radiotherapy, potentially reducing treatment related complications and therefore improving patient quality-of-life after treatment.

2.7 Acknowledgment

The HMC dataset with contours was collected at Haukeland University Hospital, Norway, and was provided to us by oncologist Svein Inge Helle and physicist Liv Bolstad Hysing. The EMC dataset with contours was collected at Erasmus University Medical Center, The Netherlands, and was provided to us by radiation therapist Luca Incrocci and physicist Mischa Hoogeman. They are gratefully acknowledged.

3

AI-Based Motion Artifact Severity Estimation in Undersampled MRI Allowing for Selection of Appropriate Reconstruction Models

This chapter was adapted from:

Laurens Beljaards, Nicola Pezzotti, Chinmay Rao, Mariya Doneva, Matthias van Osch, and Marius Staring (Jan. 2024). "AI-based motion artifact severity estimation in undersampled MRI allowing for selection of appropriate reconstruction models". In: Medical Physics 51. doi: 10.1002/mp.16918

Abstract

Background: MR acquisition is a time consuming process, making it susceptible to patient motion during scanning. Even motion in the order of a millimeter can introduce severe blurring and ghosting artifacts, potentially necessitating re-acquisition. MRI can be accelerated by acquiring only a fraction of k-space, combined with advanced reconstruction techniques leveraging coil sensitivity profiles and prior knowledge. AI-based reconstruction techniques have recently been popularized, but generally assume an ideal setting without intra-scan motion.

Purpose: To retrospectively detect and quantify the severity of motion artifacts in undersampled MRI data. This may prove valuable as a safety mechanism for AI-based approaches, provide useful information to the reconstruction method, or prompt for re-acquisition while the patient is still in the scanner.

Methods: We developed a deep learning approach that detects and quantifies motion artifacts in undersampled brain MRI. We demonstrate that synthetically motion-corrupted data can be leveraged to train the CNN-based motion artifact estimator, generalizing well to real-world data. Additionally, we leverage the motion artifact estimator by using it as a selector for a motion-robust reconstruction model in case a considerable amount of motion was detected, and a high data consistency model otherwise.

Results: Training and validation were performed on 4387 and 1304 synthetically motion-corrupted images and their uncorrupted counterparts, respectively. Testing was performed on undersampled *in vivo* motion-corrupted data from 28 volunteers, where our model distinguished head motion from motion-free scans with 91% and 96% accuracy when trained on synthetic and on real data, respectively. It predicted a manually defined quality label ('Good', 'Medium' or 'Bad' quality) correctly in 76% and 85% of the time when trained on synthetic and real data, respectively. When used as a selector it selected the appropriate reconstruction network 93% of the time, achieving near optimal SSIM values.

Conclusions: The proposed method quantified motion artifact severity in undersampled MRI data with high accuracy, enabling real-time motion artifact detection that can help improve the safety and quality of AI-based reconstructions.

3.1 Introduction

Magnetic Resonance Imaging (MRI) is a non-invasive medical imaging modality essential for visualizing the internal anatomy of a patient in the clinic, based on which a diagnosis can be made. While other modalities such as CT can also acquire anatomical images in a non-invasive manner, MRI does not expose the subject to harmful ionizing radiation. However, a major drawback of MRI is that the process of acquiring the necessary k-space data is time-consuming, and requires subjects to lie still for extended periods of time, which can be especially challenging for young or very sick subjects. Small movements can already introduce severe blurring and ghosting artifacts [17], necessitating re-acquisition. One study reports that 20% of the scans in the investigated hospital had to be reacquired as a result of motion artifacts, with an associated cost estimated at $115,000 per scanner every year [79].

If the time required to traverse k-space during acquisition can be reduced, the scan will last shorter, thereby reducing the chance that the acquired image contains motion artifacts [17]. Acceleration of MRI can be achieved by parallel imaging methods (which utilize multicoil acquisition followed by coil-combination) or by compressed sensing methods (which utilize incoherent undersampling followed by sparsity promoting reconstruction). In both cases, only a fraction of k-space is acquired and a reconstruction technique leveraging prior knowledge is used to fully reconstruct the image. Recent research employing AI-based reconstruction techniques has been successful [80, 81], but generally assumes an ideal setting without intra-scan motion.

Whereas acceleration of MRI acquisition lowers the risk of motion, residual motion artifacts may still result in images with reduced diagnostic quality. This problem is exacerbated by the fact that, when a lower percentage of k-space is sampled, the relative impact of motion becomes higher. In the era of AI-based reconstructions, however, such motion-corrupted data may confuse the network, giving rise to reconstructions that appear of sufficient quality but contain so-called hallucinations. Examples of hallucinations are the omission of existing pathology [80] or the wrongful creation of structures. As AI hallucinations may take a regular form (anatomy or pathology), they may go unnoticed and affect the subsequent clinical decision-making in contrast to traditional motion artifacts that are easily recognised by a radiologist. It is therefore of growing importance to reliably detect motion, preferably in a real-time and automated fashion. Data-driven motion detection is preferred in terms of simplicity and flexibility over MR navigators that come with increased complexity and acquisition times, as well as over external trackers that require additional hardware [17, 82].

Motion can be estimated in terms of motion parameters [83] on an inter-scan basis by e.g. estimating deformation vector fields [84, 85] and on an intra-scan basis

by e.g. aligned reconstruction [86, 87], which jointly searches for an uncorrupted multishot reconstruction and rigid-body motion parameters, with recent research inserting deep learning based components [88, 89]. Besides estimating the motion parameters, retrospective motion estimation can also focus on estimating motion in terms of artifacts in the image, which is useful as a more direct metric of image quality, and is the focus of this paper. Attempts have been made to relate motion trajectories acquired with tracking systems to image quality, such as integrating motion based on head speed [90] and comparing that to known values of motion-corrupted cases [91]. CNNs have been used to classify fully sampled MRI images as motion-free or corrupted [92], to estimate motion between artificially corrupted images and the corresponding uncorrupted images in terms of SSIM [93], or to generate probability maps for motion artifacts in MRI [94, 95].

While research has been done on estimating *motion parameters* of undersampled data, the aforementioned *motion artifact* estimation approaches all work on fully sampled images, whereas differentiation between motion and undersampling artifacts is challenging, and the focus of this paper. Moreover, undersampled imaging has become the clinical standard, which elevates the importance of differentiating motion artifacts from undersampling artifacts, that can have similar appearance. Our primary contributions are as follows:

(i) We propose a deep-learning based regressor that can accurately estimate the severity of motion artifacts in brain MRI scans. Its ability to detect motion artifacts in *undersampled* acquisitions is critical for potential application in the clinic.

(ii) We synthesize motion-corrupted raw MR data from motion-free data, for training with realistic motion-corrupted and corresponding uncorrupted ground truth image pairs. We demonstrate on a prospectively motion-corrupted test set that synthetic motion-corrupted data can be used effectively during training in the common case no labeled data with *in vivo* intra-scan motion is available.

(iii) Our model is able to detect motion corruption during acquisition before all k-space lines are sampled, enabling the possibility to alert the MR technician early, before the scan is fully completed. It could also prove valuable as a safety mechanism to prevent low-quality input data being used by AI-based approaches. Another use case is to leverage the model in a reconstruction framework, for example in an optimization strategy that uses the estimated motion artifact severity as a quality heuristic, or for the grouping criteria of adjacent k-space shots when little motion is detected between those shots in an approach like DISORDER [87], or as stopping criterion in a model-based reconstruction.

(iv) We introduce and evaluate one potential use-case for the regressor: a deep-

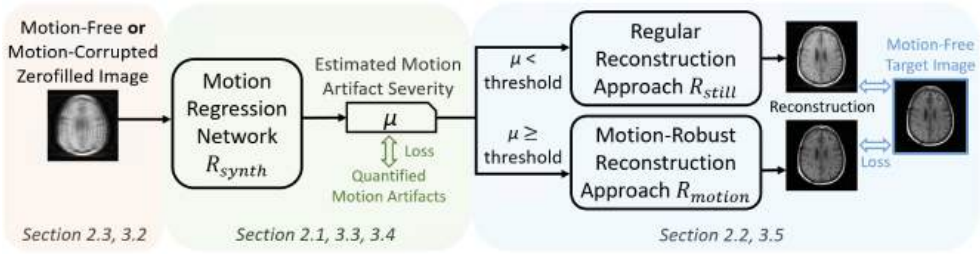


Figure 3.1: Overview of our framework. The reconstruction models also receive the coil sensitivity maps as input, which are omitted from this figure.

learning based reconstruction framework that falls back on a motion-robust solution when a considerable amount of motion is detected. This can improve quality if the motion-robust solution performs less well on motion-free cases, or improve speed if it is slower than a regular reconstruction approach.

(v) We show that our approach accurately estimates motion artifact severity on prospectively and retrospectively motion-corrupted *in vivo* MRI data of the brain with an undersampling factor between $1 - 8\times$. Additionally, we investigate the effects of rigid-body motion on AI-based reconstructions by training reconstruction models with and without motion-corrupted data. We show an improvement in performance on a mixed set of motion-free and motion-corrupted data as a result of our selector framework.

3.2 Methods

An overview of our framework is given in Figure 3.1 and of our models in Table 3.1. The motion corruptor can use one or more motion-free datasets to generate a large amount of training pairs of motion-corrupted images with corresponding motion-free ground truth. Either an undersampled motion-corrupted image or an undersampled motion-free image is provided to the networks during training and inference. The framework contains a motion-robust and a high data consistency reconstruction network that are trained to reconstruct either motion-corrupted data or still data, respectively. The severity of motion artifacts as estimated by the motion artifact regressor determines whether the high data consistency or motion-robust reconstruction is deployed.

3.2.1 Motion Artifact Regression Network

For estimating the amount of motion in a scan, we use a CNN architecture g with weights ϕ called $\mathcal{R}_{\text{synth}}$ that takes an undersampled motion-corrupted 2D zero-filled image slice $\mathbf{x}$ as input and returns the predicted motion artifact amount μ. We focused on a 2D image acquisition protocol as this constitutes the majority of scans acquired

Model	Type	Arch.	Loss	Training Data	Test Data	Unders.
$\mathcal{R}_{\text{synth}}$	Regressor	g	Eq. (3.1)	Synthetic & Still	Prospective & Still	1x to 8x
$\mathcal{R}_{\text{VGG19}}$	Regressor	VGG19	Eq. (3.1)	Synthetic & Still	Prospective & Still	1x to 8x
$\mathcal{C}_{\text{mrart}}$	Classifier	g	-	Prospective & Still	Prospective & Still	1x to 8x
$\mathcal{R}_{\text{still}}$	Reconstructor	f	Eq. (3.3)	Still	Prospective & Still	4x
$\mathcal{R}_{\text{motion}}$	Reconstructor	f	Eq. (3.3)	Synthetic & Still	Prospective & Still	4x

Table 3.1: Overview of the different models. *Arch* means architecture, *unders* means undersampling factor. *Synthetic* refers to the synthetic retrospectively motion-corrupted training set, *Still* to motion-free data, and *Prospective* to the test set with real prospective motion.

in radiological practice. We propose a loss that quantifies the average pixel-wise difference between the motion-corrupted zero-filled image and the motion-free image, thereby optimizing

$$\arg\min_{\phi} \| g_\phi(A^H A C \mathbf{x}) - \| A^H A C \mathbf{x} - A^H A \mathbf{x} \|_1 \|_1, \tag{3.1}$$

where the inner L1-norm operator provides a scalar representing the mean pixel-wise difference between the images, and the measurement operator A multiplies the image with the coil sensitivities, applies the Fourier transform, and finally applies a mask that zeroes out the k-lines that were not measured. A^H is the Hermitian transpose of A, and $A^H A \mathbf{x}$ applies k-space masking to $\mathbf{x}$. This loss aims to approximate the intensity of motion artifacts in the motion-corrupted image, by isolating the effect of the motion-corruption operator C on the uncorrupted image. We reason that comparing the zero-filled images rather than the fully sampled images in the loss may lead to more stable training as the network input is undersampled as well. For the motion-free scans, C is the identity function and thus the target is 0. We define $\mathcal{R}_{\text{synth}}$ as a CNN with seven blocks, each consisting of two convolutional with corresponding leaky ReLU layers, followed by a max pooling layer. Each block has half the width and height but double the number of feature layers compared to the previous block. The final block ends with a linear layer instead of a max pooling layer. $\mathcal{R}_{\text{synth}}$ uses 9M parameters. We compare this architecture against VGG19 [96], a popular architecture in medical imaging, trained in the same way on the same data. We used a VGG19 model with pre-trained weights (IMAGENET1K_V1), where after the original output layer we added one fully connected layer that returns the estimated motion artifact severity. This model $\mathcal{R}_{\text{VGG19}}$ uses 144M parameters.

To mitigate the effect of training variance, we employ an ensemble that combines 21 instances of the same network design that were trained with a different seed and order of data. The median prediction of the 21 networks is considered to be the prediction of the ensemble. For testing we selected the network that performed the best on the validation set.

3.2.2 Reconstruction Selector

We propose a reconstruction 'selector' framework that utilizes the regressor $\mathcal{R}_{\text{synth}}$ to estimate the level of motion artifacts in the image data to be reconstructed. If substantial motion is detected, a motion-robust approach is used for reconstruction, otherwise a regular reconstruction approach is used. This can improve quality when the motion-robust solution performs less well on motion-free cases, or improve speed when it is slower than a regular reconstruction approach. The regular reconstruction model $\mathcal{R}_{\text{still}}$ is trained to reconstruct motion-free k-space data, while the motion-robust model $\mathcal{R}_{\text{motion}}$ receives an undersampled corrupted image slice as input during training, with the target being the corresponding fully sampled uncorrupted image. Both reconstruction models use a modified version of the Adaptive-CS-Network architecture [7], which is a deep-learning unrolled iterative reconstruction scheme consisting of a sequence of blocks that each apply a reconstruction and a data consistency operation. Specifically, we lowered the number of blocks to ten for faster training. The data consistency enforces similarity between the measured k-space data and reconstructed k-space [97] as follows:

$$\mathbf{r}_{i+1} = \mathbf{x}_i - \lambda_i A^H (A\mathbf{x}_i - y), \tag{3.2}$$

where $\mathbf{x}_i$ is the output of the reconstruction step in block i, $\mathbf{r}_{i+1}$ is the 'data consistent' residual image, and λ_i is a learned data consistency modifier for block i, allowing the data consistency to be imposed less strongly if beneficial for performance. The training loss for reconstruction architecture f with weights θ, which differ per reconstruction model, is described by:

$$\arg\min_{\theta} w \left\| |f_\theta(A^H A C\mathbf{x})| - |\mathbf{x}| \right\|_1 + \sum_i \left\| |f_\theta(A^H A C\mathbf{x}_i)| - |\mathbf{x}| \right\|_1 . \tag{3.3}$$

The first term is the loss for the final predicted image and the second term is the loss for the intermediate image predicted by block i, compared against the fully sampled ground truth image. The first term is weighted stronger via w, which we set to 50.

3.2.3 Synthesizing Motion-Corrupted K-Space Data

We simulate motion-corrupted MRI acquisitions with a linear interleaved scanning protocol. We synthesize motion-corrupted scans by applying a 3D rigid motion pattern to a given still image over a series of timesteps to simulate the motion of the subject in the scanner. During each timestep, we apply a 3D shift and rotation in image space, and convert to k-space to sample relevant k-space lines according to a linear interleaved multi-slice pattern. Finally, we combine the sampled lines from each timestep into

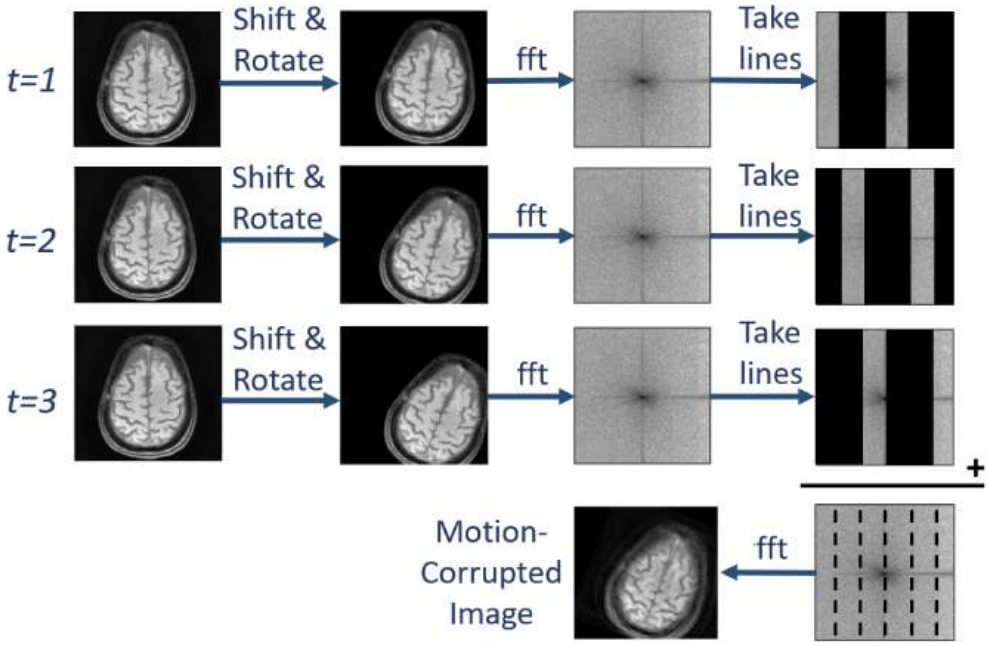


Figure 3.2: Overview of the motion-corruption process. It starts with a still brain image and a motion pattern, from which a motion-corrupted k-space and image are generated. For illustration purposes, the motion is exaggerated and the number of timesteps reduced.

one motion-corrupted k-space. This process is also illustrated in Figure 3.2. The coil sensitivity maps were kept unchanged. We used 524 rigid-body head motion patterns acquired on a scanner with an optical tracking system from a study where participants were instructed to perform shaking or nodding motion [98]. 393 patterns were used for training and 131 for validation. We multiplied each measured motion pattern by a uniformly randomly selected factor between 0 and 2 in order to generate a variety of training images with different amounts of motion. For motion-corrupted validation images, the patterns were either strengthened or weakened by a random factor of up to 2. The motion patterns were only used for training data generation, our approach does not require them during inference.

3.2.4 K-Space Undersampling Patterns

The regressor and reconstruction networks are fed images reconstructed from zero-filled k-space data that was retrospectively undersampled using Cartesian masks, with k-space lines set to zero in the phase encoding direction. As a default undersampling mask, we use the uniformly random distributed sampling pattern as used for the 2019 FastMRI challenge [80]. Given an undersampling factor R and a center fraction

parameter c that determines the fraction of lines guaranteed to be sampled from the center of k-space, the remaining lines are sampled with probability $p = (\frac{1}{R} - c)/(1 - c)$. For some experiments we also used an undersampling mask based on Poisson disk sampling [99] that reflects better undersampling patterns as used by clinical MRI scanners. It precludes the occurrence of large unsampled gaps in k-space that may be present in completely random sampling approaches. The sampling probability is higher near the center (low frequencies), and lower near the edges (high frequencies). Our implementation incorporates a center fraction parameter that guarantees a sufficient number of centermost k-space lines are sampled.

3.3 Experiments and Results

3.3.1 Training and Implementation Details

The motion artifact regression models were trained for $9{\times}10^3$ iterations with a batch size of 64 on an NVIDIA Quadro RTX 6000 GPU, while the reconstruction models were trained for $3{\times}10^6$ iterations. For the regression models we selected the retrospective undersampling factors per slice random uniformly from the range of $1\times$ to $8\times$, with a corresponding central fraction between 11% and 4%, so that the models learn to estimate motion artifact severity on scans with every possible reasonable acceleration factor. Even the low accelerations are interesting as the amount of motion still needs to be correctly estimated and be discerned from undersampling artifacts. For reconstruction we used an acceleration of $4\times$ with 8% central fraction since the low accelerations are considered to be trivial and $4\times$ acceleration is clinically the most relevant at the moment.

3.3.2 Data

We used multicoil brain T1, T2 and FLAIR scans from the NYU FastMRI brain dataset [100] as motion-free images and to generate retrospectively 3D-motion-corrupted images. We focused on a 2D image acquisition protocol as this constitutes the majority of scans acquired in radiological practice. Since 2D acquisition protocols are subject to through-plane motion as well, we used 3D motion. We discarded images with width or height smaller than 320 voxels. For training we used 4267 uncorrupted scans, and the same scans again as basis for 4267 retrospectively motion-corrupted training scans. 1389 of the training scans are T1-weighted, 2668 are T2-weighted and 210 are FLAIR. For validation, we used 1304 uncorrupted scans and 1304 derived motion-corrupted scans (427 T1, 809 T2, 68 FLAIR).

For prospectively motion-corrupted data we used the motion-related artifacts MR-ART dataset [101]. This dataset consists of 3D T1 brain images from 148 volunteers, who were asked to lie still, nod five times ('Head Motion 1'), and nod ten times ('Head Motion 2') during three separate scans, respectively. For the motion-free volunteer

task, 119/28/1 image volumes were labeled as 'Good'/'Medium'/'Bad' quality images by neuroradiologists. This changed to 7/59/75 for the first head motion task and 3/22/122 for the second head motion task. 28 sets of three scans each were used as test set. We resampled the MR-ART images from 1×1 mm to 0.6875×0.6875 mm followed by cropping to 320×320 pixels to match the resolution of the FastMRI challenge dataset. We discarded slices above the top of the head since those contain no information, as well as any slices > 85 mm lower to avoid slices where the face had been removed for anonymization purposes and to match the NYU FastMRI data that only contains the top of the head.

3.3.3 Performance of Motion Artifact Regressor on Retrospectively Motion-Corrupted Data

The motion artifact regression model $\mathcal{R}_{\text{synth}}$ was trained on 4267 and 120 motion-free and 4267 and 120 derived synthetically motion-corrupted NYU and MR-ART cases, respectively. While prospectively corrupted MR-ART data was available, only retrospectively corrupted data was used during training as we wanted to investigate the effectiveness of training exclusively on synthetically generated data. For validation, we used 1304 uncorrupted scans and 978 motion-corrupted scans.

While our approach predicts the motion amount as a scalar, for evaluation we instead report the ability to differentiate between still and motion scans based on the predicted motion amount, since the quantified motion artifact severity is a non-interpretable number and to allow for a comparison against other approaches. Figure 3.3 displays a histogram of the predictions of $\mathcal{R}_{\text{synth}}$ on retrospectively motion-corrupted NYU FastMRI data. Given a classification threshold separating the two peaks in the histogram at the lowest place in between, the model has an accuracy of 93.1%, with a false positive rate of 6.6% (incorrectly predicting a high amount of motion on a still scan) and a false negative rate of 7.3% (incorrectly predicting less motion than the threshold on a scan affected by motion). These results demonstrate that the regressor can accurately detect motion and differentiate between still and retrospectively motion-corrupted images.

3.3.4 Performance of Motion Artifact Regressor on Prospectively Motion-Corrupted Data

To assess whether the results on retrospectively motion-corrupted data are indicative of performance on prospectively motion-corrupted scans, we also evaluate the model $\mathcal{R}_{\text{synth}}$ on 28 volunteers (84 scans in total) from the MR-ART dataset. While the regressor is able to quantify motion artifacts as a scalar, only three discrete clinical artifact score labels by neuroradiologists ('good' $<$ 'medium' $<$ 'bad') and three task labels (still $<$ head motion 1 $<$ head motion 2) are available on the prospectively motion-corrupted data, as described in Section 3.3.2. We therefore evaluated the

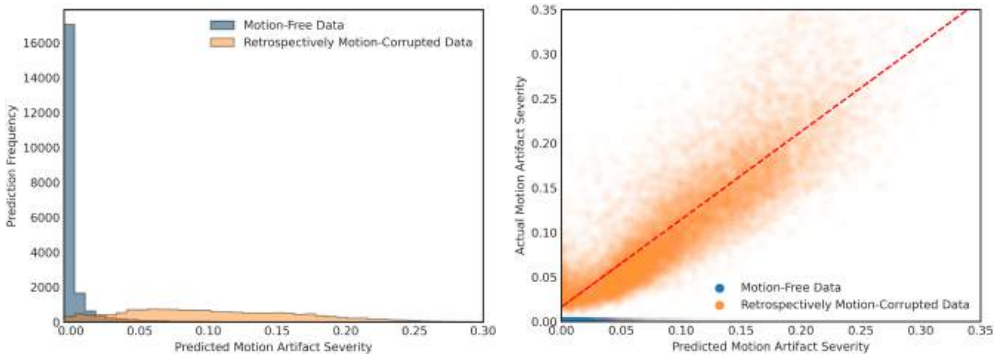


Figure 3.3: Histogram and scatterplot of motion severity predictions during the evaluation of $\mathcal{R}_{\text{synth}}$ on retrospectively motion-corrupted undersampled NYU FastMRI challenge data. The trend line for predictions on motion-corrupted data is $0.984x + 0.016$.

model based on the correspondence between its motion estimations and the artifact or task labels. First, we provided slices from the three different tasks per volunteer to the model and checked whether the difference in estimated motion artifact severity between the slices was in line with their labels, provided of course that the labels were different, i.e. were not both 'Bad'. For a given slice pair (there are three possible pairs per slice per volunteer), the model correctly orders the artifact score labels 98.1% of the time. If instead of the artifact label we consider the task (still $<$ head motion $1 <$ head motion 2), the model correctly predicted the ordering only 89.7% of the time. If we only compared the still task against head motion task 1 and 2, i.e. not including the comparison of head motion task 1 against 2, this increased to 98.0%. In other words, estimating which task was performed is challenging since the artifacts caused by head motion task 1 and 2 are hard to distinguish from each other. Figure 3.4 illustrates the average predictions of $\mathcal{R}_{\text{synth}}$ for each set of three scans per volunteer on a per-volume basis. The model predicted an increasing amount of motion 100% of the time as the label of a volunteer progressed from Good to Medium or from Medium to Bad. When looking at predictions between different volunteers, it can be seen that the model sometimes estimated a scan to have more severe motion artifacts than a scan of another volunteer despite it having a better label.

We also evaluated the model on the prospectively motion-corrupted data using the motion amount predicted by our model to bin scans into one of three artifact score categories (Good, Medium or Bad quality) based on two thresholds, which were selected to maximize accuracy on 120 MR-ART cases not included in the test set. The results on the prospectively motion-corrupted test set are shown in Table 3.2.

The VGG19 architecture did not perform as well as $\mathcal{R}_{\text{synth}}$ on the test set using the

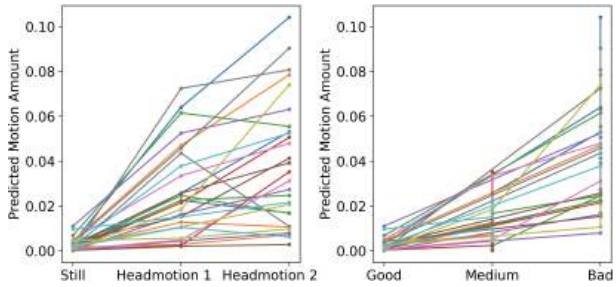


Figure 3.4: Predictions of our model $\mathcal{R}_{\text{synth}}$ on MR-ART data for each set of three scans per volunteer, with each scan grouped either by motion task (left) or artifact label (right). Each colored line corresponds to a different volunteer. Lines are vertical if the same artifact label was given to multiple tasks of the same volunteer. The data point for each scan was obtained by averaging the predictions for all its slices, with each slice having a random undersampling factor.

Table 3.2: Performance of the different models on MR-ART data with a random undersampling factor between $1-8\times$.

Data label:	Good Artifact Score			Medium Artifact Score			Bad Artifact Score			3 Label-	2 Label-
Prediction:	Good↑	Med↓	Bad↓	Good↓	Med↑	Bad↓	Good↓	Med↓	Bad↑	Accuracy	Accuracy
$\mathcal{R}_{\text{synth}}$	88%	12%	0%	30%	48%	22%	0%	14%	86%	76%	88%
$\mathcal{R}_{\text{VGG19}}$	76%	8%	16%	35%	22%	43%	0%	0%	100%	71%	83%
$\mathcal{C}_{\text{mrart}}$	96%	4%	0%	13%	57%	30%	0%	6%	94%	85%	95%

Table 3.3: Performance of the different models on MR-ART data with a random undersampling factor between $1-8\times$. HM1 and HM2 stand for head motion task 1 and 2.

Task:	Still Task			Head Motion Task 1			Head Motion Task 2			3 Task-	2 Task-
Prediction:	Still↑	HM1↓	HM2↓	Still↓	HM1↑	HM2↓	Still↓	HM1↓	HM2↑	Accuracy	Accuracy
$\mathcal{R}_{\text{synth}}$	93%	7%	0%	14%	39%	46%	7%	21%	71%	68%	91%
$\mathcal{R}_{\text{VGG19}}$	93%	7%	0%	25%	57%	18%	11%	39%	50%	67%	86%
$\mathcal{C}_{\text{mrart}}$	96%	4%	0%	4%	39%	57%	4%	11%	86%	74%	96%

same training conditions and data, potentially because its architecture is much larger (namely 7 times more parameters), making it more susceptible to overfitting. We used pre-trained weights for $\mathcal{R}_{\text{VGG19}}$ as it improved the accuracy by about 2%.

We also trained a model on MR-ART artifact labels specifically ($\mathcal{C}_{\text{mrart}}$) on 120 training cases. As this work focuses on training with synthetic data ($\mathcal{R}_{\text{synth}}$), this experiment is not intended to be a competitive comparison, but it serves to demonstrate what performance can be reached when training to predict the labels on prospectively motion-corrupted data. The model correctly ordered the artifact score labels 99.85% of the time for a given slice pair per subject, e.g. given a 'medium' and 'bad' slice, the

model correctly predicted almost always more motion for the 'bad' slice. It achieved an accuracy on inter-subject separability between Good and Medium/Bad volumes of 95.2%, though that required reducing the task of the model to a classification task and training on prospectively motion-corrupted data, both of which were not the goal of this paper.

In an alternative experiment where $\mathcal{R}_{\text{synth}}$ was trained and evaluated on fully sampled data instead of undersampled data, it correctly ordered the artifact score labels 98.5% of the time for a given slice pair per subject, and achieved an accuracy on inter-subject separability between Good and Medium/Bad volumes of 88.1%, and between still data and head motion volumes of 91.7%.

3.3.5 Reconstruction Models and Selector Performance

The reconstruction models were initially trained for 1.7M iterations with one slice per iteration on uncorrupted brain T1, T2 and FLAIR images from the NYU FastMRI dataset [100], and subsequently finetuned for 1M iterations on either 4267 uncorrupted cases (model $\mathcal{R}_{\text{still}}$) or both 4267 uncorrupted and 4267 derived retrospectively motion-corrupted cases (model $\mathcal{R}_{\text{motion}}$). The retrospectively motion-corrupted dataset was used as opposed to the MR-ART dataset as only the former contained enough data to train a reconstruction model on.

Table 3.4 shows the results of the two reconstruction models. The introduction of motion caused a large decrease in reconstruction quality of $\mathcal{R}_{\text{still}}$. In comparison, the model $\mathcal{R}_{\text{motion}}$ that was fed motion-corrupted data during training displayed better performance when evaluated on motion-corrupted data, although the reconstructions appear less sharp than the uncorrupted fully sampled images. On still data, the performance of $\mathcal{R}_{\text{motion}}$ decreased compared to $\mathcal{R}_{\text{still}}$ by 7.1%, relative to the target SSIM of 1. When inspecting the learned data consistency modifiers that allow the data consistency to be imposed less strongly, we measured a decreased weighing in the first block from 1.00 for $\mathcal{R}_{\text{still}}$ to 0.65 for $\mathcal{R}_{\text{motion}}$, and of the remaining blocks from an average of 0.71 to 0.13. The change in data consistency strength indicates that it is beneficial for the motion-induced models to have more freedom to compensate for motion in the measured k-space, and may explain the lower performance of $\mathcal{R}_{\text{motion}}$ on motion-free data.

In this experiment we investigated a way to leverage the advantages of both the standard and motion-robust reconstruction approaches, by creating a motion-adaptive reconstruction framework based on a model selection mechanism that does not compromise on quality or data consistency for cases without motion. Using the amount of motion artifacts as predicted by the developed regressor, we select between a motion-robust versus a conventional reconstruction model. If $\mathcal{R}_{\text{synth}}$ quantifies the severity of motion artifacts to be above 0.025, we considered the scan to be motion-

Table 3.4: Performance in SSIM of the individual reconstruction models and the selector framework. † denotes a significant difference from $\mathcal{R}_{\text{selector}}$. Note that the reconstruction models received $4\times$ undersampled data, thus the fully sampled motion-corrupted data should not be seen as a baseline.

Dataset:	Still Data	Motion Data
Metric:	[median] $\mu \pm \sigma$ SSIM	[median] $\mu \pm \sigma$ SSIM
Fully Sampled	$[1.000]\ 1.000 \pm 0.000$	$[0.762]\ 0.742 \pm 0.161$
Zero-Filled	$[0.761]\ 0.750 \pm 0.077$†	$[0.648]\ 0.629 \pm 0.108$†
$\mathcal{R}_{\text{still}}$	$[0.915]\ 0.892 \pm 0.094$†	$[0.773]\ 0.746 \pm 0.131$†
$\mathcal{R}_{\text{motion}}$	$[0.909]\ 0.887 \pm 0.095$†	$[0.837]\ 0.805 \pm 0.119$†
$\mathcal{R}_{\text{selector}}$	$[0.914]\ 0.891 \pm 0.096$	$[0.835]\ 0.803 \pm 0.118$

corrupted and it was thus fed into $\mathcal{R}_{\text{motion}}$ rather than $\mathcal{R}_{\text{still}}$. Table 3.4 shows the results of the selector framework. The selector framework comes very close to the performance of $\mathcal{R}_{\text{still}}$ on motion-free data and performed much better on motion-corrupted data, for which it almost matched the performance of $\mathcal{R}_{\text{motion}}$. The gold standard is for the selector framework to perform as well as $\mathcal{R}_{\text{still}}$ on still data and as well as $\mathcal{R}_{\text{motion}}$ on motion-corrupted data. The fact that the framework got close to the optimal values indicates that the regressor model was most of the time able to correctly provide $\mathcal{R}_{\text{still}}$ with motion-free scans and $\mathcal{R}_{\text{motion}}$ with scans containing motion artifacts.

On motion data, in the cases where the regressor provided accurate/inaccurate results and selected the optimal/suboptimal network, the difference between the reconstructions of $\mathcal{R}_{\text{motion}}$ and $\mathcal{R}_{\text{still}}$ was 0.0657/0.0235 SSIM on average. This indicates that classification errors on motion data are made more often when the reconstructions are similar. On still data the difference was 0.0054/0.0062, indicating that classification errors on still data are made more often when the reconstructions are different.

3.3.6 The Effect of the Sampling Scheme on Performance

We investigated the performance of different sampling strategies, under the assumption that sampling the same set of k-lines for different cases may make the undersampling artifacts more predictable and thereby motion artifacts easier to identify. Acceleration during training and evaluation was fixed at $4\times$. Using random uniform probability undersampling masks, the model achieved an accuracy of 76% when predicting the exact label (Good/Med/Bad). We compared this random approach, which samples a different set of k-lines for each case, to a similar masking strategy that always samples the same set of k-lines for every scan. This approach achieved an accuracy of 75%. The Poisson mask approach described in Section 3.2.4 achieved an accuracy of 79%, allowing the network to better estimate motion artifact severity compared to using

random uniform masking. However, note that we still used random uniform masking in the other experiments of this paper to adhere to the fastMRI-challenge setup.

3.4 Discussion and Conclusion

We developed a deep-learning based regressor that can accurately estimate the severity of motion artifacts in undersampled brain MRI scans. To the best of our knowledge, the existing retrospective motion *artifact* estimation approaches require the full set of k-space data to be available [92, 93, 94, 95]. We investigated motion artifact severity estimation on accelerated MRI data, which introduces undersampling artifacts on top of the motion artifacts that can have similar appearances and are thus challenging to be distinguished from each other. Our model is able to detect motion corruption during acquisition before all k-space lines are sampled, enabling the possibility to alert the MR technician early, before the scan is fully completed.

We simulated motion-corrupted MRI acquisitions based on uncorrupted MRI data, according to a linear interleaved scanning protocol, though different protocols can be implemented as well. When trained to quantify the severity of motion artifacts on undersampled motion-free data and undersampled synthetically motion-corrupted data, the model was able to separate data from the two classes with an accuracy of over 93%. While this regressor was trained exclusively on retrospectively motion-corrupted data, it was still able to distinguish prospectively motion-corrupted nodding data from still data 91% of the time, indicating that our motion-corruption framework generalized well to real world data. The regressor ordered the artifact score labels (Good/Medium/Bad) correctly 98% of the time for each set of scans of a given subject. When compared against VGG19, we found that our architecture performed better, potentially because the higher number of parameters of VGG19 makes it more susceptible to overfitting.

When investigating different sampling strategies, the use of a particular Poisson disk sampling pattern improved the performance compared to using random uniform sampling. We believe that using the same sampling pattern between all training and testing cases caused the undersampling artifacts to be more predictable, making it easier to distinguish them from motion artifacts. However, using such a single sampling mask instance will introduce a bias, when a different sampling scheme would be used, i.e. a compromise between generalizability and performance.

Presence of motion severely impacts reconstruction by a regular AI-based approach. By training the reconstruction model also on motion-corrupted cases, it learns to deviate from the measured k-space, which can partially alleviate the negative effects of motion and improve reconstruction quality on motion-corrupted data. However, the learned data consistency that allows the model to deviate from the measured k-space and thereby compensate for the motion, also reduces how strongly the reconstruction

is based on the measured data when no motion is present and will therefore negatively impact reconstructions of still data.

We investigated a way to leverage the advantages of both the standard and motion-robust reconstruction approaches, by creating a motion-adaptive reconstruction framework based on a model selection mechanism that does not compromise on quality or data consistency for cases without motion. This is done by selecting the optimal reconstruction network based on our motion artifact severity estimator. One can also imagine a use-case where a motion-robust approach performs as well as a regular approach, but is much more computationally expensive. In such case, the selector can be a time saving measure by only performing motion-robust reconstruction on cases with visible motion artifacts. We show a significant improvement in reconstruction performance on a mixed set of motion-free and motion-corrupted data as a result of our selector framework. For motion-corrupted cases on which the regressor made a mistake (i.e. predicted too little motion corruption), the difference in SSIM between the two reconstruction models was on average three times as small compared to when the regressor did correctly estimate much motion corruption. The cases where the difference in SSIM is larger are likely to be high motion cases on which $\mathcal{R}_{\text{still}}$ performs much worse. These high motion cases are easier for the regressor to identify correctly. Thus, classification mistakes by the regressor are most often made on low motion cases where the choice in reconstruction model is not very impactful. This is further substantiated by the fact that the selector achieves close to optimal performance.

A limitation of the used datasets is that the validation set uses raw k-space *in vivo* data but with simulated motion, and the test set uses *in vivo* image data with real prospective motion, albeit obtained by instructing participants to move, i.e. not natural motion. As basis for the retrospectively motion-corrupted training and validation sets, we used data with a 2D acquisition protocol. We took the anisotropic voxel sizes into account when performing rotation and shift calculations. The lower resolution in the z-direction can make the simulation less realistic, although we believe the effect to not be very detrimental since the performance was good on the 3D test set with real motion. For future work, the performance of 3D and 2D reconstruction models on motion-corrupted data could be compared to investigate the impact of through-plane motion, as through-plane motion requires the 2D reconstruction models to hallucinate to fill the gaps. We also would like to train and evaluate a reconstruction model on prospectively motion-corrupted cases, though a challenge is the availability of data and the fact that the still and motion-corrupted scans need to be perfectly aligned. Another direction for future research could be to more explicitly compensate for motion in k-space by integrating the quantified motion artifacts in the data consistency term of the reconstruction. Further investigations could be towards the effect of different motion patterns on model performance such as finetuning on nodding

patterns, and investigating whether it can be impactful to take the effect of motion on the coil sensitivity maps into account during synthetic motion-corruption and motion compensation.

In conclusion, the proposed motion detector showed a very high accuracy on retrospectively as well as prospectively motion-corrupted MRI data, and a motion synthesis framework can be used effectively during training in the common case when no labeled data with real intra-scan motion is available. This enables, among others, the use of our method as a safety mechanism against AI hallucinations, as a prompt for re-scanning, or as a component in a motion-robust reconstruction framework.

Acknowledgments

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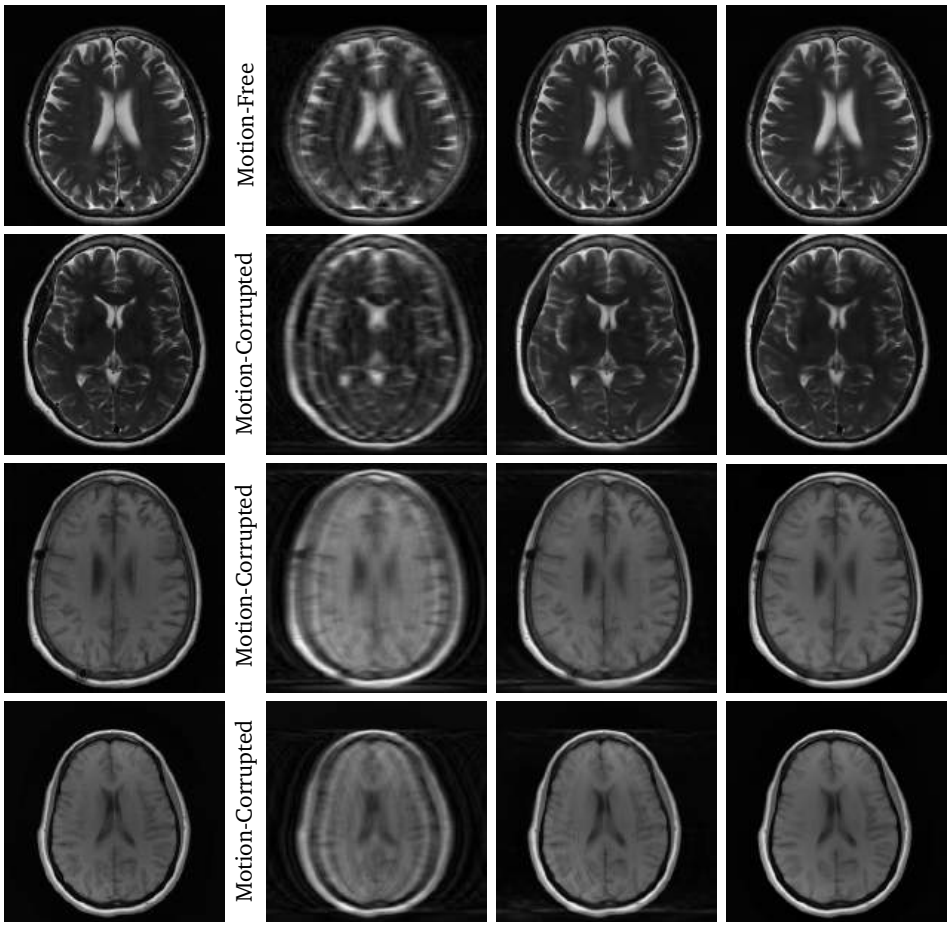


Figure 3.5: From left to right: Fully sampled uncorrupted image, the 4× undersampled zero-filled image (network input) that is uncorrupted in the first row and motion-corrupted in the other rows, the reconstruction by $\mathcal{R}_{\text{still}}$ and the reconstruction by $\mathcal{R}_{\text{motion}}$. The first row shows a case where $\mathcal{R}_{\text{still}}$ performed better, as $\mathcal{R}_{\text{motion}}$ blurred structure away in the top left and did not reconstruct detailed structures at the center top. The second row of images shows a case where $\mathcal{R}_{\text{motion}}$ performed better, as the reconstruction by $\mathcal{R}_{\text{still}}$ propagated some of the motion artifacts. The bottom images display a case where both models performed suboptimally, as the anatomy is smoothed away at the top and the bottom.

4

DEEP-DISORDER: Motion Correction in 3D MRI via Segment Reconstruction and Registration

This chapter was adapted from:

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Abstract

3D MR image acquisition is inherently time-intensive, rendering it susceptible to patient motion during scanning. This may introduce significant blurring and artifacts, potentially necessitating re-acquisition.

We propose a modular framework to retrospectively correct for intra-scan motion in 3D brain MRI, without active motion tracking. Serving as the backbone of our approach is an existing distributed and incoherent sampling scheme (DISORDER), combined with a fast network trained for highly undersampled reconstruction. This enables approximate reconstructions of anatomy after every few seconds, using only a tiny fraction of k-space data ($< 2\%$). While these reconstructions are only approximate, we postulate they are sufficient to estimate motion patterns at said temporal resolution. Groupwise registration, notable for its elimination of registration bias, is utilized for estimating rigid motion parameters, which are leveraged to reconstruct the measured data with reduced motion artifacts.

The approach was evaluated on 94 retrospectively and 3 prospectively motion-corrupted in vivo 3D T1-weighted brain MRI acquisitions. The estimated motion parameters matched the known retrospective motion with 0.06 mm and $0.13°$ accuracy, resulting in an improvement in reconstruction quality from 0.942 ± 0.026 to 0.992 ± 0.003 SSIM for the retrospective scans. The prospective scans improved from 0.915 ± 0.024 to 0.936 ± 0.014 SSIM after correction in the case of gradual motion and from 0.764 ± 0.008 to 0.923 ± 0.011 SSIM for extreme motion.

In conclusion, the proposed approach, that is free of external tracking devices or navigators, successfully estimated and corrected 3D motion between small sub-portions of a scan. This resulted in vastly improved image quality, making volumetric MRI substantially more tolerant to motion.

4.1 Introduction

Multi-shot MRI is an imaging technique used to obtain high-resolution anatomical images by acquiring k-space data over multiple excitation pulses, which is a time-intensive process that requires the subject to lie still for several minutes. This can be especially challenging for young or vulnerable patients. If the patient does move during MRI acquisition, this can introduce artifacts and distortions into the image with their severity depending on the amount of motion, and may in the worst case require the scan to be reacquired. One study estimated the costs of motion at $115,000 per scanner per year in 2015 [4], while another study in 2020 estimated the annual cost of patient head motion at $45,066 per scanner without pediatric examinations and $364,242 with pediatric examinations [102]. Thus, development of approaches that make MRI more robust to motion without increasing scan time or decreasing image quality is paramount. Retrospective intra-scan motion correction aims to reduce motion artifacts in the reconstruction of an MRI image, often requiring an estimate of the motion of the subject to do so [103, 17, 20, 21, 104, 105, 106]. One way of obtaining motion parameters is via tracking devices, though depending on the approach there may be additional effort that complicates the workflow [17], increased cost, and also the accuracy of the external tracking devices may be insufficient. On the other hand, data-driven approaches are preferred as they do not require external hardware.

Retrospective motion correction is deeply intertwined with image reconstruction, thus addressing these two problems jointly confers synergistic benefits. After all, a better motion estimate can enable better reconstruction and vice versa. Such an approach may involve incorporating a motion term in the forward model and iteratively estimating the motion parameters and motion-corrected image [107]. An example of such a work is the DISORDER [87] framework, which additionally introduces a specialized sampling scheme. This distributed and incoherent sampling scheme maximizes the sensitivity to motion, which improves the ability to detect motion and solve the aligned reconstruction problem. While these approaches have the benefit of being data consistent, the iterative process can be computationally expensive and difficult to converge. One work proposed integrating a CNN trained to remove motion artifacts to improve the conditioning and convergence of the joint search [89]. Some approaches, on the other hand, avoid the repeated image updates in the joint optimization, for example using a scout scan to replace the image estimate, thereby speeding up the reconstruction but slowing down the acquisition [108], or a deep-learning based reconstruction to replace the image estimate [109], optimizing only the motion parameters using a data consistency loss.

AI-based reconstruction end-to-end models have fast inference times and can

be trained to reconstruct images that appear free of motion artifacts, often using a physics-informed motion-corruption simulation to generate paired training data [20]. Approaches that do not explicitly model the motion parameters have been proposed using 3D CNNs [110], foundation models [111], Fourier aggregation [112] and GANs [113]. However, if the underlying motion is not modeled explicitly, deviating from the measured k-space data (data inconsistency) is required to correct the motion artifacts. In the case of severe motion, such a model may be prone to hallucinations, potentially resulting in missing anatomical details or the generation of non-existent anatomy in the reconstruction [26, 109].

We therefore propose a framework that retrospectively corrects for motion in 3D MRI requiring no external tracking device, reference scan, or fully sampled center of k-space. The proposed approach generates reconstructions of small scan segments that are registered in a groupwise fashion, thereby estimating a time series of rigid motion parameters to be used in a final motion-corrected reconstruction. Only the estimated motion parameters are used in the final motion-corrected reconstruction, enabling the enforcement of data consistency and avoiding hallucinations. The estimation of motion parameters by registering some form of low quality reconstructions for each scan segment has been applied before in self-navigated approaches, for example by repeatedly sampling rotated blocks of k-space and using the overlapping central data from each to generate low-resolution images [114], or by apodization of the repeatedly sampled central k-space data with a Fermi filter [115]. However, these approaches rely on a fully sampled center of k-space for each segment, requiring a slower sequence [17] or a sequence that already oversamples the k-space center [115]. We instead postulate that deep learning enables the reconstruction of small scan segments without a fully sampled k-space center at a sufficient quality for motion estimation. To ensure each scan segment contains a few lines from the center of k-space, we use the distributed incoherent sampling introduced for the DISORDER [87] method. We have therefore named our framework DEEP-DISORDER.

In the following, we detail our approach, and compare it to the original DISORDER reconstruction on retrospectively motion-corrupted data, as well as on prospectively motion-corrupted and motion-free 3D T1-weighted in vivo brain scans.

4.2 Methods

4.2.1 Problem Definition

The data acquisition process is modeled by the forward equation $y = Ax$, with y denoting the measured k-space data, x denoting the underlying image, and the forward operator $A = MFSC_\theta$ describing the acquisition process. Here, C_θ is the segment-wise motion corruption operator that projects the image into different motion states

with different motion parameters θ. In the proposed pipeline, trilinear interpolation is used in the operator C_θ. S represents the coil sensitivity encoding operator, F is the Fourier transform, and M is the masking operator that selects k-space lines by applying different sampling masks to the different segments. We define a segment as a fraction of k-space that represents one estimated motion state. As our acquisition process (a 3D-T1-weighted TFE sequence) comprises multiple shots, it follows naturally that we partition the k-space at shot boundaries with each segment consisting of a whole number of shots. Estimated motion parameters $\hat{\theta}$ can be used to invert the displacements and rotations that occurred during the acquisition for each segment, yielding the motion-corrected reconstruction problem

$$\hat{x} = \underset{x}{argmin} \|MFSC_{\hat{\theta}}x - y\|_2^2, \tag{4.1}$$

which can be solved using CG-SENSE [116] adapted for motion compensation [113, 117]. Given the estimated forward operator $A = MFSC_{\hat{\theta}}$, the least-squares solution for a given A can be found by solving the equation:

$$A^H Ax = A^H y. \tag{4.2}$$

Since the matrix $A^H A$ is too large to be inverted or stored explicitly, the system is solved iteratively using the Conjugate Gradient (CG) algorithm [118].

4.2.2 DISORDER Reconstruction

The original DISORDER reconstruction approach [87] solves for x by iteratively updating the estimated motion parameters θ for a fixed image, and updating the image using fixed motion parameters:

$$\hat{x}^{(i+1)} = \underset{x}{argmin} \|MFSC_{\hat{\theta}^{(i)}}x - y\|_2^2, \tag{4.3}$$

$$\hat{\theta}^{(i+1)} = \underset{\theta}{argmin} \|MFSC_{\theta}\hat{x}^{(i+1)} - y\|_2^2. \tag{4.4}$$

DISORDER introduced a distributed incoherent sampling scheme that for each shot acquires a broad variety of frequencies from all over k-space, making the motion easier to detect and making the problem easier to solve. We feed the final motion parameters estimated by DISORDER to the same motion-corrected CG-SENSE algorithm described in Equation 4.1 to obtain a motion-corrected reconstruction comparable to the proposed approach.

4.2.3 DEEP-DISORDER Reconstruction

An overview of the proposed retrospectively motion-corrected MRI reconstruction approach is given in Figure 4.1. The framework estimates 3D motion by reconstructing

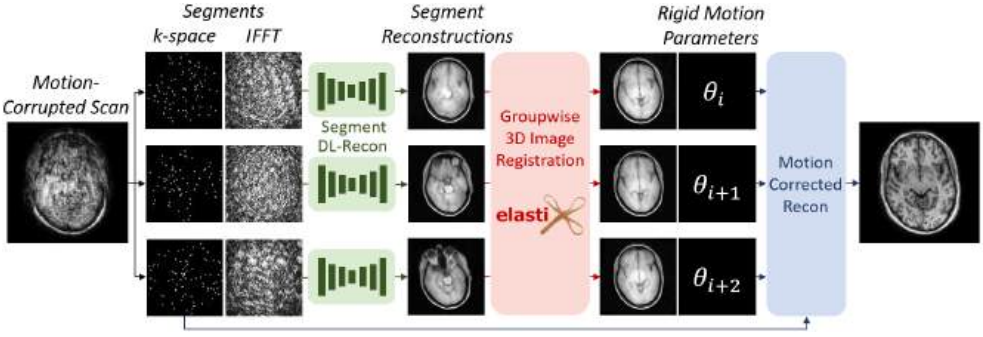


Figure 4.1: *Overview of the proposed motion correction approach. The acquired motion-corrupted k-space is split into segments based on acquisition time. A zero-filled fast Fourier transform is applied and an unrolled 3D U-Net performs reconstructions for each extremely undersampled segment. These reconstructions are aligned using groupwise registration, yielding one set of estimated motion parameters per segment. Finally, these motion parameters and the measured data are used in the forward operator of a classical reconstruction approach to produce a motion-corrected image. The coil sensitivities are omitted from the figure for clarity.*

and registering segments of the acquired MRI data. We use the DISORDER sampling scheme to ensure a balanced distribution of low and high frequencies in every shot. As a single shot may contain very little anatomical information, we group multiple shots into a segment to reach an image quality that is sufficient for the subsequent registration process. Reconstructing the segments by zero-filling does not yield sufficient image quality for registration as these reconstructions lack clear anatomical features due to the extremely high acceleration factor, and each segment has different undersampling artifacts, as seen in Figure 4.1. Moreover, the segment reconstruction problem is too ill-posed for classical reconstruction algorithms. Therefore, we pass the segments through a deep learning reconstruction network that was trained to reconstruct extremely highly undersampled data. The segment reconstruction network f_{recon} reconstructs the nth k-space segment as

$$\hat{x}_n = f_{\text{recon}}\left(S^H F^{-1} M_n^H y; \phi\right). \tag{4.5}$$

During training, a random segment index r and random rigid transformation C_r are generated to obtain a synthetic motion-corrupted k-space segment $y_r = M_r F S C_r x$ from uncorrupted image x. The network parameters ϕ are learned according to the optimization objective

$$\phi^* = \underset{\phi}{argmin} \mathcal{L}(C_r x, f_{\text{recon}}(S^H F^{-1} M_r^H y_r; \phi)) \tag{4.6}$$

with loss function $\mathcal{L}$. These segment reconstructions are limited in quality due to the high undersampling factor and resulting lack of information, but we postulate

that they are sufficiently accurate for registration purposes. The magnitude images of the segment reconstructions are registered with groupwise registration [23] using the medical image registration toolbox Elastix [119, 120]. By employing groupwise registration, we avoid having to select an arbitrary fixed image, eliminating bias in the selection by optimizing the group objective across all images via a consensus template representative of the entire group. This could be especially relevant if the quality of the segment reconstructions varies, which could potentially happen in the case of intrasegment motion, or variations in coil sensitivity due to the position of the subject. Additionally, groupwise registration avoids the accumulation of errors inherent in sequential pairwise registrations since transformations are jointly estimated.

A registration mask was used to ignore the neck region, which is often affected by non-rigid motion, and instead focus on head registration. During inference, the segment reconstructions are registered to estimate the rigid motion parameters

$$\{\hat{\theta}_1, \ldots, \hat{\theta}_N\} = f_{\text{reg}}(\{\hat{x}_1, ..., \hat{x}_N\}). \tag{4.7}$$

The groupwise registration objective function of f_{reg} is described by

$$\{\hat{\theta}, \hat{T}\} = \underset{\theta, T}{argmin} \sum_{s=1}^{N} \left\{ \sum_{v} \left\| \theta_s\Big(\hat{x}_s(v)\Big) - T(v) \right\|_2^2 + \text{dist}\Big(e,\, \theta_s\Big) \right\}. \tag{4.8}$$

Here, θ_s is the transformation for segment s, $T(v)$ is the common template for voxel v, and $\text{dist}(e,\, \theta_s)$ is a regularization term enforcing closeness to the identity transformation e [121].

Finally, as described in Equation 4.1, the resulting registration parameters are used as motion parameters in the motion-corrected CG-SENSE reconstruction algorithm to reconstruct the measured data with greatly reduced motion artifacts.

4.2.4 Training and Implementation Details

For segment reconstruction we trained an unrolled 3D U-Net [122, 123] with four resolution levels with 16, 32, 64, and 128 feature maps per level. Each layer of the model consists of group normalization, 3D convolution and ReLU operators, respectively. The model received a highly undersampled zero-filled 3D image as input and was tasked to estimate the corresponding uncorrupted fully sampled 3D image. A blend of SSIM [124] loss and smooth-L1 loss was used during training. Both the input and target were randomly rotated and translated during training to account for different patient positions.

Unrolled or iterative networks are commonly used in MRI reconstruction [125, 7], as they enable iterative reconstruction in which data consistency is continually enforced. Since each segment has few k-space lines, the benefit of data consistency that an iterative approach provides is small. Nonetheless, we observed a minor

improvement in performance over a single-stage approach from 0.043 to 0.033 NMSE on 90 retrospectively corrupted scans, albeit at the cost of an increased GPU memory footprint. We thus use a segment reconstruction approach with six consecutive residual U-Nets that were trained end-to-end. The reconstruction models were pretrained for 16 segments on an NVIDIA RTX A6000 GPU for 1×10^6 iterations with a batch size of 2 and a group normalization size of 8, and fine-tuned on the desired number of segments for 4×10^5 iterations, which was 64 for most experiments. We used a leave-one-out approach to reduce out-of-distribution issues on the test set, by further fine-tuning multiple models on a subset of motion-free volunteer scans and evaluating each on the unseen volunteer.

For groupwise registration we used Elastix with a multi-resolution pyramid of 4 levels and 2500 iterations per resolution. The used registration pipeline and settings differ from DISORDER, as the proposed approach performs only a final groupwise registration while DISORDER performs many quick motion update steps alternating with reconstruction steps. We selected a high number of registration iterations and three repetitions of the registration schedule to ensure convergence. We did not optimize these settings for runtime performance. The registration took about 10 minutes for R=2 on the CPU.

4.2.5 Number of Segments

Besides the hyperparameters of the segment reconstruction and registration, there are only a few factors of influence for our method, most notably the number of segments. There is a natural tradeoff between the achievable image quality for each segment and the temporal resolution for the motion correction. To find a balanced configuration for which the proposed approach performs best, we trained and evaluated our approach on acquisition patterns with different numbers of segments, which were constructed using random-checkered sampling [87]. Figure 4.2 shows the reconstruction performance for a given number of segments on 90 retrospectively sampled motion-corrupted scans with shaking or nodding-like motion with 320 motion states, demonstrating that the ideal number of segments is around 64 for our current setup.

4.2.6 Generalizability

An experiment is included in the appendix that investigates generalizability to different retrospectively motion-corrupted data, specifically two spoiled gradient echo 3D-scans with different echo-times, with and without fine-tuning. The segment reconstruction model can handle unseen data that is moderately similar to the training data, but the performance lowers as the contrast gets further from T1w data which the model was trained on. It is demonstrated that this issue can be solved by including a number of scans of the unseen contrast in the training set. Additionally, successful correction of motion in higher resolution scans (384^3 voxels) was demonstrated when the motion

estimation is run on a lower resolution (192^3 voxels) to reduce the need for compute resources.

4.2.7 Data

Retrospectively motion-corrupted data: We focus our experiments on T1-weighted 3D MRI scans. We used an in-house dataset of 361 3D T1-weighted turbo field echo scans from our radiological clinic without intentional motion for training and validation, and 94 for the synthetic test set. 416 of the scans were acquired on Philips Ingenia 3.0T scanners, and 39 on an Ingenia Elition 3.0T X scanner. All participants provided informed consent and the study was approved by the institutional review board. The scans were performed in a transverse orientation. The flip angle was $8°$, the repetition time (TR) around 9.9 ms, and the echo time (TE) around 4.6 ms. The scans were acquired with $1.8\times$ acceleration. Coil-combined images were obtained using the scanner's built-in compressed sensing reconstruction without post-processing. These scans were retrospectively motion-corrupted by simulating the k-space acquisition process, giving us paired motion-corrupted and ground-truth data. For each motion state, we applied a translation and rotation to the coil-combined image and we sampled k-space lines from this transformed image according to the acquisition pattern. The acquisition matrices of the original data had different sizes, which we cropped or zero-padded to 240^3. For intersegment motion, we used one motion state per segment, meaning 64 motion states with multiple shots sharing one motion state. This is less realistic, but matches the number of motion states our approach can estimate, allowing us to see how well it resolves correctable motion. To simulate intrasegment motion, we used 320 motion states, giving each shot its own motion state. We used a TFE factor of 180 in our simulated acquisitions, meaning 180 k-space lines per shot and 320 shots, resulting in 900 k-space lines per segment for 64 segments. We obtained coil sensitivity maps from the SENSE reference scan that is automatically acquired for sequences that require coil sensitivity maps for image reconstruction. The coil sensitivity maps were compressed to 8 coils by a singular value decomposition [126] and normalized using root-sum-of-squares normalization. The scans were retrospectively undersampled by discarding a number of shots corresponding to the desired acceleration factor. Different motion patterns were used in the different experiments. To create random intersegment motion, we sampled 64 translations and rotations from a normal distribution with a standard deviation of 2 mm and $1°$ for each axis ('spatially medium motion') or 12 mm and $6°$ ('spatially extreme motion'). This resulted in an average distance between the centers of two segments of 4.5 mm or 27.1 mm respectively. We optionally added intrasegment motion on top of this motion, randomly generated in a similar fashion but with 320 motion states and only $1/16^{\text{th}}$ as severe, as we expected such motion to be less severe

due to the shorter timeframe. Besides the random motion patterns, we also used more realistic rigid head motion patterns from a previous work [98] where a scanner with an optical tracking system was used to track participants who were instructed to perform shaking or nodding-like motion.

Prospectively motion-corrupted data: We acquired additional 3D T1-weighted prospectively motion-corrupted scans and still reference scans from 3 healthy volunteers for testing and 1 for validation, who all provided informed consent. The volunteers were instructed to move according to a specified motion pattern in each scan. For the gradual motion pattern, the volunteers were asked to occasionally slightly rotate their head as if slowly looking around, mimicking the drift often seen in clinical settings by patients who gradually relax or tire. For the jittery motion pattern, the volunteers were asked to abruptly rotate their head in a random direction approximately once every 15 seconds, such that their nose moves by approximately 2 cm each time. For the spatially extreme motion pattern, the volunteers were asked to look as far as possible to the bottom left at the start of the acquisition and rotate their head until they faced the top right at the end of the acquisition. The volunteers were informed in advance of the length of the scan and were asked to perform this motion in a stepwise fashion by moving every 15 seconds as it can be straining to slowly move continuously. This results in a reproducible large motion pattern with visible progression, which enables visually checking the reconstructed segments and registration for correctness while no ground truth motion is available. The first, middle and last segment reconstructions of the volunteer used for validation can be seen in Figure 4.1. We did not apply padding in the head coil, to enable more severe motion and test the limits of what motion can be corrected. The acquisitions used the MP-RAGE [127] sequence and were acquired on a 1.5T scanner (Ingenia, Philips Healthcare, Best, The Netherlands). The acquisitions were of size $191 \times 191 \times 384$ and consisted of 133 or 68 shots for prospective accelerations R=1 and R=2 with a TFE factor of 216. Using 3 shots per segment, this leads to 648 k-space lines per segment. We used flip angles of $8°$, TR of 4.6 ms and TE of 2.1 ms. The acquisitions lasted 6 minutes and 44 seconds for R=1 and 3 minutes and 29 seconds for R=2, and were performed in a sagittal orientation. Coil sensitivity maps were again obtained from the automatically acquired SENSE reference scan, compressed to 8 coils and normalized.

4.3 Results

For evaluation, we compare our method against DISORDER, which is closest to our work and an excellent example of another navigator-free and retrospective approach. In Section 4.3.1, we evaluate both the DISORDER and proposed DEEP-DISORDER approach on retrospectively motion-corrupted data. Section 4.3.2 shows the performance of both approaches on deliberately prospectively motion-corrupted

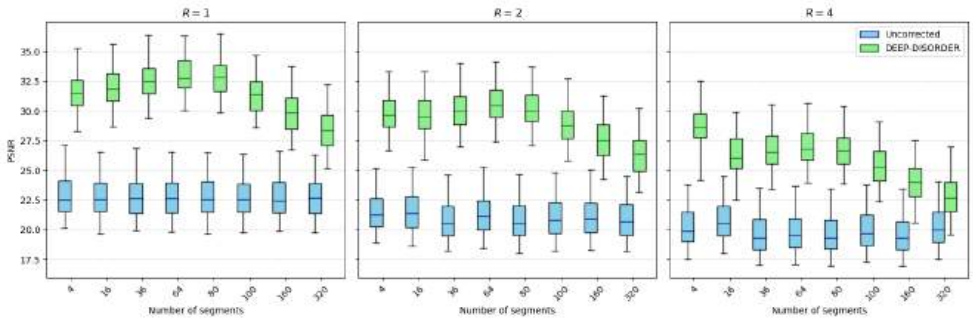


Figure 4.2: Motion-corrected reconstruction performance on synthetic data with different numbers of shots, with one shot per segment.

MRI acquisitions of healthy volunteers. We selected the segment size to be 3 shots for both approaches on the retrospectively corrupted data. The final reconstructions were obtained by using the estimated motion parameters in the exact same motion-corrected CG-SENSE reconstruction algorithm with the same settings for both DISORDER and DEEP-DISORDER. Thus, the only difference in reconstruction quality between the approaches comes from the estimated motion. As is well known [128], running the CG algorithm for too many iterations can lead to amplified noise in the final reconstruction, so we selected 10 and 3 iterations for the retrospectively and prospectively corrupted data, respectively. The reconstruction quality was evaluated using two measures: the Structural Similarity Index (SSIM) [124], which quantifies perceptual similarity by comparing luminance, contrast and structural information; and the Peak Signal-to-Noise Ratio (PSNR), which quantifies reconstruction fidelity via the ratio of the maximum possible signal power to the error power.

4.3.1 Evaluation on Retrospectively Motion-Corrupted Data

Table 4.1 shows a comparison on the retrospectively motion-corrupted test set between a baseline uncorrected approach that uses a zero-motion estimate, DISORDER and DEEP-DISORDER. Each approach estimated 64 motion states. The first row evaluates DISORDER and DEEP-DISORDER on motion-free data. DISORDER estimates motion closer to zero due to the nature of its optimization loop, but neither approach deteriorates image quality. For medium random motion and nodding motion, both approaches correct the motion similarly well enough to produce near-perfect reconstructions. Of these, the pattern with the lowest SSIM after correction (R=2 intrasegment nodding motion) is shown in the top row of Figure 4.3. The motion-corrected image is almost indistinguishable from the motion-free image, bar a slight increase in noise. The remaining images show the performance for extreme motion. Once again a major improvement by both approaches over the uncorrected image can be seen, but unlike

Simulated Motion Pattern	Motion Estimation	Motion-Corrected Reconstruction SSIM ($\mu \pm \sigma$) $R = 1$	$R = 2$	$R = 4$	Mean Motion Error mm	degrees	Segment Recon SSIM
Still	None	1	0.999 ± 0.000	0.968 ± 0.005	–	–	–
	DISORDER	1.000 ± 0.000	0.999 ± 0.000	0.968 ± 0.005	0.01 ± 0.00	0.01 ± 0.00	–
	DEEP-DISORDER	0.999 ± 0.000	0.998 ± 0.001	0.967 ± 0.006	0.05 ± 0.01	0.09 ± 0.01	0.823 ± 0.023
Random intersegment	None	0.791 ± 0.021	0.768 ± 0.024	0.743 ± 0.026	–	–	–
	DISORDER	$1.000 \pm 0.000^{\dagger}$	$0.996 \pm 0.001^{\dagger}$	$0.957 \pm 0.006^{\dagger}$	0.03 ± 0.01	0.06 ± 0.02	–
	DEEP-DISORDER	$0.999 \pm 0.000^{\dagger}$	$0.995 \pm 0.001^{\dagger}$	$0.955 \pm 0.007^{\dagger}$	0.06 ± 0.01	0.13 ± 0.02	0.839 ± 0.021
Random intrasegment	None	0.790 ± 0.022	0.768 ± 0.024	0.744 ± 0.027	–	–	–
	DISORDER	$0.997 \pm 0.000^{\dagger}$	$0.991 \pm 0.001^{\dagger}$	$0.951 \pm 0.007^{\dagger}$	0.21 ± 0.01	0.12 ± 0.01	–
	DEEP-DISORDER	$0.996 \pm 0.001^{\dagger}$	$0.990 \pm 0.002^{\dagger}$	$0.950 \pm 0.007^{\dagger}$	0.20 ± 0.01	0.16 ± 0.02	0.838 ± 0.021
Nodding from Tracking Device intersegment	None	0.941 ± 0.026	0.915 ± 0.035	0.844 ± 0.054	–	–	–
	DISORDER	$1.000 \pm 0.000^{\dagger}$	$0.997 \pm 0.001^{\dagger}$	$0.959 \pm 0.007^{\dagger}$	0.04 ± 0.01	0.07 ± 0.02	–
	DEEP-DISORDER	$0.999 \pm 0.000^{\dagger}$	$0.996 \pm 0.001^{\dagger}$	$0.957 \pm 0.008^{\dagger}$	0.06 ± 0.01	0.13 ± 0.02	0.838 ± 0.021
Nodding from Tracking Device intrasegment	None	0.942 ± 0.026	0.917 ± 0.035	0.847 ± 0.054	–	–	–
	DISORDER	$0.993 \pm 0.003^{\dagger}$	$0.987 \pm 0.006^{\dagger}$	$0.940 \pm 0.017^{\dagger}$	0.25 ± 0.10	0.42 ± 0.11	–
	DEEP-DISORDER	$0.992 \pm 0.003^{\dagger}$	$0.986 \pm 0.006^{\dagger}$	$0.938 \pm 0.018^{\dagger}$	0.25 ± 0.09	0.44 ± 0.11	0.834 ± 0.022
Spatially Extreme Random intersegment	None	0.708 ± 0.025	0.697 ± 0.028	0.684 ± 0.030	–	–	–
	DISORDER	$0.945 \pm 0.020^{\dagger}$	$0.918 \pm 0.029^{\dagger}$	$0.873 \pm 0.032^{\dagger}$	7.14 ± 2.01	2.32 ± 0.81	–
	DEEP-DISORDER	$0.992 \pm 0.003^{\dagger}$	$0.982 \pm 0.005^{\dagger}$	$0.942 \pm 0.010^{\dagger}$	0.16 ± 0.04	0.32 ± 0.05	0.832 ± 0.021
Spatially Extreme Random intrasegment	None	0.707 ± 0.025	0.697 ± 0.028	0.684 ± 0.030	–	–	–
	DISORDER	$0.893 \pm 0.018^{\dagger}$	$0.866 \pm 0.022^{\dagger}$	$0.827 \pm 0.027^{\dagger}$	7.67 ± 1.74	2.63 ± 0.80	–
	DEEP-DISORDER	$0.927 \pm 0.009^{\dagger}$	$0.905 \pm 0.012^{\dagger}$	$0.865 \pm 0.017^{\dagger}$	1.24 ± 0.05	0.83 ± 0.08	0.803 ± 0.024

Table 4.1: Mean and standard deviation of the motion-corrected reconstruction quality on the retrospectively motion-corrupted test set of 94 subjects. The mean registration distance and segment reconstruction quality were calculated for R=1 as it contains all shots. The mean motion error was obtained by averaging the distance between the 64 estimated and 64 or 320 known rigid motion parameters after adding a motion that registers the corrected result to the target. The standard deviation expresses the variation between the population of 94 scans, not between individual segments. † denotes a significant improvement over the uncorrected reconstructions.

before, DEEP-DISORDER corrects it more accurately, both in quantitative measures and in visual appearance. For the extreme intrasegment motion, neither approach sufficiently corrects for the motion. After all, the motion that happens within a segment is not estimated and therefore uncorrected. While this extreme motion pattern effectively illustrates the breaking point of both approaches, the displacements are too large to typically occur in practice. In practice, the breaking point may be at less severe motion since not all effects of motion can be accounted for in the forward model.

4.3.2 Evaluation on Prospectively Motion-Corrupted Data

Table 4.2 evaluates the baseline motion-agnostic reconstruction approach, the DISORDER approach and DEEP-DISORDER on prospectively motion-corrupted in vivo data. The motion-corrected reconstructions are registered to a motion-free reference acquisition for evaluation. The lower neck area was ignored during this registration via a mask as there may be physical deformations between the reconstructed scans and the reference. The mouth and lower neck area were masked out for quantitative evaluation since the brain is the region of interest. The first row of Figure 4.4 shows the reconstructions of both approaches on volunteer acquisitions without intentional motion. The higher noise level compared to the R=1 reference scan can

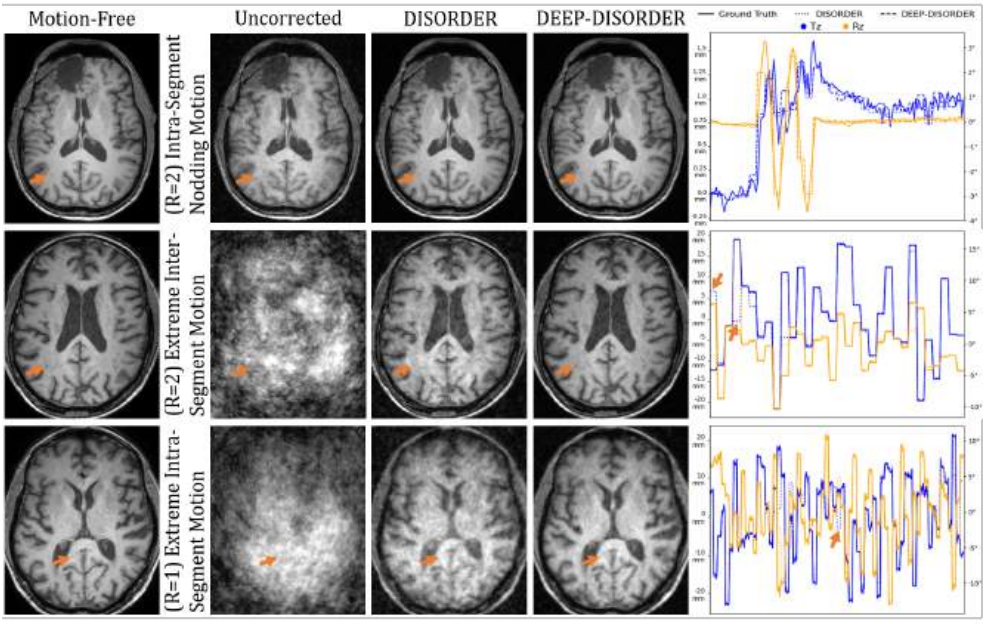


Figure 4.3: Reconstructions of retrospectively motion-corrupted and retrospectively undersampled data. The arrows highlight tiny structures that are rendered imperceptible by motion artifacts. DISORDER and DEEP-DISORDER correct the majority of motion artifacts, making the structures visible again for all but the most extreme motion patterns. The estimated and ground truth motion parameters are shown for a selected axis of translation (T) and rotation (R), with arrows highlighting outlying estimations by DISORDER.

be attributed in large part to the fact that the data is two times undersampled, as the uncorrected reconstruction also shows increased noise. The results demonstrate that both approaches do not meaningfully deteriorate image quality even in cases without motion. On scans where the volunteers performed gradual motions, we see major blurring in the uncorrected reconstructions, especially at the front of the head, as highlighted by the arrows. The values in Table 4.2 do not show a large increase in SSIM for the gradual motion, but the reconstructions of DISORDER and DEEP-DISORDER are greatly deblurred as seen in the figure. Some amplified noise and minor artifacts remain visible, particularly in the R=2 case. The remaining images show examples of severe jittery and extreme motion. The uncorrected reconstructions have lost all detail, while DISORDER and DEEP-DISORDER produce clearly improved reconstructions. The R=1 jittery and R=1 extreme motion-corrected reconstructions no longer contain structural artifacts that hinder diagnostic ability, but the noise is highly amplified relative to the reference scan. Finally, the bottom example with

Volunteer Motion	Motion Estimation	Motion-Corrected Reconstruction Quality ($\mu \pm \sigma$)			
		$R = 1$, SSIM	$R = 2$, SSIM	$R = 1$, PSNR	$R = 2$, PSNR
Still	None	1	0.977 ± 0.002	∞	37.1 ± 0.1
	DISORDER	0.993 ± 0.003	0.979 ± 0.001	40.8 ± 1.8	37.5 ± 0.1
	DEEP-DISORDER	0.994 ± 0.002	0.977 ± 0.001	41.7 ± 1.8	37.3 ± 0.2
Gradual	None	0.915 ± 0.024	0.901 ± 0.032	30.1 ± 1.4	29.7 ± 1.8
	DISORDER	0.937 ± 0.013	0.921 ± 0.027	31.0 ± 0.9	30.6 ± 1.7
	DEEP-DISORDER	0.936 ± 0.014	0.920 ± 0.027	31.0 ± 1.0	30.6 ± 1.7
Jittery	None	0.837 ± 0.034	0.850 ± 0.051	27.3 ± 1.8	27.7 ± 2.3
	DISORDER	0.944 ± 0.009	0.923 ± 0.021	32.2 ± 1.0	30.8 ± 1.6
	DEEP-DISORDER	0.941 ± 0.011	0.919 ± 0.024	31.9 ± 1.2	30.5 ± 1.6
Extreme	None	0.764 ± 0.008	0.775 ± 0.017	24.6 ± 0.7	24.9 ± 1.0
	DISORDER	0.927 ± 0.009	0.910 ± 0.016	30.9 ± 0.8	29.9 ± 0.7
	DEEP-DISORDER	0.923 ± 0.011	0.904 ± 0.014	30.7 ± 0.9	29.6 ± 0.6

Table 4.2: Mean and standard deviation of motion-corrected reconstruction performance on prospectively motion-corrupted volunteer scans with intentional motion.

R=2 extreme motion could not be sufficiently corrected and contains both structural artifacts and severe noise. Interestingly, the same artifacts can be seen here in both the DISORDER and DEEP-DISORDER reconstructions on the left side of the brain, indicating that this motion could potentially be uncorrectable with the used number of segments.

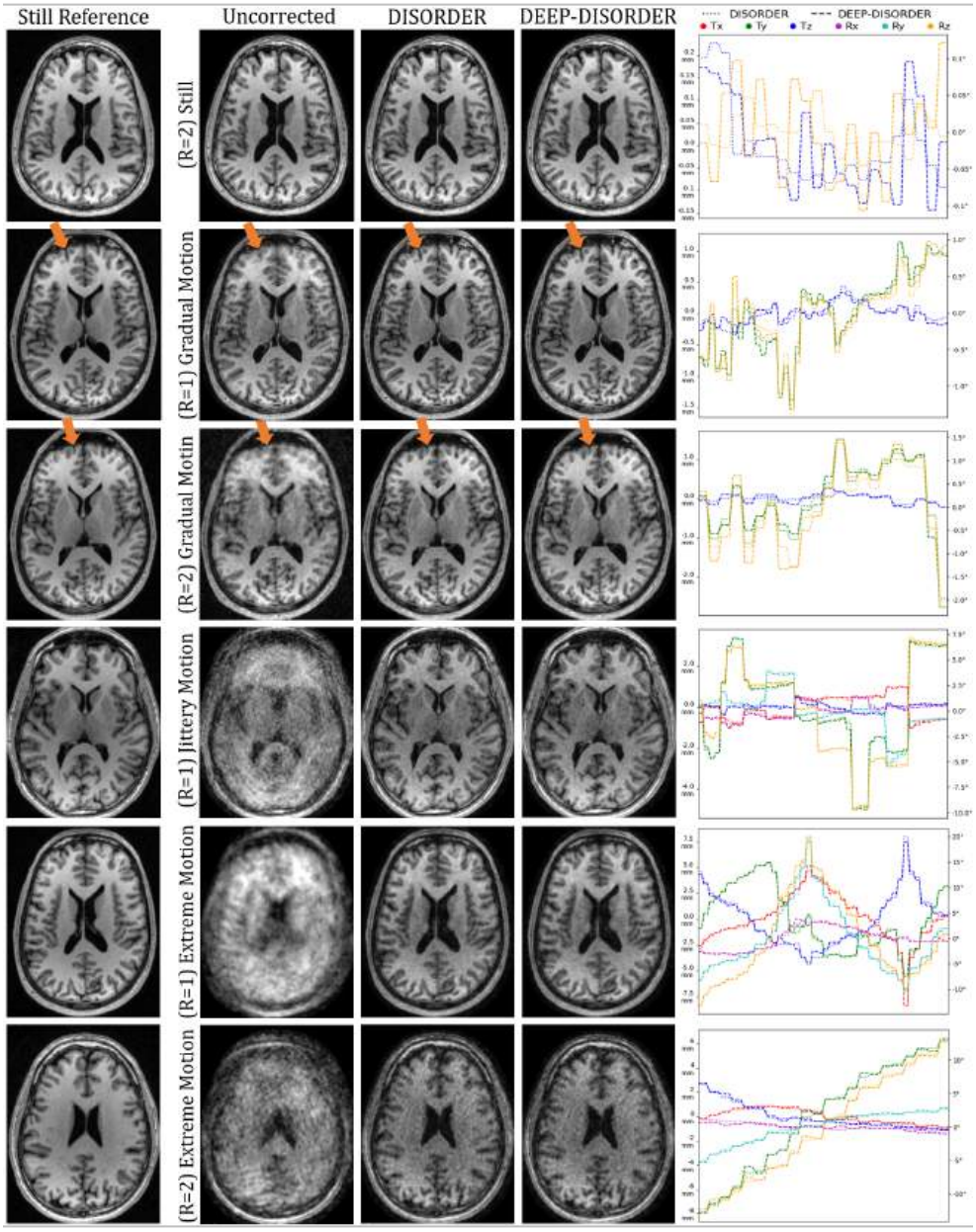


Figure 4.4: Reconstructions of prospectively motion-corrupted data, and corresponding R=1 still reference scans. The arrows highlight blurring caused by volunteer motion, most of which is corrected by DISORDER and DEEP-DISORDER. Arbitrarily selected translations (T) and rotations (R) are plotted on the right.

4.4 Discussion and Conclusion

We presented a framework for motion estimation and motion-corrected reconstruction. The proposed approach reconstructs fractions (segments) of k-space and registers them to estimate the motion of the subject. The estimated parameters can be used in a motion-corrected reconstruction, such as a CG-SENSE reconstruction, to remove the majority of motion artifacts.

On the retrospectively motion-corrupted dataset, both the proposed DEEP-DISORDER and the reference approach (DISORDER) produced visually flawless images in the case of mild intersegment motion. Neither approach perceptually degraded the image quality when no motion was present, indicating there is no downside to applying the motion correction when patients lie still. For spatially extreme intersegment motion, DISORDER saw a drop in performance while DEEP-DISORDER successfully corrected for it with 0.16 mm accuracy. While DISORDER implicitly aligns the segments by optimizing only for the k-space fit, DEEP-DISORDER performs explicit image-domain registration. As DEEP-DISORDER's segment reconstruction approach operates at full resolution, it can generate images with improved low-frequency information that is preserved when downscaling the images. This enables the use of a Gaussian multi-resolution pyramid strategy, which potentially increases the capture range and accuracy. A multi-resolution pipeline for DISORDER on the other hand would entail cropping k-space, reducing data to 4 to 5 k-space lines per shot for the lowest resolution, and it is not clear if DISORDER would still work with this little information. DEEP-DISORDER's explicit registration also enables the use of similarity metrics in the spatial domain, which potentially enriches or regularizes the problem definition, thereby making the optimization easier and more flexible. The experiments with retrospective intrasegment motion demonstrated that rapid motion occurring within a segment will elude correction, introducing mild blurring artifacts. However, in a realistic scenario, the brevity of each segment's acquisition window probably limits the severity of such intrasegment motion. Nevertheless, future research could investigate the possibility of intrasegment correction, potentially in combination with lower resolution registrations or deep-learning based registration to lower the registration time. Another direction for future work could be to explore hybrid solutions where DEEP-DISORDER is used to initialize a DISORDER reconstruction, for example the estimation of large motion by DEEP-DISORDER followed by refinement of the motion parameters at finer temporal scales by DISORDER.

For the prospectively motion-corrupted volunteer data, DISORDER and the proposed approach vastly reduced motion artifacts in the images, particularly for realistic gradual motion. The artifacts and noise amplification from the most extreme motion could not be completely corrected, though these voluntary motions were more severe

than can be expected in practice as a result of our instructions to the volunteers since we wanted to find the breaking point of the correction approaches, and the fact that we deliberately scanned without padding in the head coil. The fact that the motion-corrected reconstructions of the prospectively motion-corrupted data still contain some motion artifacts and noise compared to the motion-free reference scans can be attributed to multiple factors. Firstly, not all effects of motion can be accounted for in the forward model, such as the intricate effects of motion on the coil sensitivities, B0 inhomogeneities and spin history effects. This directly hampers the final motion-corrected reconstruction quality, and potentially indirectly the segment reconstruction quality through less realistic training data. Mismatches between the coil sensitivities and the undersampled data due to motion are known to lead to aliasing and artifacts [129]. Motion also results in suboptimal k-space sampling, thereby increasing the effective undersampling factor. Secondly, we used the basic conjugate gradient solver in our motion-corrected iterative reconstruction. While an optimized reconstruction algorithm with post-processing or denoising may produce better images, this was not studied by us as the focus was on the motion estimation. The approach performed slightly worse on the volunteer dataset when the segment reconstruction model was not fine-tuned on volunteer scans, mostly as a result of different acquisition settings between the test set that was sampled sagittally and the training set that was sampled transversely, negatively impacting the motion estimation. This seems to highlight a downside of our approach, as it involves training a 3D U-Net, which requires enough training data to be available and may limit generalizability if the data is not varied. However, deep learning-based products that can reconstruct a wide variety of contrasts (T1w, T2w, FLAIR, etc) with different acquisition settings and k-space coverage are in the portfolio of every major scanner vendor nowadays. It stands to reason that a segment reconstruction model can similarly learn to reconstruct segments from a wide range of contrasts and different k-space coverages if supplied with enough varied data during training. While we used 64 segments in the experiments, the optimal number of segments could vary depending on factors like the spatio-temporal scale of the acquisition, acquisition geometry, coil profiles and levels of motion. This optimal number is however unknown at scan time, as for example subject motion is not known in advance. Another point of attention is that for inference the motion-corrupted scans should be sampled with DISORDER-like sampling, which samples the same k-space lines as conventional sampling but in a different order to optimize the detectability of the motion.

Registered segments could be potentially checked automatically for quality issues as all segments show the same anatomy. For example, image difference metrics between the registered segment reconstructions could be used to estimate potential issues by corrupted shots within a segment. Additionally, corrupted shots at the border between

two adjacent segments could be detected by checking if the absolute distance between the corresponding motion parameters in the estimated motion trajectory exceeds a certain threshold. During the final motion-corrected reconstruction, individual shots or segments could be rejected or reduced in weight. Furthermore, data consistency metrics during the final reconstruction may help to identify outliers among the shots to reduce their influence on the final image quality. Moreover, when estimated continuously, feedback of estimated motion could help improve the efficiency of MRI examination. For example, the first two segments could already be used to detect motion early in the scan in less than 4% of total scan-time, enabling the MR technician to provide feedback to the patient, or acquiring additional data in an on-line manner when motion is detected during scanning.

In conclusion, this work outlined a framework for retrospectively correcting motion artifacts in 3D MR images. We introduced a deep learning-based segment reconstruction model that can estimate the anatomy of *in vivo* subjects on a fine temporal resolution (1/64th of k-space). These approximate reconstructions enable estimating subject motion using conventional groupwise registration. Finally, the estimated motion parameters can be leveraged in a motion-corrected reconstruction algorithm. In an evaluation on scans from volunteers, the approach did not deteriorate motion-free scans while removing the vast majority of motion artifacts from motion-corrupted scans, making volumetric MRI substantially more tolerant to motion. The proposed approach is modular and flexible in nature. For example, the rigid registration component could be replaced with deformable registration to enable non-rigid motion correction, which can be challenging for conventional iterative approaches. This may be explored in future work.

4.5 Author Contribution Statement

Laurens Beljaards: Conceptualization, Data Curation, Formal Analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – Original Draft Preparation, Writing – Review & Editing. **Martijn Nagtegaal:** Data Curation, Investigation, Methodology, Writing – Review & Editing. **Chinmay Rao:** Data Curation, Software, Writing – Review & Editing. **Yiming Dong:** Data Curation, Software, Writing – Review & Editing. **Matthias J.P. van Osch:** Conceptualization, Funding Acquisition, Methodology, Supervision, Writing – Review & Editing. **Nicola Pezzotti:** Conceptualization, Funding Acquisition, Methodology, Software, Supervision, Writing – Review & Editing. **Mariya Doneva:** Conceptualization, Methodology, Supervision, Writing – Review & Editing. **Marius Staring:** Conceptualization, Funding Acquisition, Methodology, Project Administration, Resources, Supervision, Writing – Review & Editing.

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4.7 Conflict of Interest Statement

The authors declare the following financial interests or personal relationships which may be considered as potential competing interests: Laurens Beljaards and Chinmay Rao declare the grant and research support they receive from Philips. Mariya Doneva declares employment at Philips. Nicola Pezzotti declares former employment at Philips. The C.J. Gorter MRI center receives research support from Philips. The other authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

4.8 Appendix

Table 4.3 shows motion-corrected reconstruction quality on 6 retrospectively motion-corrupted scans with 2 spoiled gradient echo 3D-scans with different echo-times. The echo times of the acquisitions were 7.2 and 13.4 ms with a TR of 31 ms and a flip angle of $17°$, making the first appear closer to the T1w scans that the model was trained on (which had an echo time of 4.6 ms), while the second adds a bit more T2*-contrast into the images. The scans had an acquisition matrix of around size $368 \times 366 \times 70$ with a slice thickness of 2 mm and 1 mm spacing between slices. We center-padded the isotropic reconstructed k-space of each scan to obtain data with a uniform size of 384^3 voxels. Motion estimation was performed with DEEP-DISORDER on center-cropped data with size 192^3, i.e. on a lower resolution, while the final reconstructions were performed on size 384^3 data. The table shows the performance of DEEP-DISORDER using the segment-reconstruction model that was trained on T1w data exclusively, and a version that was fine-tuned on 21×2 spoiled gradient echo training scans.
The results demonstrate that the model performed well on the first echo time even without fine-tuning, but performed worse for the second echo time that is further from T1w data and has a less pronounced skull. Fine-tuning the T1-model on a limited number of spoiled gradient echo scans re-enabled accurate motion correction, restoring reconstruction performance even for the scans at 13.4 ms echo time. Figure 4.5 shows example reconstructions. It can be seen that DEEP-DISORDER can accurately correct

Fine-tuning strategy	7.2 ms echo time SSIM	7.2 ms echo time PSNR	13.4 ms echo time SSIM	13.4 ms echo time PSNR
Uncorrected	0.826 ± 0.015	30.88 ± 1.28	0.805 ± 0.015	32.35 ± 1.06
No fine-tuning	0.988 ± 0.002	45.57 ± 1.35	0.967 ± 0.013	43.54 ± 2.60
Fine-tuning	0.992 ± 0.001	47.50 ± 1.17	0.989 ± 0.003	48.49 ± 1.59

Table 4.3: DEEP-DISORDER motion-corrected reconstruction quality ($\mu \pm \sigma$) on retrospectively motion-corrupted multi-echo spoiled gradient echo data with an echo time of 7.2 ms (left) and 13.4 ms (right).

for motion on higher-resolution data, even when the motion estimation is run on a lower resolution.

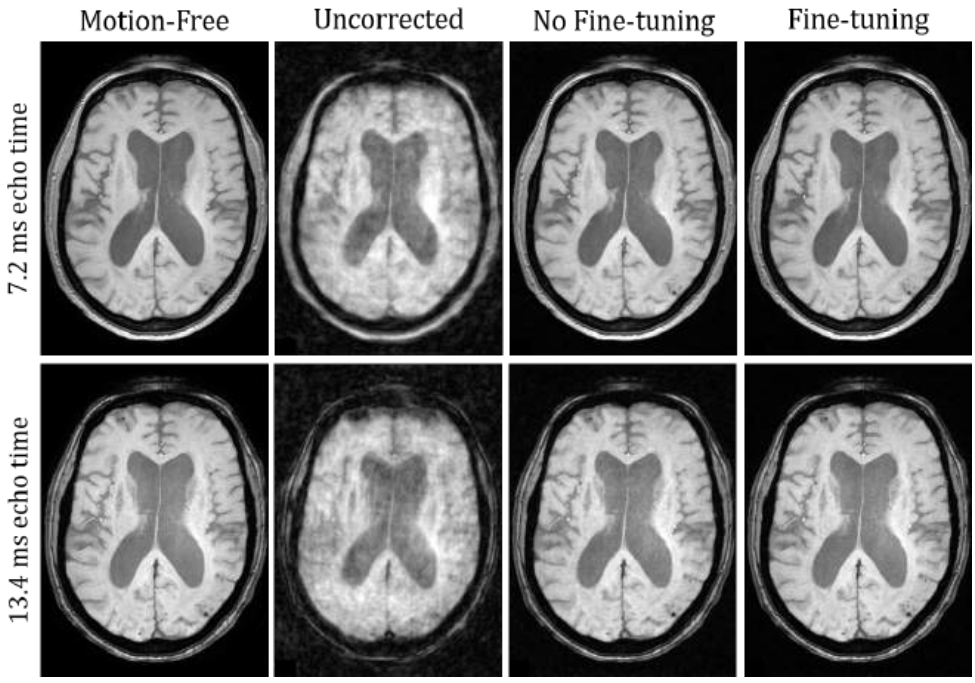


Figure 4.5: Motion-corrected reconstructions by DEEP-DISORDER of retrospectively motion-corrupted multi-echo spoiled gradient echo data with an echo time of 7.2 ms (top) and 13.4 ms (bottom) without and with fine-tuning.

5

Deep-Reorder: Refined Motion Correction for 3D Brain MRI via Iterative Data Regrouping

This chapter was adapted from:

Laurens Beljaards, Matthias J.P. van Osch, Martijn Nagtegaal, Jochen Keupp, Chinmay Rao, Nicola Pezzotti, Mariya Doneva, and Marius Staring (2026). "Deep-Reorder: Refined Motion Correction for 3D Brain MRI via Iterative Data Regrouping". In: Magnetic Resonance in Medicine. doi: 10.1002/mrm.70634

Abstract

Purpose Accurate and temporally high-resolution estimation of subject motion is paramount for effective motion-correction in MRI. This work proposes a novel approach for estimating subject motion within a 3D MRI scan with high temporal resolution.

Methods Starting from an incoherent sampling scheme, temporally ordered k-space fractions are reconstructed via deep learning, enabling motion estimation via registration between each fraction. Postulating that the initial motion-correction sufficiently aligns temporally non-adjacent k-space fractions, it is proposed to iteratively regroup arbitrary fractions of k-space for further correction. This effectively increases the temporal resolution of the approach, enabling individual correction of all shots.

Results When applied to 3D T1w TFE MRI of the brain, the proposed approach "Deep-Reorder" estimated a unique motion state for all shots, producing significantly better estimations of retrospective motion compared to estimation without iterative data regrouping. On prospectively motion-corrupted scans with intentional head movement by volunteers, the proposed approach consistently improved image quality, producing less noisy images with reduced motion artifacts.

Conclusion In conclusion, the proposed approach, which is free of external tracking devices or navigators, effectively estimated 3D motion and corrected motion artifacts with increased temporal resolution.

5.1 Introduction

Patient motion during MRI acquisition can result in artifacts typically seen as ghosting and blurring that degrade diagnostic quality and may in severe cases necessitate a repeat scan. This disproportionately affects populations who find it difficult to remain still, such as children, elderly patients, and those with movement disorders. The effects of motion have measurable financial and throughput consequences for imaging centers, and estimates of annual per-scanner costs attributed to motion can run into the hundreds of thousands of dollars [4, 102]. Consequently, methods that mitigate motion artifacts can partially relieve these problems.

Retrospective motion-correction operates after data collection, seeking to remove or reduce motion-induced artifacts during the reconstruction stage. In recent years, end-to-end AI-driven reconstruction frameworks that learn mappings from motion-corrupted to motion-free images have gained traction due to rapid inference, relatively straightforward implementation, and the increasing availability of training data. These models are often trained on synthetic datasets in which motion is introduced to motion-free images through physics-based simulations of k-space corruption, allowing the network to learn to reduce motion artifacts on paired data in a supervised manner [20, 110, 21]. Recent work has expanded beyond supervised regression models toward generative approaches [113], particularly diffusion models [130, 131], which have shown promise in reconstructing high-fidelity images by iteratively denoising motion-corrupted images. While such methods can produce visually convincing results, their lack of an explicit motion model means that they must deviate from the acquired k-space data to remove artifacts. Under substantial motion, the reliance of these methods on learned priors increases the risk of hallucinations, namely omitting true anatomy or introducing features that are not present in the underlying data [26, 130, 109].

Data-consistent retrospective motion correction approaches are preferred for this reason, but they must solve an ill-posed inverse problem that couples image content and motion, generally requiring a parameterized estimation of the motion.

External tracking hardware can supply motion parameters but often adds complexity to the workflow, raises cost, and may not always deliver sufficiently accurate measurements. Data-driven correction strategies that infer motion directly from the imaging data are attractive as they avoid extra equipment and integrate more naturally into clinical protocols [17].

Many approaches jointly search for the motion parameters and motion-corrected image by alternately solving for one given the other [107, 109, 89, 108]. Notable among these techniques is the DISORDER framework [87], which incorporates a dedicated k-space sampling strategy designed to make the motion easier to detect by

incoherently distributing low and high frequencies across all shots.

In order to accurately describe the motion, one needs to consider sufficiently small time intervals that are at the time scale of the motion or smaller. However, the shorter the intervals, the less data is available per interval and the more difficult it is to estimate the motion parameters. For example, it is not feasible to infer the subject's orientation from a single line in k-space. Naturally, attempting to estimate motion from too short time frames can lead to inaccurate estimations that would even deteriorate the motion-corrected reconstruction. Existing approaches therefore often opt to estimate motion over longer time intervals, guaranteeing that each interval contains enough data to reliably estimate the motion state, though the effective temporal resolution of the motion estimation is then per definition compromised, leaving some motion artifacts.

We propose a novel approach, Deep-Reorder, for estimating subject motion with increased temporal resolution. We demonstrate its effectiveness by combining this idea with DEEP-DISORDER, our previously proposed motion estimation technique that can reliably estimate motion for k-space fractions of only ∼2.3% of the data. Rather than using even smaller k-space fractions, which may contain too little information and result in inaccurate estimations, the proposed approach iteratively recombines data to generate new k-space fractions and corrects the residual motion therein. This effectively increases the temporal resolution to the shot level, at ∼0.8% of the data for the investigated 3D T1w TFE sequence.

Finally, we demonstrate that after sufficient correction with the proposed strategy, a cold-diffusion-based refiner model can be effective in diminishing noise and tiny residual motion artifacts that cannot be completely accounted for in a standard motion-corrected CG-SENSE approach.

5.2 Methods

5.2.1 Problem Definition

The acquisition of k-space y is described by the forward model $y = Ax$, with anatomical image x and the forward operator $A = MFSC_\theta$. S denotes the coil-sensitivity encoding operator, F represents the Fourier transform and M denotes the masking operator that selects the k-space lines measured in each shot. The C_θ operator describes motion-corruption of x in a shotwise manner by projecting the image into different motion states, using rigid motion parameters θ with 3 translations and 3 rotations per shot. Estimated motion parameters $\hat{\theta}$ can be used to reverse displacements and rotations that occurred during acquisition on a per-shot basis. Specifically, we use CG-SENSE adapted for motion compensation [116, 132, 118, 128] to solve the motion-

corrected reconstruction problem

$$\hat{x} = \underset{x}{argmin} \|MFSC_{\hat{\theta}}x - y\|_2^2. \tag{5.1}$$

The main challenge lies in estimating the motion parameters $\hat{\theta}$ from the measured k-space data y.

5.2.2 DEEP-DISORDER

In a previous work, we developed an approach that reconstructs fractions of k-space and registers them to estimate rigid motion parameters for each [27]. This approach, termed DEEP-DISORDER, can be summarized as follows. A 3D-Unet [123] was trained for this reconstruction task, which produces low-quality images due to the short time window. The images are nonetheless accurate enough to estimate the position of the subject at that time using groupwise registration, for example with Elastix [119, 120, 23]. Depending on the sequence, there can be too little information in a single shot for reliable motion estimation, in which case multiple shots are grouped into a "segment" for which a joint motion state is estimated. Specifically, in the case of 3D T1-weighted TFE for which 133 shots are acquired, 3 shots are grouped into one segment. Figure 5.1b illustrates the DEEP-DISORDER approach. The method uses incoherent DISORDER sampling to ensure a mix of low and high frequencies in every shot, making the motion easier to estimate [87]. In the previous work, this approach was demonstrated to have similar performance to the aligned reconstruction of DISORDER [87], a state of the art motion estimation and correction algorithm.

5.2.3 Iterative Regrouping

With DEEP-DISORDER, motion that occurs within the segments is uncorrected. Ideally, a motion state would be estimated for each shot, but several shots need to be grouped into one segment to have enough information for accurate motion estimation [27]. The conventional algorithm makes use of temporal adjacency in the measured data through contiguous segments, by assuming that shots which are adjacent in time are likely to be the closest in motion state and, therefore, can be grouped together. We postulate that initial motion-correction makes temporally non-adjacent k-space fractions sufficiently aligned, such that any shot can hereafter be regrouped with any other shot. After all, assuming that before correction there was an acceptable level of intershot motion for segment reconstruction between time-adjacent shots, a similar acceptable level of intershot motion is expected between all shots after removing intersegment motion via initial correction. After initial correction, we propose to continually regroup the shots into different segments, and estimate residual motion parameters for these precorrected segments, which can be used to further refine the previous motion parameters on a per-shot basis. While the number of motion

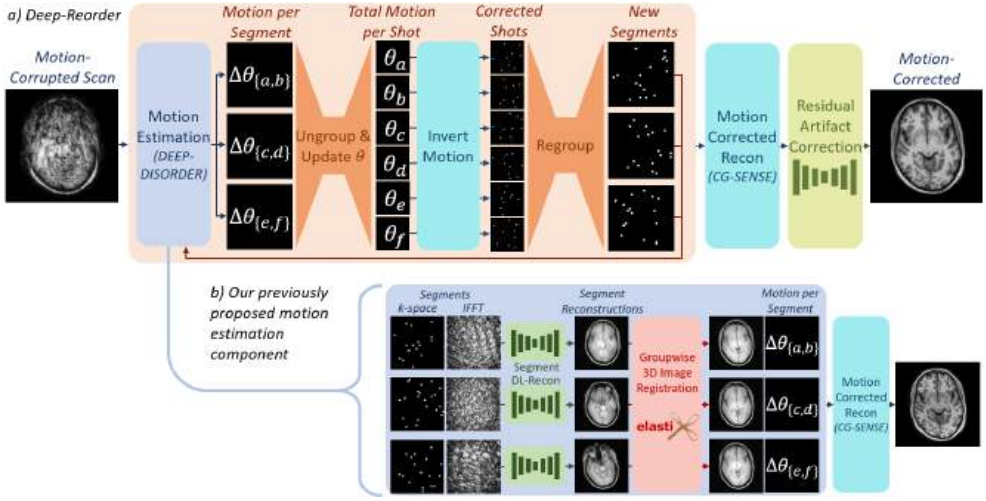


Figure 5.1: **a:** The proposed Deep-Reorder approach iteratively re-uses an existing motion estimation method. After motion is estimated for a segment, which consists of multiple shots (e.g. shots a and b), the individual motion parameters of those shots are updated accordingly. Next, the total parameters are used to invert the motion in the measured data, and new segments are made from different combinations of shots, for which more residual motion may be removed in the next iteration. In the end, each shot can be corrected with its own motion parameters in a final motion-corrected reconstruction. Finally, a refiner network can correct for remaining noise or tiny motion artifacts. **b:** The existing approach DEEP-DISORDER estimates motion on a per-segment basis.

states estimated during a single iteration is still limited to the number of segments, after multiple iterations there will be a unique estimated motion state for each of the shots. This concept is visualized in Figure 5.2. Figure 5.1a illustrates the proposed approach. Algorithm 1 describes conventional motion estimation and subsequent motion-corrected reconstruction, while Algorithm 2 describes the proposed iterative regrouping.

Naturally, the residual motion can be reduced most quickly when shots are grouped that are similar in terms of motion state. Determining which shots are the closest in terms of motion state is by no means trivial, as the premise of our problem is estimating the motion states in the first place. We use a random segment regrouping strategy, which, compared to a heuristic selection of segments, is algorithmically straightforward, may generalize well, and is computationally inexpensive. Additionally, random selection makes it statistically likely that occasionally shots are grouped that are relatively similar in motion state.

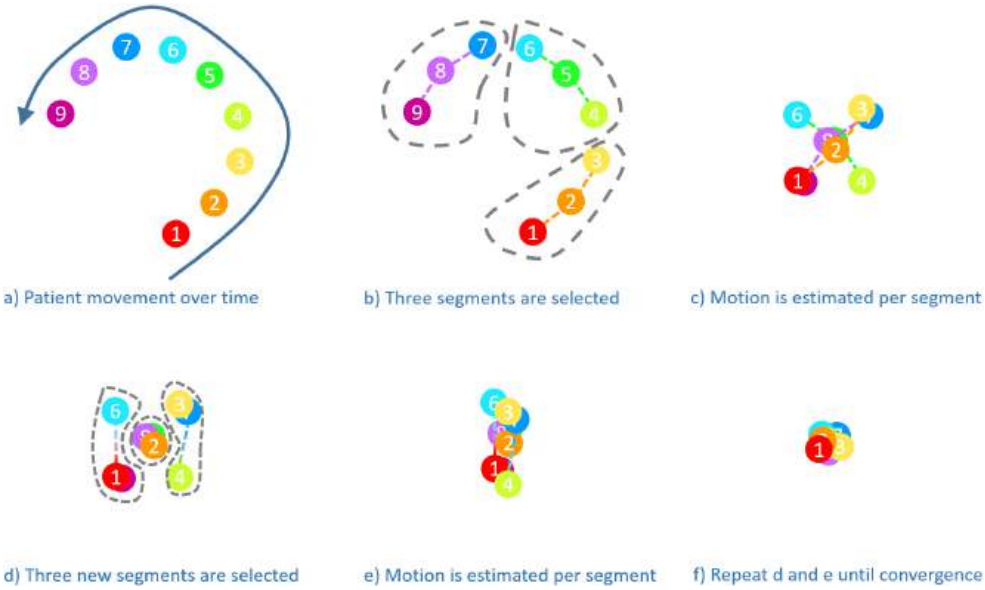


Figure 5.2: Given a simplified motion-corrupted scan with 9 motion states (a), an existing motion correction algorithm accurately estimates motion for 3 segments, but cannot solve intrasegment motion (b, c). The proposed iterative regrouping algorithm improves the temporal resolution of the existing motion correction algorithm from 3 to 9 estimated motion states (d, e, f).

Algorithm 1 Conventional Motion Correction

Data: Motion-corrupted image x, motion estimation function f, motion-corrected reconstruction algorithm $\mathcal{R}$.
Result: Corrected image $\hat{x}$, estimated motion parameters $\hat{\theta}$.

1: $\hat{\theta} \leftarrow f(x)$
2: $\hat{x} \leftarrow \mathcal{R}(x, \hat{\theta})$
3: **return** $\hat{x}, \hat{\theta}$

5.2.4 Maximum Motion Update

Rather than directly adding the newly estimated residual motion parameters, we limit updates of the motion to at most $\Delta^k_{\max}$ via vector-norm clamping, to reduce the impact of potential outlying incorrect estimates and to reduce potential fluctuations when the remaining motion is as small as the expected intersegment error of the used motion estimator. This limit is initially high to enable motion states that were poorly corrected due to low initial temporal resolution to be pulled into the fold, and then decreases according to

$$\Delta^k_{\max} = c/(k + 1/\mu), \tag{5.2}$$

Algorithm 2 Proposed Iterative Regrouping Algorithm

Data: Motion-corrupted image x, motion corruption operator C, motion estimation function f_G using grouping G defining which segment each shot belongs to, n segments and m shots with $n < m$, rearrangement function ρ, motion-corrected reconstruction algorithm $\mathcal{R}$.
Result: Corrected image $\hat{x}$, estimated motion parameters $\hat{\theta}$.

$\hat{\theta} \leftarrow 0$
$G \leftarrow [\underbrace{1, 1, \ldots}_{m/n \text{ times}}, \underbrace{2, 2, \ldots}_{m/n \text{ times}}, \ldots, \underbrace{n, n, \ldots}_{m/n \text{ times}}]$
while stopping criterion not reached **do**
    $\bar{x} = C_{\hat{\theta}}^{H} x$
    $\hat{\theta} \leftarrow \hat{\theta} + f_G(\bar{x})$
    $G \leftarrow \rho(G)$
$\hat{x} = \mathcal{R}(x, \hat{\theta})$
return $\hat{x}, \hat{\theta}$

where k is the current iteration. During the first iteration, which is equivalent to conventional correction, large intersegment motion needs to be corrected and therefore no maximum motion limit is used. Based on this initial estimation of intersegment motion, $\mu = \Sigma_j \Sigma_i |\theta_{i+1,j} - \theta_{i,j}|$ quantifies the estimated motion severity given segment index i and six degrees of freedom j. Thus, larger initial estimated motion leads to a higher maximum motion update limit. The underlying rationale is that a large initial motion estimate is indicative of large intrasegment motion, in which case the iterative algorithm benefits from a larger maximum motion update for faster convergence. Conversely, if the initial estimated motion is low, sudden large residual motion estimations are more likely to be mispredictions and are clamped more strongly. We selected $c = 10$. For two hypothetical examples of $\mu = 0.05$–0.5, this yields maximum translations and rotations $\Delta_{\max}^{k}$ of 0.4–1.4 mm and degrees at iteration $k = 5$ and 0.2–0.4 mm and degrees at iteration $k = 25$.

As a stopping criterion for our algorithm, we use the reference-less Laplacian variance sharpness metric. This metric evaluates image focus by computing the variance of the Laplacian response, which increases with high-frequency detail [133]. If the latest solution has not improved over the best solution in terms of Laplacian variance for a defined number of iterations, the algorithm is stopped and the best solution is accepted. In other words, given the Laplacian variance metric V, stop at iteration k and accept the best corrected image $\hat{x}_{\text{best}}$ if

$$\forall\, t \in \{k - 9, \ldots, k\} :\ V(\hat{x}_t) \leq V(\hat{x}_{\text{best}}). \tag{5.3}$$

where

$$\hat{x}_{\text{best}} = \arg \max_{0 \leq s \leq k} V(\hat{x}_s). \tag{5.4}$$

5.2.5 Cold-diffusion-based Refiner

After correction, in case of severe motion, there may remain residual motion artifacts and amplified noise that cannot be completely accounted for in a motion-corrected CG-SENSE reconstruction. As a final step, we employ a cold-diffusion-based model to correct for these effects. We reason that since the residual motion artifacts are not adequately described by pure Gaussian noise, deep-learning-based cold diffusion could benefit from learning the characteristic morphology of these artifacts and would therefore be more suitable than classical model-based denoising methods.

We trained a 3D-Unet model [123] with 4 levels to estimate motion artifacts in images with light motion-corruption. Specifically, we define the motion artifacts for training to be the difference between an uncorrupted image and a synthetically lightly motion-corrupted counterpart. The motion was sampled independently along each axis from a normal distribution with $\sigma = t \cdot 0.12$ mm and degrees for t between 0 and 10. An equally weighted sum of SSIM, Huber and Sobel loss was used during training. For data augmentation, the training data was randomly translated and rotated. The images were normalized linearly to a range of -1 to 1, while clamping the lowest and highest percentile of voxels in terms of intensity. During evaluation, the motion artifacts were iteratively estimated four times and subtracted from the motion-corrected reconstruction.

5.2.6 Training and Implementation Details

The segment reconstructions of DEEP-DISORDER are performed by an unrolled 3D U-Net [122, 123, 7] with 16, 32, 64, and 128 feature maps per level. The model uses group normalization, 3D convolution and ReLU operators in each layer. An equally weighted sum of SSIM and smooth-L1 loss was used during training, which was calculated between a motion-free fully sampled 3D image as target and the estimated segment reconstruction. The latter was estimated by the network based on the highly undersampled zero-filled 3D image that it received as input. To account for the various positions the subject may be in, the training data was randomly translated and rotated. Different reconstruction models were trained on retrospectively and on prospectively motion-corrupted data for each acceleration factor. Finally, accounting for the fact that the prospectively motion-corrupted scans may be out-of-distribution, the models were finetuned using leave-one-out cross-validation on motion-free volunteer scans. Training was performed on an NVIDIA RTX A6000 GPU for 3×10^5 iterations, using a batch size of 2 and a group normalization size of 8. Groupwise registration was performed using the Elastix registration toolbox. A multi-resolution pyramid of 4 resolutions was used, and was run for 2500 iterations per resolution. Trilinear interpolation was used in the motion-corruption operator C_θ.

5.2.7 Evaluation Metrics

To assess the quality of the reconstructed images, we employ two widely used reference-based image quality metrics: Structural Similarity Index Measure (SSIM) and Peak Signal-to-Noise Ratio (PSNR) [124, 134]. SSIM evaluates perceptual similarity by comparing luminance, contrast and structural information between the reconstructed image and the ground truth. PSNR quantifies the pixelwise fidelity between the reconstructed image and the ground truth. The ground truth is either the uncorrupted image in the case of retrospective motion, or a separately acquired motion-free reference scan in the case of prospective motion.

5.2.8 Data

Retrospectively motion-corrupted data: An in-house collection of 403 volumetric brain scans without intentional motion was used, acquired with a 3D T1-weighted turbo field echo (TFE) sequence. All participants provided written informed consent and the study protocol was approved by the institutional review board. The dataset was partitioned into a synthetic training and validation set of 361 cases and a synthetic test set of 42 cases. All scans were acquired on 3T scanners (Ingenia 3.0T and Ingenia Elition 3.0T X, Philips Healthcare, Best, The Netherlands). The sequence parameters included a flip angle of 8°, repetition time TR ≈ 9.9 ms, and echo time TE ≈ 4.6 ms. Data were collected with an effective acceleration of $R = 1.8$. Coil-combined images were obtained using the vendor's compressed sensing reconstruction software. As the original acquisition matrices varied across subjects, each volume was resampled by cropping or zero-padding in k-space to a uniform size of 192^3 voxels. Coil sensitivity maps were extracted from the SENSE reference scan that is automatically acquired with sequences requiring sensitivity information. Sensitivity maps were compressed to 8 virtual coils using singular value decomposition–based array compression and were normalized using root-sum-of-squares normalization [126].

Paired motion-free and retrospectively motion-corrupted k-space datasets were produced by forward-simulating the MR acquisition from the motion-free coil-combined images, applying different 3D rigid motion to each of the 133 shots. The "nodding" motion pattern uses intentional head nodding and shaking trajectories captured on a scanner with an optical tracking system in a previous study [98], from which we sampled 133 motion states at uniform temporal intervals. The "Perlin random" motion pattern uses smooth trajectories generated using Perlin noise with a base amplitude of 8 and four noise periods along each spatial axis, producing coherent low-frequency motion. The "uniform random" motion pattern uses 133 motion states sampled independently along each axis from a uniform distribution with $[-4, 4]$ mm and degrees per axis, while the "normal random" pattern is sampled from a normal distribution with $\sigma = 4$ mm and degrees. The simulation emulated the TFE

acquisitions with 216 k-lines per shot, matching the TFE factor of the prospectively motion-corrupted data.

Prospectively motion-corrupted data: Additional 3D T1-weighted prospectively motion-corrupted datasets were obtained from our previous study [27], consisting of four healthy volunteers who provided written informed consent, three of whom were used for testing and one for validation. The scans used a 3D TFE [127] sequence on a 1.5T scanner (Ingenia, Philips Healthcare, Best, The Netherlands), using DISORDER sampling [87]. The sequence parameters included a flip angle of $8°$, repetition time TR ≈ 4.6 ms, and echo time TE ≈ 2.1 ms. Volumes were acquired with a matrix size of $191 \times 191 \times 384$ and a TFE factor of 216. Coil sensitivity maps were obtained from the automatically acquired SENSE reference scan, compressed to eight virtual coils, and normalized prior to use in reconstruction. Data were collected using an elliptical shutter with prospective accelerations of $R = 1.0$ and $R = 2.0$, consisting of 133 and 68 shots, respectively. During reconstruction, we discarded the last shots to make the total number divisible by 3 and keep the algorithm simpler. This reduced the number of shots used to 132 and 66, thereby slightly increasing the undersampling factor of the data to 1.01 and 2.06. Alternatively, it would be possible to correct the remaining shots by randomly leaving different shots out per iteration or by adding a few duplicate shots, but in our case the acceleration is low enough that the CG-SENSE reconstruction can recover the discarded data. Total acquisition times were 6 min 44 s for $R = 1$ and 3 min 29 s for $R = 2$.

During each scan, volunteers were instructed to follow one of three predefined motion patterns so that characteristic forms of head motion could be reproduced under controlled conditions. In the "gradual" motion condition, volunteers occasionally performed slow head rotations to mimic the drift commonly observed in clinical examinations as subjects relax or fall asleep. In the "jittery" motion condition, volunteers were instructed to make abrupt head rotations in a random direction at intermittent intervals. In the "extreme" motion condition, volunteers began the scan looking as far as possible toward the lower left and progressed in discrete steps until they were facing the upper right. Several volunteers completed the instructed motion early and then reversed direction toward the starting pose. The acquisitions were performed without padding in the head coil to allow larger head excursions. Finally, motion-free scans were acquired for reference. In total, 8 scans were acquired per volunteer.

5.3 Results

5.3.1 Evaluation on Retrospectively Motion-Corrupted Data

Figure 5.3 shows DEEP-DISORDER and the proposed approach on retrospectively motion-corrupted 3D T1w TFE scans. As illustrated by the plots of the estimated

Motion Pattern	Reconstruction Approach	Motion-Corrected Reconstruction Quality ($\mu \pm \sigma$) Motion Error	SSIM	PSNR
Nodding	Uncorrected	2.59 ± 1.05	0.898 ± 0.026	28.2 ± 1.6
	DEEP-DISORDER	$0.76 \pm 0.27*$	$0.973 \pm 0.010*$	$35.7 \pm 2.0*$
	Deep-Reorder	$0.27 \pm 0.09*\dagger$	$0.992 \pm 0.005*\dagger$	$42.1 \pm 2.5*\dagger$
Perlin Random	Uncorrected	6.38 ± 0.12	0.794 ± 0.016	23.0 ± 1.0
	DEEP-DISORDER	$0.63 \pm 0.03*$	$0.952 \pm 0.006*$	$32.9 \pm 1.1*$
	Deep-Reorder	$0.25 \pm 0.06*\dagger$	$0.992 \pm 0.004*\dagger$	$42.2 \pm 1.9*\dagger$
Uniform Random	Uncorrected	1.92 ± 0.04	0.837 ± 0.014	25.8 ± 0.9
	DEEP-DISORDER	$1.56 \pm 0.03*$	$0.865 \pm 0.013*$	$27.2 \pm 0.9*$
	Deep-Reorder	$0.21 \pm 0.05*\dagger$	$0.994 \pm 0.003*\dagger$	$43.4 \pm 2.2*\dagger$
Normal Random	Uncorrected	3.30 ± 0.08	0.805 ± 0.015	23.7 ± 1.0
	DEEP-DISORDER	$2.76 \pm 0.04*$	$0.822 \pm 0.015*$	$24.9 \pm 0.9*$
	Deep-Reorder	$0.25 \pm 0.06*\dagger$	$0.991 \pm 0.003*\dagger$	$40.6 \pm 2.5*\dagger$

Table 5.1: Reconstruction quality and accuracy of the estimated motion on retrospectively motion-corrupted data. The motion error was calculated as the mean difference between the estimated and ground truth motion over all shots and 3 degrees of freedom, in degrees. The standard deviation reflects the interscan variance of this error. Significant improvements over the uncorrected approach and DEEP-DISORDER are denoted by $*$ and $\dagger$, respectively (Wilcoxon signed-rank test, $p < 0.001$).

motion parameters, DEEP-DISORDER cannot correct 132 states since it needs to group 3 shots per segment to have enough information to make an accurate estimation of the anatomy. The proposed approach is capable of making a unique motion estimation for each of the 132 states, and is consequently able to correct most of the motion artifacts. The random motion patterns in particular, while unlikely to occur in in vivo subjects, aptly demonstrate the increased temporal resolution of Deep-Reorder.

Figure 5.4 shows the performance of the proposed approach per iteration. For these experiments, the algorithm was run for a fixed number of iterations, regardless of the stopping criterion. It can be seen that the solution gradually improves as more iterations are performed up to a certain lower bound. Since motion estimation realistically has limited accuracy, the iterative solution will also contain non-perfectly aligned states, and therefore the error does not reach zero. Nonetheless, the final motion estimation by Deep-Reorder reached sub-voxel accuracy and is sufficiently accurate so that the resulting images appear virtually free of motion artifacts. Figure 5.5 illustrates the quality of the corrected image as more iterations are performed. Table 5.1 shows the quantitative performance of both approaches.

5.3.2 Evaluation on Prospectively Motion-Corrupted Data

Table 5.2 compares uncorrected reconstructions, DEEP-DISORDER and the proposed Deep-Reorder approach on prospectively motion-corrupted 3D T1w TFE scans of

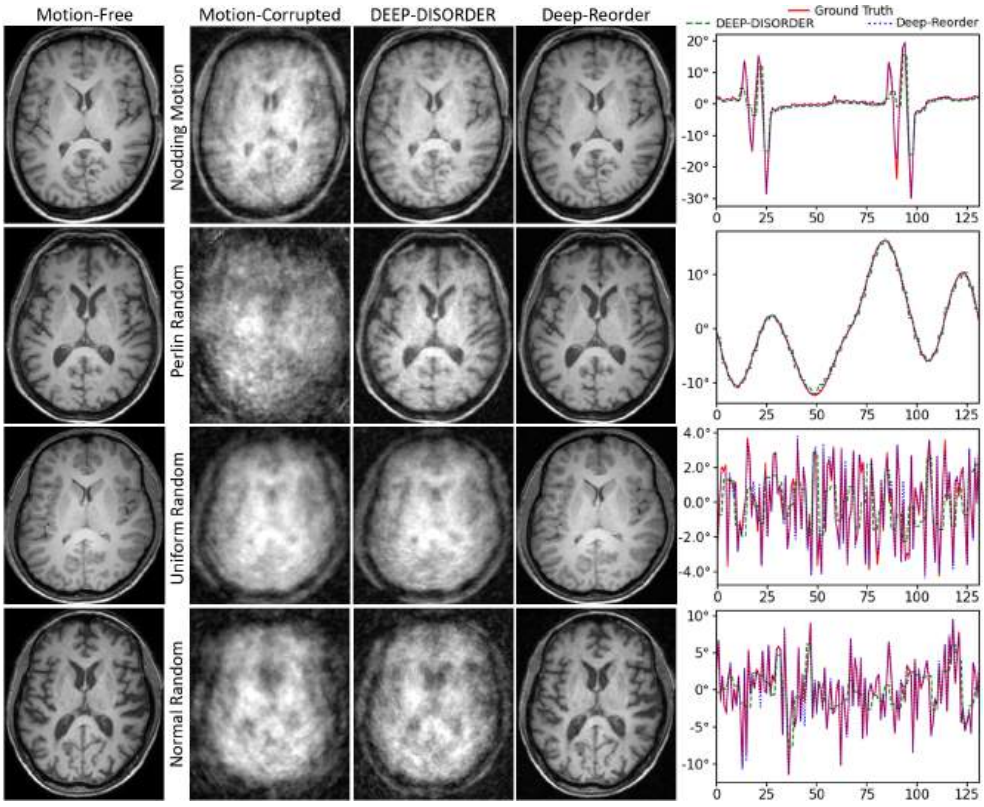


Figure 5.3: Reconstructions of retrospectively motion-corrupted data with 132 motion states. The first row displays nodding motion from a tracking device. The second row displays a pattern sampled from Perlin noise. The pattern in the third row was generated using 132 motion states sampled uniformly over $[-4, 4]$ mm or degrees per degree of freedom, while the fourth pattern was generated using a normal distribution with $\sigma = 4$. The plots show the estimated and known rotational motion about the z-axis (feet-head).

volunteers with intentional motion. The metrics were computed after registering the reconstructed scans to corresponding fully sampled motion-free reference scans. Figure 5.6 displays examples of motion-corrected images. The top row shows a case in which DEEP-DISORDER already performed well, leaving little room for improvement by Deep-Reorder. The other rows provide examples of cases in which the reconstructions still show signs of motion-corruption after correction with DEEP-DISORDER. Deep-Reorder further reduced the corruption across all such cases, yielding noticeably sharper and less noisy reconstructions. The bottom rows illustrate worst-case examples of corruption induced by particularly severe motion, where despite Deep-Reorder considerably improving over DEEP-DISORDER, substantial motion-

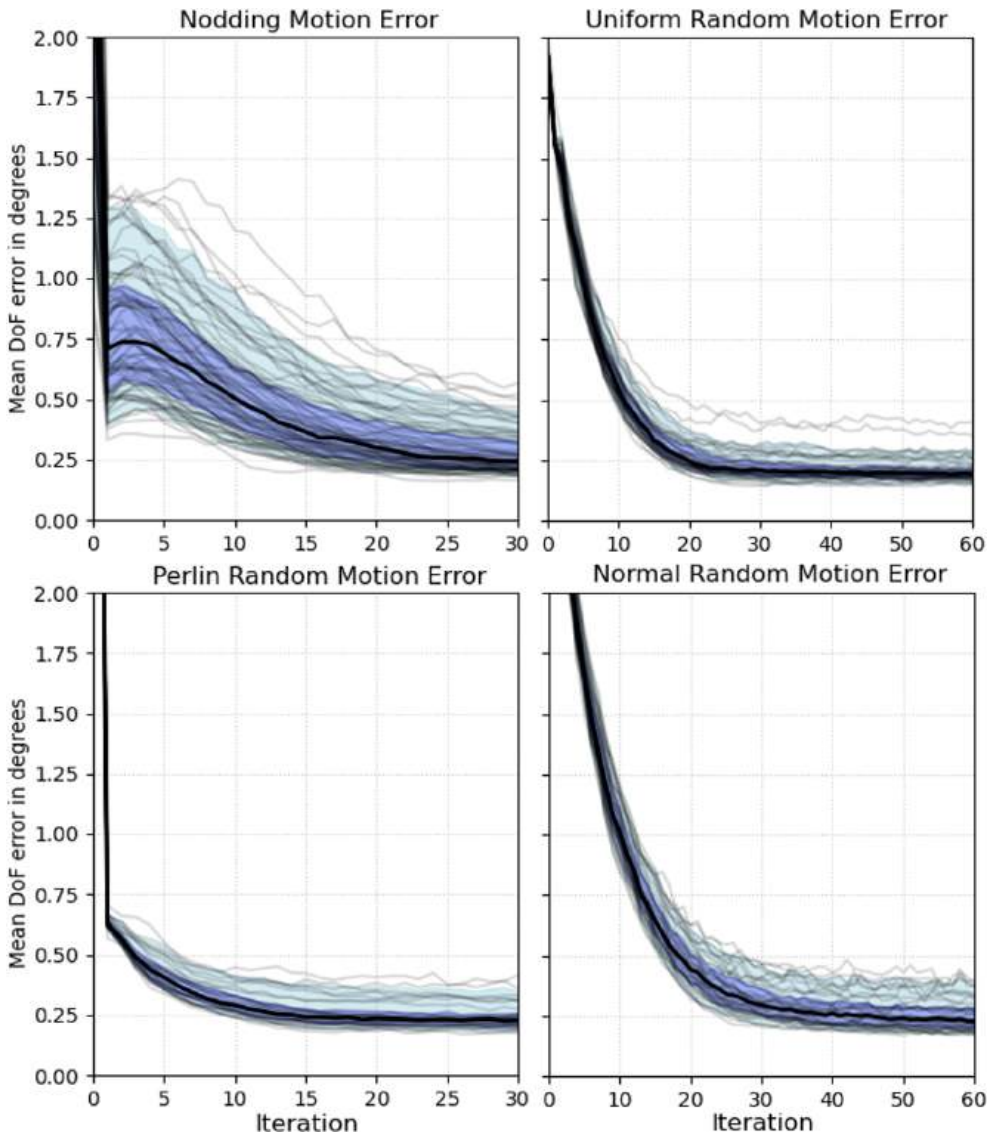


Figure 5.4: Mean rotation error per degree of freedom over iterations between ground-truth motion patterns and estimations by Deep-Reorder for retrospectively corrupted data. Zero iterations corresponds to no correction while one iteration corresponds to the previously proposed approach DEEP-DISORDER.

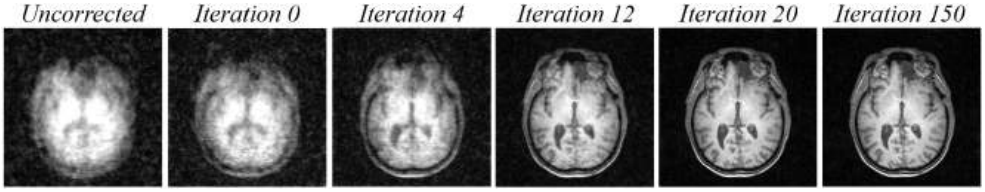


Figure 5.5: Correction quality per iteration on retrospectively motion-corrupted data with random motion.

Volunteer Motion	Reconstruction Approach	Motion-Corrected Reconstruction Quality ($\mu \pm \sigma$)			
		$R = 1$, SSIM	$R = 2$, SSIM	$R = 1$, PSNR	$R = 2$, PSNR
Gradual	Uncorrected	0.929 ± 0.023	0.893 ± 0.027	30.7 ± 1.7	28.3 ± 1.5
	DEEP-DISORDER	0.940 ± 0.013	0.925 ± 0.026	31.2 ± 1.1	30.7 ± 1.7
	Deep-Reorder	0.951 ± 0.003	0.943 ± 0.010	31.8 ± 0.4	31.6 ± 0.9
	Deep-Reorder + refiner	0.952 ± 0.003	0.945 ± 0.009	31.7 ± 0.3	31.6 ± 0.8
Jittery	Uncorrected	0.857 ± 0.032	0.851 ± 0.039	27.7 ± 1.9	26.8 ± 1.8
	DEEP-DISORDER	0.947 ± 0.009	0.928 ± 0.018	32.1 ± 1.0	30.8 ± 1.5
	Deep-Reorder	0.955 ± 0.001	0.944 ± 0.005	32.6 ± 0.5	31.7 ± 0.8
	Deep-Reorder + refiner	0.956 ± 0.000	0.946 ± 0.004	32.4 ± 0.3	31.7 ± 0.7
Extreme	Uncorrected	0.774 ± 0.005	0.789 ± 0.016	24.8 ± 0.7	24.8 ± 1.0
	DEEP-DISORDER	0.940 ± 0.005	0.928 ± 0.006	31.5 ± 0.5	30.5 ± 0.2
	Deep-Reorder	0.947 ± 0.004	0.936 ± 0.009	31.8 ± 0.5	31.0 ± 0.5
	Deep-Reorder + refiner	0.949 ± 0.003	0.939 ± 0.010	31.8 ± 0.4	31.0 ± 0.6

Table 5.2: Reconstruction quality on prospectively motion-corrupted data with intentional motion by volunteers.

related degradation remains in the final reconstruction.

5.4 Discussion

We presented Deep-Reorder, a framework for motion correction in 3D MRI that enables motion estimation with an increased number of estimated motion states. The approach considerably increased image quality in cases with severe motion, which previously could not be corrected due to the limited temporal resolution.

A previously proposed approach DEEP-DISORDER only has enough information when it estimates motion between segments of 3 shots (1/44th of the data), and thus is less effective when motion occurs within those segments. The proposed Deep-Reorder approach on the other hand, gradually estimated a unique set of rigid motion parameters for each shot (1/132nd of the data), enabling individual motion-correction for each, as demonstrated on retrospectively motion-corrupted 3D T1w TFE scans with a random motion state per shot. On retrospectively motion-corrupted scans with nodding and shaking motion from a previous study applied to each shot, this led to an improvement in accuracy of estimated motion parameters from $2.59°$, via $0.76°$ with DEEP-DISORDER, to $0.27°$ with Deep-Reorder. The reconstruction quality improved from 0.898 SSIM when uncorrected, via 0.973 with DEEP-DISORDER, to 0.992 SSIM

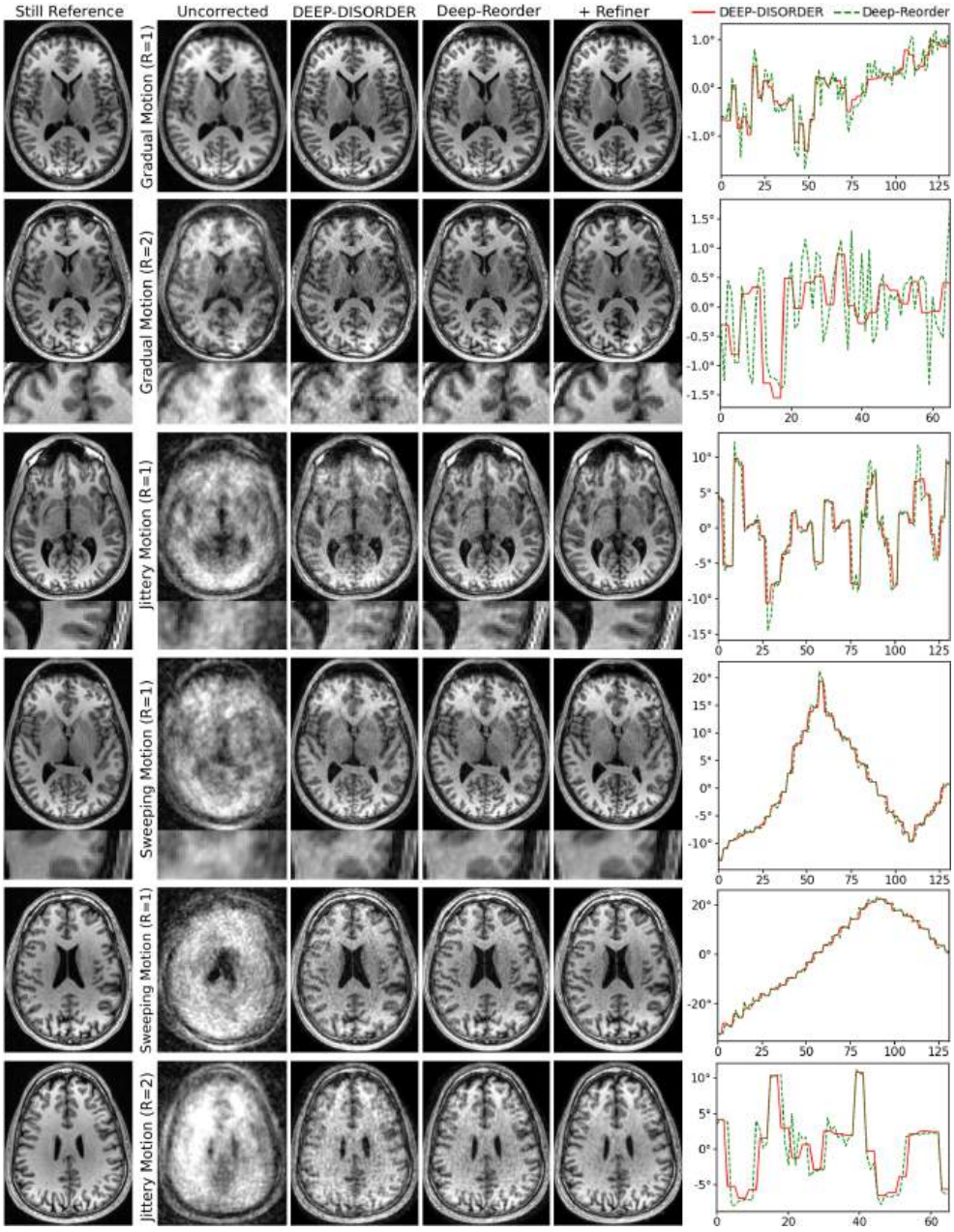


Figure 5.6: Reconstructions of motion-corrupted 3D MRI scans from volunteers performing intentional motion. The rightmost column shows the Deep-Reorder reconstructions after applying the refiner. The estimated rotational motion about the z-axis (feet-head) is displayed.

with Deep-Reorder.

On prospectively motion-corrupted scans with intentional volunteer motion, we observed two situations. In some cases, often with light motion, the previously proposed approach DEEP-DISORDER sufficiently corrected for motion artifacts, in which case the scans could naturally not be further corrected. In the remaining cases where motion artifacts remained after correction by DEEP-DISORDER, the proposed Deep-Reorder considerably improved image quality, producing sharper and less noisy images. For scans corrupted with jittery motion, DEEP-DISORDER increased the SSIM from 0.857 to 0.947, while the proposed Deep-Reorder approach further increased it to 0.955 SSIM. Scans corrupted with gradual motion patterns and spatially extreme motion patterns displayed a similar trend. Additionally, after Deep-Reorder removed the majority of motion-artifacts, a cold-diffusion-based refiner model contributed to an additional reduction in noise and tiny motion artifacts that could not be accounted for in the motion-corrected CG-SENSE reconstruction.

In the end, the proposed approach yielded robust correction across most cases, though a fraction of highly motion-degraded scans retained moderate noiselike residual artifacts. Future work could examine whether intrashot motion estimation can increase the temporal resolution of Deep-Reorder to further correct these cases, or investigate whether the method could be applied to sequences that do not have an inherent shot-like structure, such as gradient echo sequences. In the former scenario, if the underlying motion estimation relies on DISORDER sampling, attention should be paid to grouping sub-shots only in such a manner that a balanced distribution of frequencies is ensured for each segment. Additionally, an increase in temporal resolution will correspond to an increase in computational cost. The runtime of the underlying motion estimation algorithm could be a limitation for the proposed Deep-Reorder framework. Using DEEP-DISORDER as a motion estimation module, one iteration took about 8 minutes for $R = 2$. Thus, while an initial motion-corrected image is visible after this time, a thorough correction of intrasegment motion would take considerably longer depending on the severity of the motion and required number of iterations. This time could be substantially reduced by switching the registration component in DEEP-DISORDER to GPU-based registration, potentially using deep learning. Another option would be to combine the proposed approach with a neural network that directly estimates the motion parameters for each segment, which would also be interesting for investigating the generalizability of the proposed iterative regrouping approach regarding different motion estimation components. In a future work, the choice of new segments in each iteration could be guided by further criteria from the image data or motion trajectories estimated so far, not only by random choice. After all, selecting shots with the most similar current motion state enables the largest amount of correction, potentially speeding up convergence. As another topic for future work,

instead of DISORDER sampling, which we used to ensure a balanced distribution of low and high frequencies in each shot, the approach could be implemented using other compressed sensing incoherent sampling patterns such as CASPR with golden angle profile ordering [135]. A final direction could be to investigate whether the approach could enhance the temporal resolution for non-rigid motion correction as well, a substantially more challenging task for which many more motion parameters need to be estimated.

In conclusion, the iterative regrouping approach Deep-Reorder introduced in this work improved the temporal resolution of an existing motion-correction approach called DEEP-DISORDER, enabling individual motion-correction of all 132 shots in 3D T1w MRI.

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6

Summary, Discussion and Future Work

6.1 Summary

The growing demand for healthcare services has placed increasing pressure on radiological imaging as one of the main diagnostic pathways. In this thesis we leveraged artificial intelligence to reduce inefficiencies in MR imaging workflows by accelerating examinations through the reconstruction of undersampled scans, as well as automatically detecting and correcting patient motion in the scans. A general introduction to the field is provided in Chapter 1. In Chapter 2, we first developed a robust motion estimation approach, based on a multi-task formulation for joint registration and segmentation, outperforming single-task registration. Next, in Chapter 3 we developed a method to automatically detect motion in undersampled MR imaging already during scanning, to allow alerting the MR technician while the patient is still in the scanner. While awareness of patient motion is important, resiliency to its effects on image quality would be preferable. Therefore, in Chapter 4, we developed an approach that estimates the position of the subject at regular intervals during scanning via groupwise registration, leveraging an AI method that enables reconstruction of tiny fractions of the data. The estimated motion parameters are used in a data-consistent motion-corrected reconstruction, greatly reducing the motion artifacts. Building on this approach, in Chapter 5 we proposed an extension that iteratively estimates and corrects remaining motion by randomly re-ordering the raw MR data, resulting in motion estimation at a higher temporal resolution, producing clear images even from severely motion-corrupted MRI acquisitions.

In **Chapter 2**, we developed an inter-scan motion estimation approach for contour propagation. We combined a deformable image registration task, namely propagating manual contours to undelineated follow-up scans, with an image segmentation task, namely delineating the follow-up scans from scratch. On their own, the registration task struggled with propagating contours from organs that had changed in volume between visits, such as the bladder, while the segmentation task was inferior at detecting small organs, such as the seminal vesicles. By combining the approaches in a multi-task cross-stitch network, the accuracy was greatly increased.

Chapter 3 describes a method for automatic motion detection in undersampled MRI. The model was able to differentiate undersampling artifacts from motion artifacts, and accurately quantified the severity of the motion artifacts. This enables alerting the MR technician of motion while the patient is still in the scanner. Additionally, an unrolled reconstruction network was trained on motion-corrupted MRI data. To enable correction of motion in the measurements, the model is able to deviate from the measured data. This enables the model to correct for motion artifacts, though the motion-corrected output appears smoothed and may omit small anatomical details. If no motion is detected, a data-consistent reconstruction model is used instead.

Chapter 4 proposes a novel approach for motion estimation and correction in 3D MRI called DEEP-DISORDER. It estimates the position of the subject at regular intervals (segments) during scanning, via deep learning-based reconstruction and conventional groupwise registration of these segments. The estimated motion parameters are used in a data-consistent motion-corrected reconstruction, greatly reducing the motion artifacts while minimizing the chance of AI-induced hallucinations. An incoherent sampling pattern called DISORDER was used to ensure a balanced distribution of frequencies in each shot for consistent segment reconstruction. The approach requires segments to be large enough, estimated as approximately 2% of the total scan, to enable sufficient quality reconstructions to allow accurate intersegment registration.

Chapter 5 builds upon the previous chapter by proposing a method that increases the number of estimated motion states. The technique, called Deep-Reorder, iteratively estimates and corrects remaining motion artifacts by randomly regrouping segments, thereby enabling individual correction of each shot, which is only 0.75% of a scan. The increase in temporal resolution enables most artifacts to be corrected, even in severely motion-corrupted 3D MRI.

6.2 Discussion and Future Work

Intrashot motion estimation: Deep-Reorder substantially improved the image quality of scans affected by pronounced motion, yet in a number of cases, substantial artifacts and noise persisted after correction. Future work could explore whether intrashot motion estimation can address these remaining issues by further increasing Deep-Reorder's temporal resolution. If the motion-estimation strategy continues to rely on DISORDER sampling, care should be taken to group sub-shots in a way that preserves a balanced frequency distribution within each segment. For instance, when dividing shots into three segments, each segment should include exactly one positive high frequency, one central, and one negative high frequency k-space sub-shot to maintain a balanced frequency distribution. Assuming intrashot motion can be successfully estimated raises an important question: what is the finest temporal resolution that can be achieved with Deep-Reorder? We speculate that as the number of states per

segment increases, convergence becomes disproportionately slower. After all, consider a spatial distribution: as more elements are sampled per group, the likelihood that the groups are substantially different in terms of mean position decreases exponentially. A more deliberate segment selection than random selection could potentially ensure quicker convergence. For example, the choice of new segments in each iteration could be guided by further criteria from the image data or motion trajectories estimated so far.

Foregoing computational limitations, another potential consideration is that intrashot correction creates more boundaries along which inaccuracies can arise by motion correction. After all, if k-lines are placed in different segments, inaccuracies in motion estimation may place these k-lines further apart in terms of motion state than they were originally. However, we expect that in practice this will not have a more detrimental effect on image quality than intershot correction, as the cumulative error across all k-lines will stay approximately in the same range.

Runtime: The runtime of the motion-estimation module may pose a limitation for the Deep-Reorder framework. With DEEP-DISORDER used for motion estimation, a single iteration required roughly 8 minutes at $R = 2$. While intersegment motion is already addressed after one iteration, fully addressing intrasegment motion would demand substantially more time, depending on motion severity and the number of iterations needed. A substantial speed-up may be possible by replacing the current registration module in DEEP-DISORDER with a GPU-accelerated alternative, with GPU-based deep learning registration models being especially fast.

Multi-task segment reconstruction: We could combine the ideas of Chapters 2, 4 and 5, by developing a deep-learning multi-task registration and segment reconstruction approach. Extrapolating the results from Chapter 2, we expect that such a multi-task approach would work better than a single-task network that estimates motion parameters directly from undersampled segments. Moreover, since this conceptual model could be completely deep learning-based, the runtime of the approach would be substantially lower. A point of attention would be the memory requirements to fit all data on the GPU. This could be addressed by performing the motion estimation at a reduced resolution by center-cropping the k-space, or pairwise registration to a common template. Additionally, this approach would demonstrate generalizability of Deep-Reorder regarding different motion estimation components.

Non-rigid motion correction: Another promising avenue would be to explore whether DEEP-DISORDER can be extended to handle non-rigid motion, which requires estimating a substantially larger set of non-rigid motion parameters. A crucial factor is

whether there is sufficient contrast in for example abdomen scans, as proper depiction of internal anatomical structures will be needed to guide non-rigid registration. DEEP-DISORDER was validated on brain MRI where an approximate reconstruction of the skull against an empty background is easier, both for reconstruction from highly undersampled data as well as for registration. Potentially, more shots may need to be combined to ensure sufficient information for each segment. Continuing along this direction, it could be investigated whether Deep-Reorder's iterative regrouping approach could enhance the temporal resolution of non-rigid motion correction as well.

Generalization across modalities: DEEP-DISORDER leverages prior knowledge via the segment reconstruction network to make approximate estimations of the anatomy with less than 2% of the data. The training of the network requires enough training data to be available and may limit generalizability if the data is not diverse enough. However, deep learning-based products that can reconstruct a wide variety of contrasts (T1w, T2w, FLAIR, etc) with different acquisition settings and k-space coverage are nowadays in the portfolio of every major scanner vendor. By the same reasoning, a reconstruction model of an individual segment should likewise be able to learn to reconstruct segments across diverse contrasts and k-space coverages, provided it is trained on a sufficiently large and diverse dataset.

Non-DISORDER sampling: We developed DEEP-DISORDER to work with MR data that was acquired using a specific incoherent sampling pattern, ensuring a balanced distribution of low and high frequencies in each shot. For future work, the approach could be extended to other compressed sensing incoherent sampling patterns such as CASPR with golden angle profile ordering [135], which also ensures a balanced distribution of frequencies. We combined our segment reconstruction and registration approach with DISORDER sampling because we reasoned that shots containing exclusively high frequencies would lack the necessary information to estimate the brain's shape and position. However, with Deep-Reorder, it might be possible to start from larger segments for the higher frequencies and gradually refine their motion estimates, with the segment reconstruction network being trained appropriately for varied segment sizes. If this approach works on conventionally sampled scans without incoherent sampling, the applicability of the approach would increase substantially.

6.3 General Conclusion

In conclusion, this thesis discussed the detection and correction of inter-scan and intra-scan motion using deep learning. Inter-scan registration for contour propagation can be considerably improved by reformulating the problem as a joint registration and

segmentation problem, addressing both tasks jointly in a multi-task learning model. Neural networks can accurately detect and quantify the severity of motion in both fully sampled as well as undersampled MRI. AI-based MRI reconstruction approaches enable proper image quality while less data needs to be acquired, accelerating the scanning process, though end-to-end deep learning approaches may be prone to hallucinations when data is out of distribution or affected by severe motion. Data consistent motion-robust reconstruction, which does not deviate from the measured data, is resilient against hallucinations but requires an estimation of the motion parameters. To obtain motion parameters at regular intervals, deep learning can be leveraged to generate approximate reconstructions of the anatomy for tiny fractions of an MRI scan (2% of the data), which are subsequently registered using groupwise registration. The proposed iterative regrouping approach Deep-Reorder improves the temporal resolution of the motion estimation, enabling individual correction of all 132 shots, demonstrably making in vivo 3D T1w MRI with severe motion by volunteers substantially robust to motion.

7

Nederlandse Samenvatting

7.1 Samenvatting

De toenemende vraag naar zorg leidt tot een steeds grotere druk op radiologische beeldvorming, een van de belangrijkste diagnostische methoden. In dit proefschrift is onderzocht hoe kunstmatige intelligentie (AI) kan worden ingezet om inefficiënties in MRI-onderzoeken te verminderen. Hierbij ligt de nadruk op het versnellen van MRI-onderzoeken door reconstructie met minder gegevens (undersampling), en op het automatisch detecteren en corrigeren van patiëntbewegingen tijdens het onderzoek.

Hoofdstuk 1 geeft een algemene introductie tot het onderzoeksgebied.

In Hoofdstuk 2 is een methode ontwikkeld voor het afleiden van beweging tussen beelden uit verschillende vervolgonderzoeken. De methode is geëvalueerd op CT-beelden, en zou ook voor MRI-beelden gebruikt kunnen worden. De methode combineert het automatisch overbrengen van handmatig getekende contouren naar vervolgonderzoeken (beeldregistratie) met het hersegmenteren van de anatomie (beeldsegmentatie). Hierdoor kunnen zowel organen die sterk in volume veranderen tussen vervolgonderzoeken, waar registratie moeite mee heeft, als kleine organen waar segmentatie weer moeite mee heeft, nauwkeuriger worden gedetecteerd. Het gezamenlijk uitvoeren van deze taken in een multi-task cross-stitch netwerk bleek accurater dan het afzonderlijk uitvoeren van segmentatie of registratie.

Hoofdstuk 3 beschrijft een methode voor het automatisch detecteren van patiëntbeweging tijdens versnelde MRI-onderzoeken waarvoor minder metingen gedaan worden. Het model kan onderscheid maken tussen artefacten die ontstaan door de versnelling en artefacten die door beweging worden veroorzaakt, en kan de graad van bewegingsartefacten nauwkeurig bepalen. Hierdoor zou de MRI-laborant al tijdens het onderzoek kunnen worden gewaarschuwd wanneer een patiënt beweegt. Daarnaast is een reconstructienetwerk getraind op MRI-data waarop synthetische beweging is toegepast. Dit netwerk kan bewegingsartefacten corrigeren, maar omdat afgeweken moet worden van de gemeten gegevens om de artefacten te kunnen corrigeren, kan dit soms leiden tot verlies van kleine anatomische details en een minder scherp beeld. Wanneer geen beweging is gedetecteerd, wordt daarom een reconstructiemethode gebruikt die consistent blijft met de daadwerkelijk gemeten gegevens.

In Hoofdstuk 4 is een nieuwe methode voor het schatten en corrigeren van beweging in 3D-MRI geïntroduceerd, genaamd DEEP-DISORDER. Hierbij wordt de positie van de patiënt tijdens het onderzoek op regelmatige tijdstippen geschat. Dit gebeurt door fracties van de scan (segmenten) eerst met behulp van deep learning te reconstrueren en vervolgens de gereconstrueerde beelden onderling te registreren. De geschatte beweging wordt vervolgens meegenomen in een bewegingsgecorrigeerde reconstructie die consistent blijft met de gemeten gegevens. Hierdoor worden bewegingsartefacten sterk verminderd en wordt de kans op door AI veroorzaakte onjuiste anatomische structuren (hallucinaties) beperkt. Om ervoor te zorgen dat de verschillende delen van de scan voldoende informatie bevatten, is gebruikgemaakt van een onregelmatig MRI-uitleespatroon, DISORDER. De afzonderlijke segmenten moeten genoeg informatie bevatten om beelden van voldoende kwaliteit te kunnen reconstrueren, wat in de praktijk neerkwam op ongeveer 2% van de totale scan per segment.

Hoofdstuk 5 bouwt voort op deze methode met Deep-Reorder, een techniek waarmee beweging op een hogere temporele resolutie kan worden geschat en gecorrigeerd. Door de MRI-data herhaaldelijk op willekeurige wijze opnieuw te groeperen, kunnen resterende bewegingsartefacten iteratief worden geschat en gecorrigeerd. Hierdoor kan uiteindelijk de beweging per afzonderlijke shot (meting tussen opeenvolgende excitatiepulsen) worden gecorrigeerd, waarbij iedere shot slechts ongeveer 0.75% van de totale scan vertegenwoordigt. Deze hogere temporele resolutie maakt het mogelijk om het merendeel van de bewegingsartefacten te corrigeren, zelfs bij 3D-MRI-onderzoeken waarin sprake is van sterke beweging.

Samenvattend laat dit proefschrift zien dat AI kan bijdragen aan efficiëntere en robuustere MRI-onderzoeken. De ontwikkelde methoden maken het mogelijk om beweging tijdens het onderzoek automatisch te detecteren, de beweging nauwkeurig te schatten en de gevolgen ervan voor de beeldkwaliteit te beperken. Hiermee wordt een stap gezet richting snellere, betrouwbaardere en patiëntvriendelijkere MRI-onderzoeken.

7.2 Algemene conclusies

De registratie tussen scans voor het overbrengen van contouren kan aanzienlijk worden verbeterd door het probleem te formuleren als een gezamenlijke registratie- en segmentatietaak, waarbij beide taken gezamenlijk worden uitgevoerd in een multi-task-leermodel.

Neurale netwerken kunnen beweging in zowel onversnelde als versnelde MRI nauwkeurig detecteren en de ernst ervan kwantificeren.

AI-gebaseerde MRI-reconstructiemethoden maken een goede beeldkwaliteit mogelijk terwijl minder gegevens hoeven te worden verzameld, waardoor het scanproces

wordt versneld. End-to-end deep learning-methoden kunnen echter gevoelig zijn voor hallucinaties wanneer de gegevens buiten de trainingsverdeling vallen of door ernstige beweging zijn beïnvloed.

Bewegingsrobuuste reconstructie die consistent blijft met de gemeten gegevens wijkt niet af van de metingen en is daardoor bestand tegen hallucinaties, maar vereist wel een schatting van de bewegingsparameters. Om deze parameters op regelmatige intervallen te verkrijgen, kan deep learning worden gebruikt om benaderende reconstructies van de anatomie te genereren uit zeer kleine delen van een MRI-scan (2% van de data), en deze segmenten te registreren met behulp van groepsgewijze registratie.

Ten slotte kunnen alle 132 shots in in vivo 3D T1-gewogen MRI afzonderlijk worden gecorrigeerd door MRI-data herhaaldelijk op willekeurige wijze opnieuw te groeperen en corrigeren. Dit maakt 3D T1-gewogen MRI zelfs bij sterke beweging aanzienlijk robuuster tegen bewegingsartefacten.

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List of publications

L. Beljaards, M. S. Elmahdy, F. Verbeek, and M. Staring. “A Cross-Stitch Architecture for Joint Registration and Segmentation in Adaptive Radiotherapy”. In: *Proceedings of the Third Conference on Medical Imaging with Deep Learning*. Volume 121. Proceedings of Machine Learning Research. PMLR, 2020, pages 62–74.

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L. Beljaards, M. J. P. van Osch, M. Nagtegaal, J. Keupp, C. Rao, N. Pezzotti, M. Doneva, and M. Staring. “Deep-Reorder: Refined Motion Correction for 3D Brain MRI via Iterative Data Regrouping”. In: *Magnetic Resonance in Medicine* (2026). DOI: 10.1002/mrm.70634.

C. Rao, M. van Osch, N. Pezzotti, J. de Bresser, M. van Buchem, **L. Beljaards**, J. Meineke, E. de Weerdt, H. Lu, M. Doneva, and M. Staring. “A plug-and-play method for guided multi-contrast MRI reconstruction based on content/style modeling”. In: *Medical Image Analysis* 113 (Sept. 2026), page 104160. DOI: https://doi.org/10.1016/j.media.2026.104160.

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Curriculum Vitae

After graduating from the gymnasium at Sancta Maria Lyceum in Haarlem, Laurens obtained both his Bachelor's degree and Master's degree in computer science cum laude at the Leiden Institute of Advanced Computer Science (LIACS). During his Master's, he specialized in Algorithms and Software Technology, with his Master's thesis focusing on the application of artificial intelligence for clinical image registration and segmentation. In 2021, he started his PhD at the Laboratorium voor Klinische en Experimentele Beeldverwerking (LKEB), focusing on novel deep learning approaches for medical image processing, particularly motion-robust accelerated MRI.

CRediT table for the thesis of Laurens Beljaards

Ch.	Type*	Short Title	Conceptualization	Data Curation	Formal Analysis	Funding Acquisition	Investigation	Methodology	Project Administration	Resources	Software	Supervision	Validation	Visualization	Writing – Original Draft	Writing – Review & Editing	Preregistered	Preprinted	Published with Peer Review
1	Introduction	Introduction	■												■	■	☐	☐	☐
2	PhD project chapter	Joint Registration and Segmentation	■	■	■	■										■	☐	☐	☑
3	PhD project chapter	Motion Artifact Severity Estimation	■	■			■	■								■	☐	☐	☑
4	PhD project chapter	DEEP-DISORDER	■	■			■				■	■	■	■	■	■	☐	☐	☑
5	PhD project chapter	Deep-Reorder	■	■			■				■	■	■	■	■	■	☐	☐	☑
6	Discussion	Discussion	■												■	■	☐	☐	☐

**PhD project chapters are the direct result of the PhD project of the PhD candidate.*
Some theses also include Collaboration Chapters, to which the PhD candidate has contributed but fall outside the PhD project.